



Image Data Management and Stewardship

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¹ University of Dundee, ² EMBL-EBI

Bristol, 20 May 2026



Building a national imaging data stewardship skills base



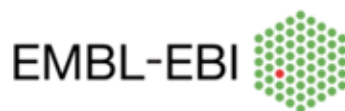
Image Data Resource



BioImage Archive



University
of Dundee



Biotechnology and
Biological Sciences
Research Council



Building a national imaging data stewardship skills base

- Direct training of research technical professionals
- Develop a comprehensive training curriculum
- Train-the-trainer events
- Support role recognition and career pathways

Training and support

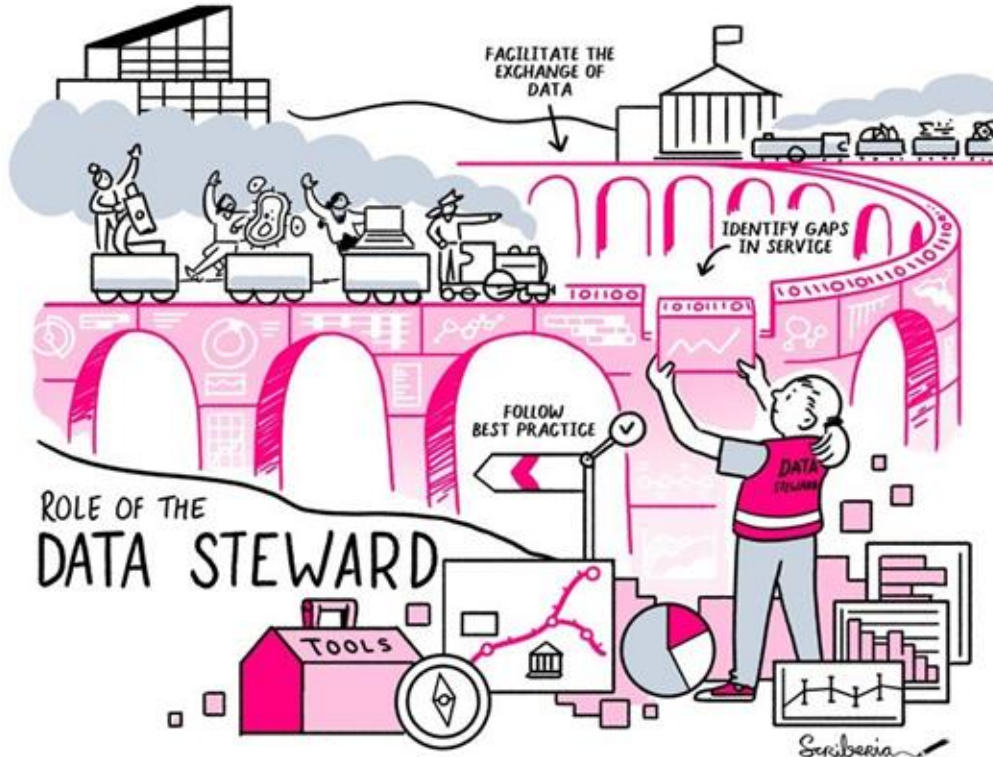
- Training events: combining direct training with train-the-trainer
 - Liverpool, Cambridge, London, Bristol, Glasgow
- Consultation sessions
- Online training materials
 - <https://github.com/BiolImagingUK/Training>
- Join our new community slack channel



Agenda

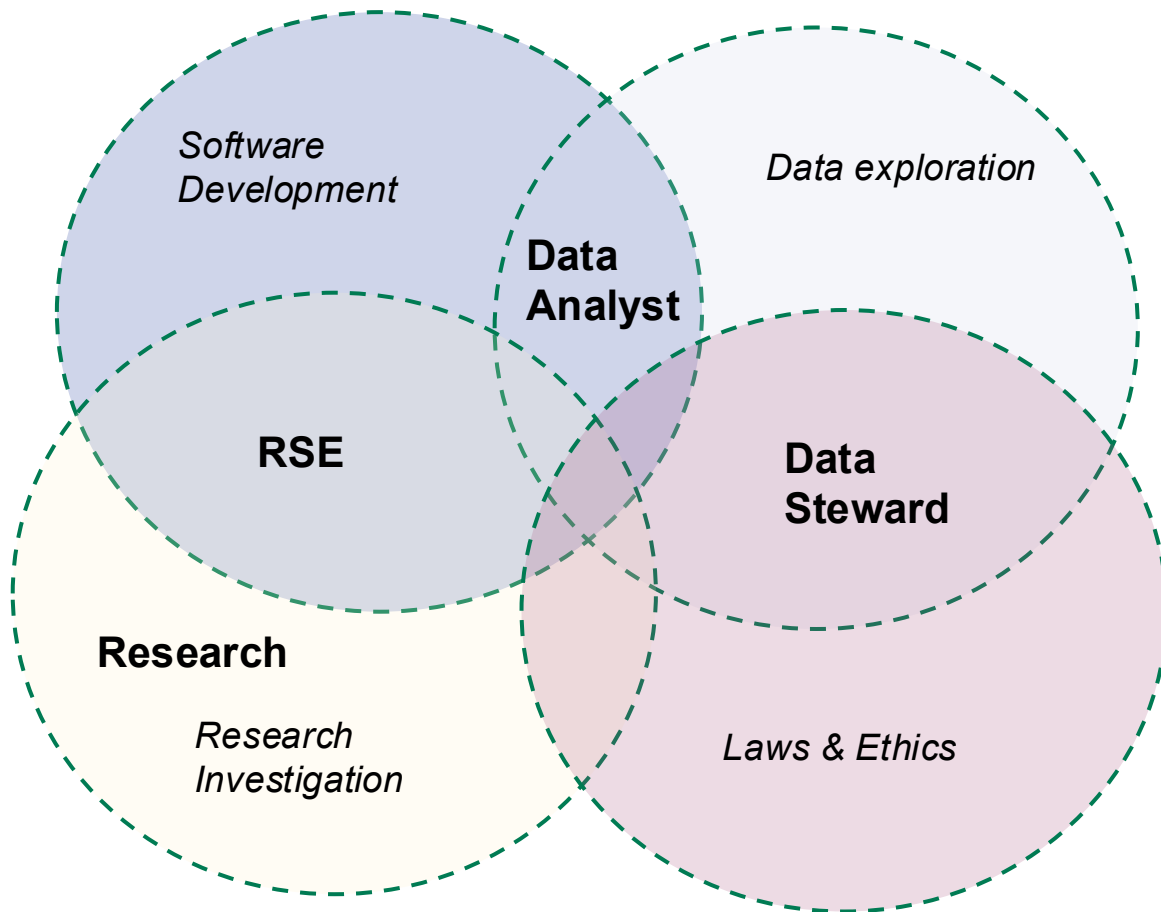
- 09:00 Arrival
- 09:30 - 10:45 Morning session 1
 - Data Stewardship
 - FAIR data
 - Data Life Cycle and planning
- 10:45 - 11:00 Coffee break
- 11:00 - 12:30 Morning session 2
 - Metadata standardisation
 - Tools for Data Management
 - OMERO
- 12:30 - 13:10 Lunch
- 13:10 - 14:30 Afternoon session 1
 - Data management: successful examples and end user tips
 - Storage and sharing strategies and tools
 - Data standardisation: bioformats and OME-NGFF/OME-Zarr
 - Visualisation strategies
- 14:30 - 14:45 Coffee break
- 14:45 - 16:45 Afternoon session 2
 - Preserving raw and processed data, qc checks
 - What do public repositories offer: Success stories
 - Submitting data to public repositories
- 16:45 Moving forward and feedback
- 17:00 End

Data Steward... What does it entail?



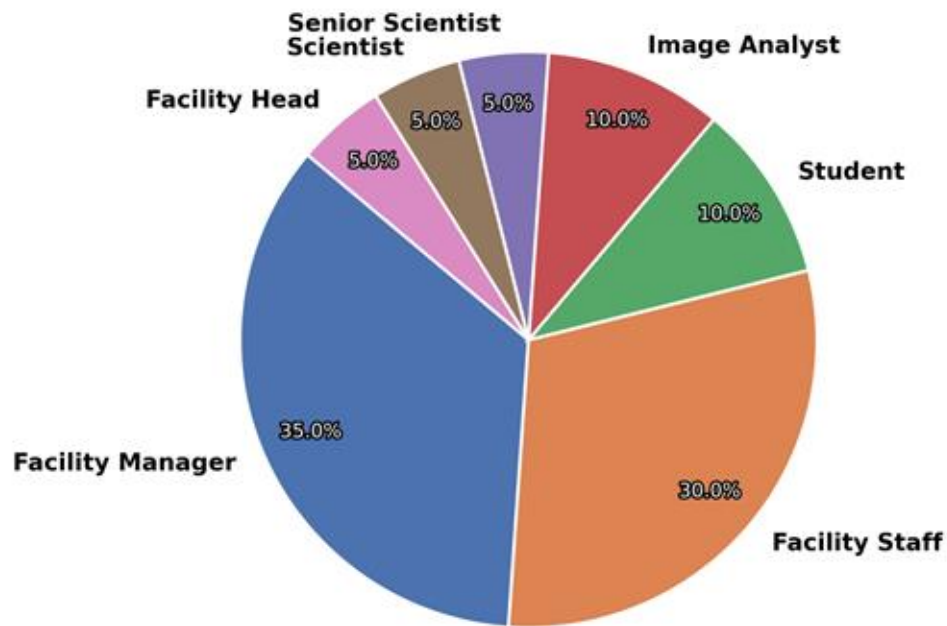
- Complex role
- Interacts with many parts
- Tie together many aspects

Data Steward... Where does it fit?

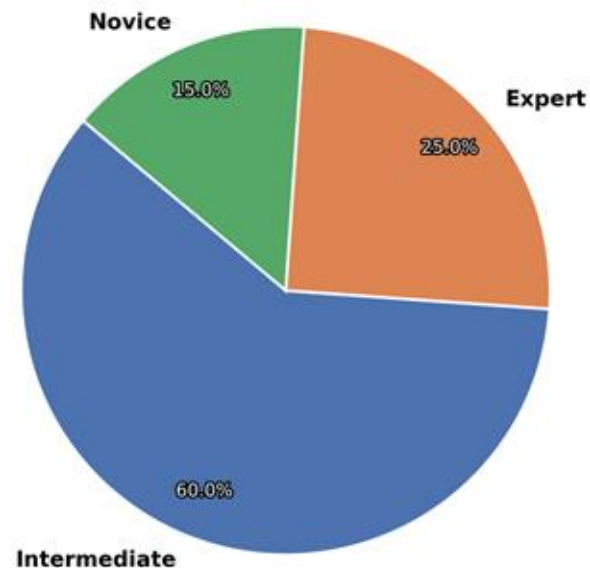


Participants Overview

Role



Level of Research Management



Why do we need image data management?

→ **Data is big and complex, or many!**

- ◆ **Not possible to remember everything about the experiment**; sample preparation, imaging parameters, other variables (antibodies, genes, chemicals, dyes etc.).
 - ◆ **Not possible to remember where all the images are** for all of the experiments. Sometimes one person doesn't have all the images.
- A single person's mind cannot be your data management system.

Exercise (3 minutes)

What do you need to know?

What do you need to know to be able to do your responsibilities?



menti.com
3770 5336

Why do we need image data management?

- **Facility managers:** Need to know where user's data are, what they are about (handling samples fit for facility/instrument use), if the instruments are up and running producing quality images, which instrument is allocated to which user/experiment when.
- **Researchers:** Need to know what the experiments are about and what they are looking for. Therefore need to know how the sample was prepared, how the experiment was conducted (acquisition parameters, conditions etc). If there are different sets they should compare (different acquisition parameters, different samples, different preparation methods, different variables such as genes, drugs, time etc).

They don't always work on a single experiment at a time; you may be writing the paper of an experiment you did a year ago, while collecting data of another experiment.

- **Image Analysts:** Need to know what the experiment is about and what they are looking for. Need to know the image size, what they're running and whether the analysis is working as expected.

Why do we need image data management?

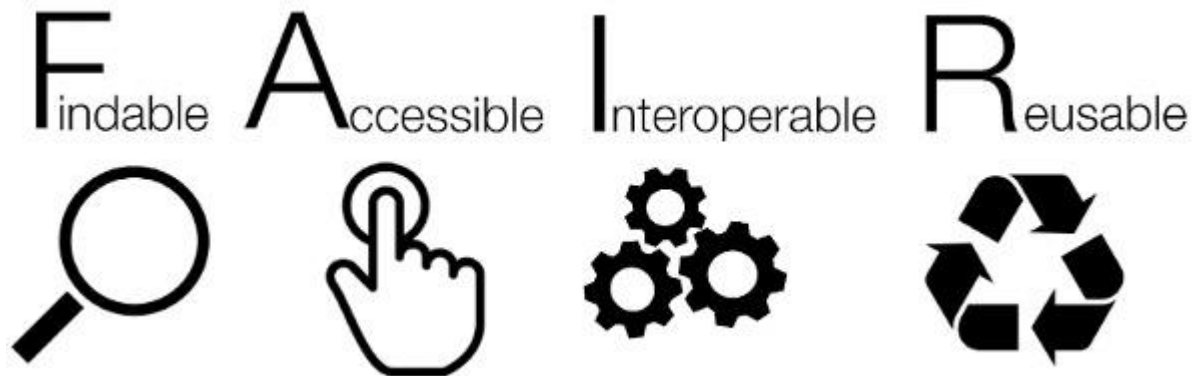
→ Data is big and complex, or many!

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→ A single person's mind cannot be your data management system.

→ **Science is collaborative and therefore different people will need to access and understand the data.**

→ **Using software/code or AI for automating the processing or analysis requires standardised metadata and data.**



The FAIR Guiding Principles for scientific data management and stewardship

<https://doi.org/10.1038/sdata.2016.18>

FAIR: “What does it mean for me?”



Where is my data? Can I search and find it?



Can I access it? Can I download it? *Is it on a disk or server I no longer have access? Is it password protected and I forgot the password?*



Can I read it/open and view it? *Or is the format no longer supported?* Are the data and metadata machine readable?



Can my collaborators analyse it if I give the data file to them? What other information do they need? *Where is that information?* Can they reproduce the images if they do the experiment?

FAIR data vs Open data vs FAIR open data

- FAIR data: Data is FAIR but not open - not shared publicly, or “available upon request”!
- Open data:
 - Find my data on google drive or Cloud
 - Here’s my data as zip file on Zenodo (doi)
- FAIR open data: Here’s my data at BiolImage Archive, EMPIAR or IDR (accession id, doi)

Where can I find Images?

FAIR Bioimage Repositories



BioImage Archive

BioImage Archive is a free, publicly available resource for life sciences images from any imaging modality, at any scale. It supports deposition, search, access and reuse of data that are either associated with a peer-reviewed publication, or of value beyond a single experiment.



EMPIAR (Electron Microscopy Public Image Archive), is a public resource for raw images underpinning 3D cryo-EM maps and tomograms, 3D images obtained with volume EM techniques, and soft and hard X-ray tomography imaging.



IDR (Image Data Resource) is a public repository of reference image datasets from published scientific studies, enabling search, access and reuse of these highly annotated datasets.

What about analytical workflows?



<https://workflowhub.eu/>

A registry for workflows

- Standardised
- Searchable
- Citable
- Interoperable
- Rich metadata

BiaPy 1.0

Visit source

Download RO-Crate

Run on Galaxy

Overview

Files

Related items

Workflow Type: Galaxy

This workflow is executing a BiaPy workflow using a YAML file where the model selection is predefined.

Associated Tutorial

This workflow is part of the tutorial [Execute a BiaPy workflow in Galaxy](#), available in the [GTN](#)

Features

- Includes [Galaxy Workflow Tests](#)
- Includes a [Galaxy Workflow Report](#)
- Uses [Galaxy Workflow Comments](#)

Thanks to...

Workflow Author(s): Riccardo Massei

Tutorial Author(s): Riccardo Massei, Daniel Franco Barranco

Tutorial Contributor(s): Beatriz Serrano-Solano, Leonid Kostrykin, Saskia Hiltemann, Riccardo Massei, Daniel Franco Barranco



1 Creators and Submitter

Creators

Not specified

Submitter

GTN Bot

Discussion Channel

[GTN Matrix](#)

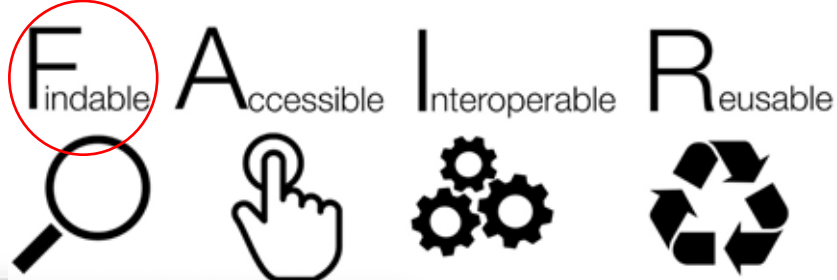
License

Creative Commons Attribution 4.0
International (CC-BY-4.0)

Activity

Views: 718 Downloads: 117 Runs: 0

Created: 16th Feb 2026 at 13:26



BioImage Archive

Search results for **brain**

1 - 10 of 143 results

Collection

- BioImages - Core
- BioStudies - ZCB
- EMPAAR

Released

- 2024
- 2023
- 2022
- 2021
- 2020
- 2019
- 2018
- 2015
- 2014
- 2013

Link Type

- DOI
- PMID
- external
- external link
- github
- image data resource
- bioRxiv

File Type

- raw
- dicom
- dic
- dicm
- img
- imgl
- img2

Region-Specific Protective Effects of Monomethyl Fumarate in Cerebellar and Hippocampal Organotypic Slice Cultures Following Oxygen-Glucose Deprivation

Release Date: 5 July 2024 • Modified: 5 July 2024

Data files

Name	Size	Section	OGD	Treatment	Concentration (μM)	Tissue
DuplicateAndDisplayed DAPI.tif	588.3 MB	Study Component	Yes	MMF	25	Cerebellum
DuplicateAndDisplayed PI.tif	588.3 MB	Study Component	Yes	MMF	25	Cerebellum
KH1-Montage2_wScanDAPI.tif	392.3 MB	Study Component	Yes	MMF	25	Cerebellum
KH1-Montage4_wScanPI.tif	392.3 MB	Study Component	Yes	MMF	25	Cerebellum
KH1.nd	2 KB	Study Component	Yes	MMF	25	Cerebellum

Showing 1 to 5 of 1,337 entries (Filtered from 1,980 total entries) [Show all files](#)

ORCID: Data claiming

You can sign in to ORCID to claim your data

☐ Remember me on this computer

Similar Studies

- Source Data Images for "DOT1L activity affects neural stem cell division mode and differentiation by regulating ASNS expression"
- Transcription profiling of rat intact CNS tissue in vitro using rat organotypic hippocampal slice cultures after anti-Flag-A antibody treatment

[Download all files](#)

EMPIAR-11313

Notice:

Due to changes in Aspera, please use user emp_ext2 instead of emp_ext3 for any command line (ascp) downloads

Principles of mitoribosomal small subunit assembly in eukaryotes

Entry authors:

Harper NJ, Burnside C, Klinge S

Publication:

Principles of mitoribosomal small subunit assembly in eukaryotes

Harper NJ, Burnside C, Klinge S

Nature **614** 175-181 (2022)

PMID: 36482135

DOI: 10.1038/s41586-022-05621-0

Experiment type:

EMDB

Related PDB entries:

8d8k, 8d8j, 8csr, 8css, 8d8l, 8cst, 8csp, 8csu, 8csq

Related EMDB entries:

EMD-27250, EMD-27249, EMD-26968, EMD-26969,

EMD-27251, EMD-26970, EMD-26966, EMD-26971,

EMD-26967

Deposited:

2022-10-21

Released:

2023-02-17

Last modified:

2023-02-17

Dataset size:

73.6 TB

Dataset DOI:

10.6019/EMPIAR-11313

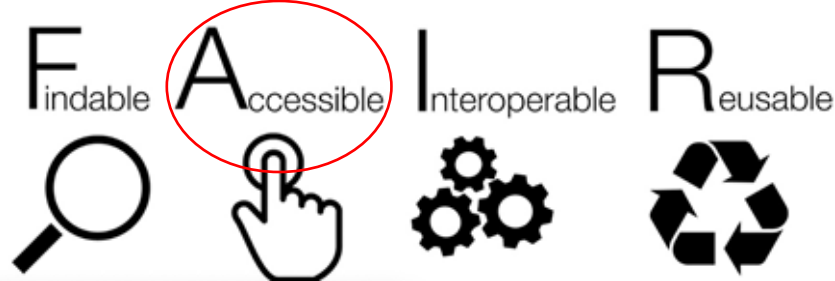
Experimental metadata:

[Download xml](#)

Contains:

 micrographs





BioImage Archive

Home Browse Submit Galleries Help Metadata Help Policies About us Feedback Login

Search results for brain

1 - 10 of 143 results

Collection

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Released

- 2024
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- 2021
- 2020
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- 2018
- 2015
- 2014
- 2013

Link Type

- DOI
- PMID
- external
- external link
- github
- image data resource
- url

File Type

- img
- csv
- txt
- json
- imgml
- imgz

3D light-sheet microscopy data for SELMA3D 2024 challenge - Training subset with no annotations - whole brain images

Ying Chen¹, Johannes C. Paetzold²

¹ Ludwig-Maximilians-Universität München; ² Imperial College London

Accession S-BIAD1197

DOI 10.6019/S-BIAD1197

Description This dataset is the training set containing whole-brain images without annotations for the SELMA3D challenge. The SELMA3D challenge focuses on self-supervised learning for 3D light-sheet microscopy image segmentation. Its objective is to encourage the development of self-supervised learning methods for general segmentation of various structures in 3D light-sheet microscopy images. The dataset contains 3D whole-brain microscopy image of different labeled biological structures, including blood vessels, c-Fos labeled brain cells involved in neural activity, cell nuclei, and Alzheimer's disease plaques.

Keywords light-sheet microscopy, fluorescence microscopy, 3D microscopic image, image segmentation, deep learning

License CC BY 4.0

Publication Self-supervised learning for 3D light-sheet microscopy image segmentation
doi: 10.5281/zenodo.10991463

Biosamples for whole-brain blood vessel images

Organism Mus musculus (mouse)

Description Mice were housed under a 12-h light/12-h dark cycle for 3 months.

Biological entity mouse brain

Biosamples for brain cell nucleus images

Organism Homo sapiens (human)

Images (brain) cell nucleus images of brain cell nucleus images corresponding to the **brain** cell nucleus images. Biosamples for brain cell nucleus...

Release Date: 6 June 2024 · Modified: 22 June 2024

[1CiteIt!](#) [1.75GB](#) [PageTablr](#) [MTP](#) [FTP](#) [Globoz](#)

Data files

Name	Size	Section
COO_0000.tif	9.2 MB	Study Component
COO_0001.tif	9.2 MB	Study Component
COO_0002.tif	9.2 MB	Study Component
COO_0003.tif	9.2 MB	Study Component
COO_0004.tif	9.2 MB	Study Component

Showing 1 to 5 of 43,411 entries

Previous 1 2 3 4 5 ... 12683 Next

Select files to download

Linked Information

Show 5 entries Search: Name

IDR Browse Data

The screenshot shows the IDR Browse Data interface. On the left, there is a list of samples under the 'Explore' tab. The main area displays a grid of microscopy images. The samples listed include:

- idr0125-georgi-influenta
- idr0125-georgi-influenta/screenA
- idr0125-georgi-influenta/screenB
- idr0125-georgi-influenta/screenC
- idr0125-georgi-influenta/screenD
- idr0125-georgi-influenta/screenE
- idr0125-georgi-rhinovirus
- idr0125-georgi-rhinovirus/screenA
- idr0125-georgi-rhinovirus/screenB
- idr0125-georgi-rhinovirus/screenC
- idr0125-georgi-rhinovirus/screenD
- idr0130-oliznenko-herpes
- idr0130-oliznenko-herpes/screenA
- idr0130-oliznenko-herpes/screenB
- idr0130-oliznenko-herpes/screenC
- idr0133-dahlin-cellpainting/screenA
- idr0134-peters-bryophytes/experimentA

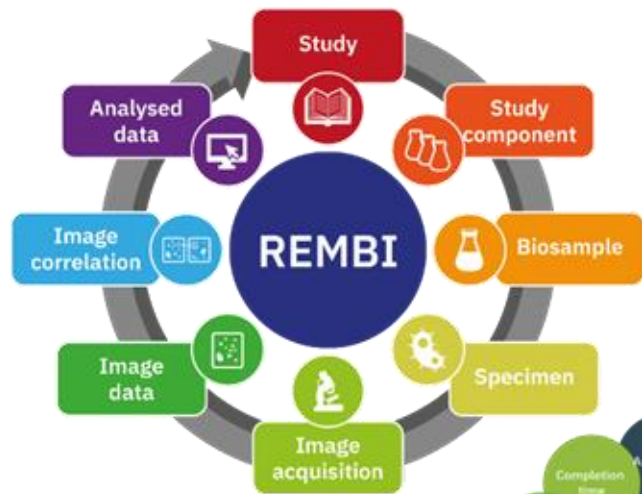
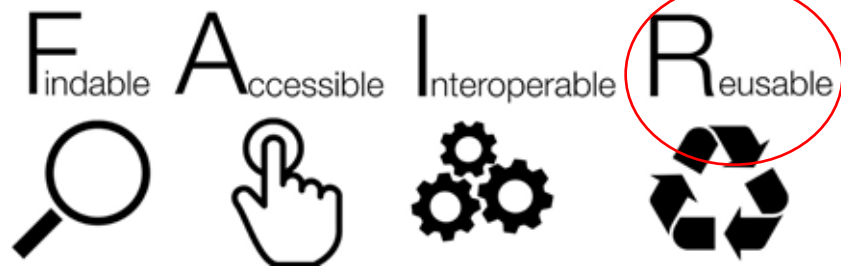
Attributes 1	
Imaging Method	multi-loc
Imaging Method	image stitching
Imaging Method	differential interference contrast
Imaging Method	microscopy
Imaging Method	macroscopy
Publication Title	Reference bioimaging to assess the phenotypic trait diversity of bryophytes within the family Scapaniaceae
Publication Authors	Peters K, König-Ries B
PubMed ID	36195605
PMCID	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC36195605/
Publication DOI	10.1038/s41597-022-01691-x
Release Date	2022-08-25
License	CC BY 4.0
Copyright	https://creativecommons.org/licenses/by/4.0/
Data Publisher	Peters et al
Data DOI	University of Dundee
BioStudies Accession	10.17867/10000183
	https://doi.org/10.17867/10000183
	S-BIAD188
	https://www.ebi.ac.uk/biostudies/studies/S-BIAD188
	idr0134-experimentA-annotation.csv
	https://www.ebi.ac.uk/biostudies/studies/S-BIAD188

Idr0134 Peters K et al Reference bioimaging to assess the phenotypic trait diversity of bryophytes within the family Scapaniaceae [Scientific Data](#)

Bioimages - Core	100%	S-00400191 • 8 October 2019 • 7434 Hits
☐ BioStudies - JCR	100%	Tissue-based map of the human proteome Placeholder submission
☐ EMPAA	100%	
Released	100%	S-00400191 • 8 October 2019 • 7434 Hits
☒ 2024	100%	High-Content Live-Cell Multiplex Screen for Chemogenomic Compound Annotation; EUBOPEN wave 6
☐ 2023	100%	
☐ 2022	100%	
☐ 2021	100%	S-00400190 • 7 December 2019 • 80 Hits
☐ 2020	100%	Effects of PCB52 (2,2',5,5'-Tetrachlorobiphenyl) on the Rat Brain After Subacute Nose-Only Inhalation Exposure
☐ 2019	100%	
☐ 2018	100%	
☐ 2017	100%	
☐ 2016	100%	S-00400190 • 7 December 2019 • 7434 Hits
☐ 2015	100%	High-Content Live-Cell Multiplex Screen for Chemogenomic Compound Annotation; EUBOPEN wave 5
Link Type	100%	S-00400192 • 8 October 2019 • 100 Hits
☐ BioStudies	100%	Images for "The role of the essential GTPase ObgE in regulating lipopolysaccharide synthesis in Escherichia coli"
☐ DOI	100%	
☐ EMPAAR	100%	
☐ PMID	100%	
☐ External link	100%	S-00400193 • 8 October 2019 • 30 Hits
☐ Image data resource (idr)	100%	Noisy nuclei dataset for testing deep learning-based denoising tools
☐ Null	100%	
☐ PubMed	100%	S-00400192 • 8 October 2019 • 100 Hits
☐ Taxonomy	100%	Whole-organism imaging of zebrafish development focussing on cadherin expression and tissue diversity
☐ Uniprot	100%	
File Type	100%	S-00400192 • 8 October 2019 • 80 Hits
☐ gif	100%	A cytoplasmic form of EHMT1(N) methylates viral proteins to enable inclusion body maturation and efficient viral replication
☐ czi	100%	
☐ jpe	100%	
☐ tif	100%	
☐ png	100%	S-00400190 • 7 December 2019 • 7434 Hits
☐ vms	100%	Pathological ultrastructural alterations of myelinated axons in normal appearing white matter in progressive multiple sclerosis
☐ tiff	100%	

- Lots of different data formats, including many proprietary formats
- Not possible to visualise the images without downloading

- Lots of different data formats, including many proprietary formats
- Not possible to visualise the images without downloading



Metadata Schemas, Ontologies and controlled vocabularies



Image Data Management is not always FAIR



"Clara shares" by Henning Falk, ©2022 NumFOCUS, is used under a CC BY 4.0 license.
<https://github.com/zarr-developers/zarr-illustrations-falk-2022#clara-shares>



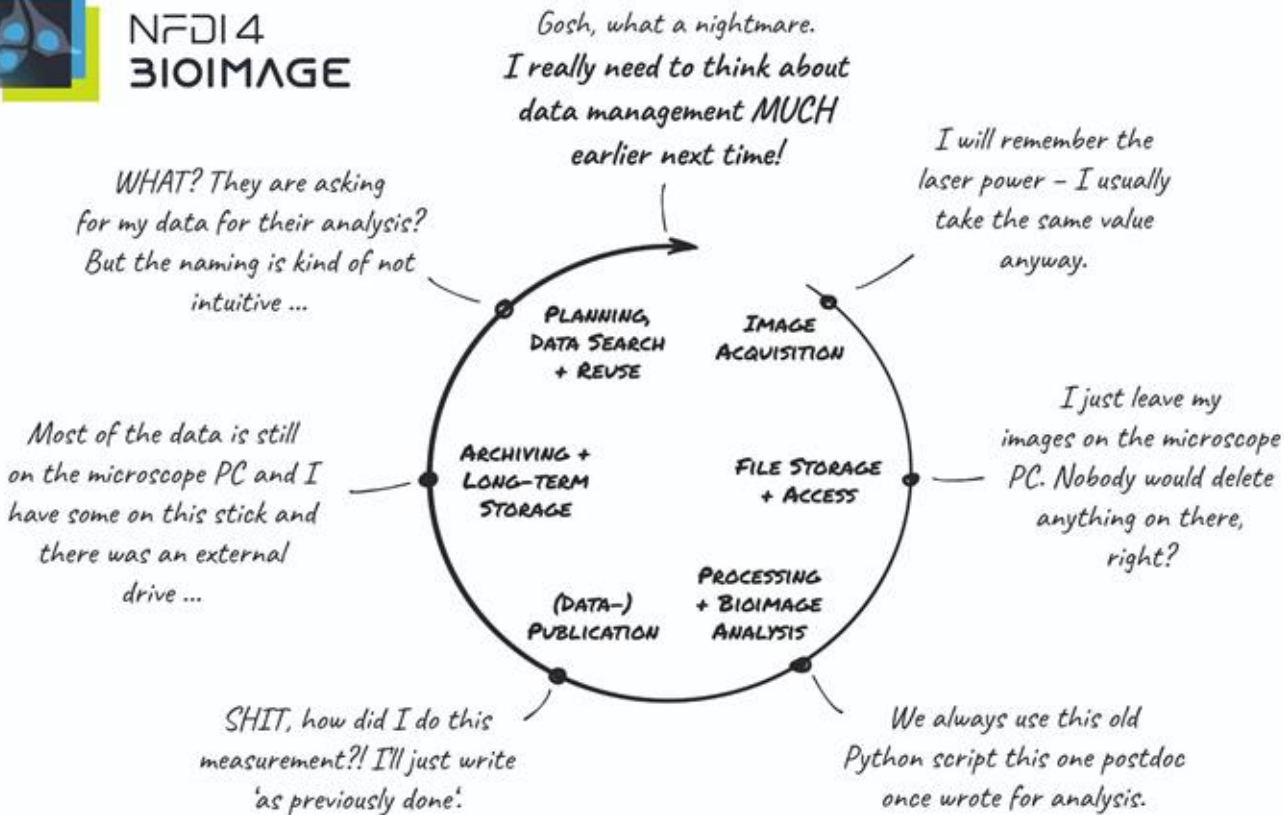
Image: Simon Li
<https://twitter.com/penguinoops/status/643812318699225088>

Data Life Cycle





NFDI4
BIOIMAGE



BEEN THERE, DONE THAT.

HENNING FALK

Ideal image data life cycle/management scenario?

The one where you need to do the minimal effort?

1. The Scientist & AURA at Command

"AURA, find high-res volumetric microscopy of neuracoy of neurodegenerative cell models... run the 3D morphological analysis pipeline."



2. Reviewing the Analysis



3. Initiating External Requests

AURA LOGISTICS & PLANNING

Submitting requests to external facilities:
(1) SYNAPSE BIOLAB - BIOSAMPLE PREP
(2) ADVANCED IMAGING CENTER - MICROSCOPY TIME

"OK, AURA. Request the cleared organoid samples from Biolab... and book 48 hours on the light sheet microscope at the Imaging Center."



4. The Experiment - Remote Execution



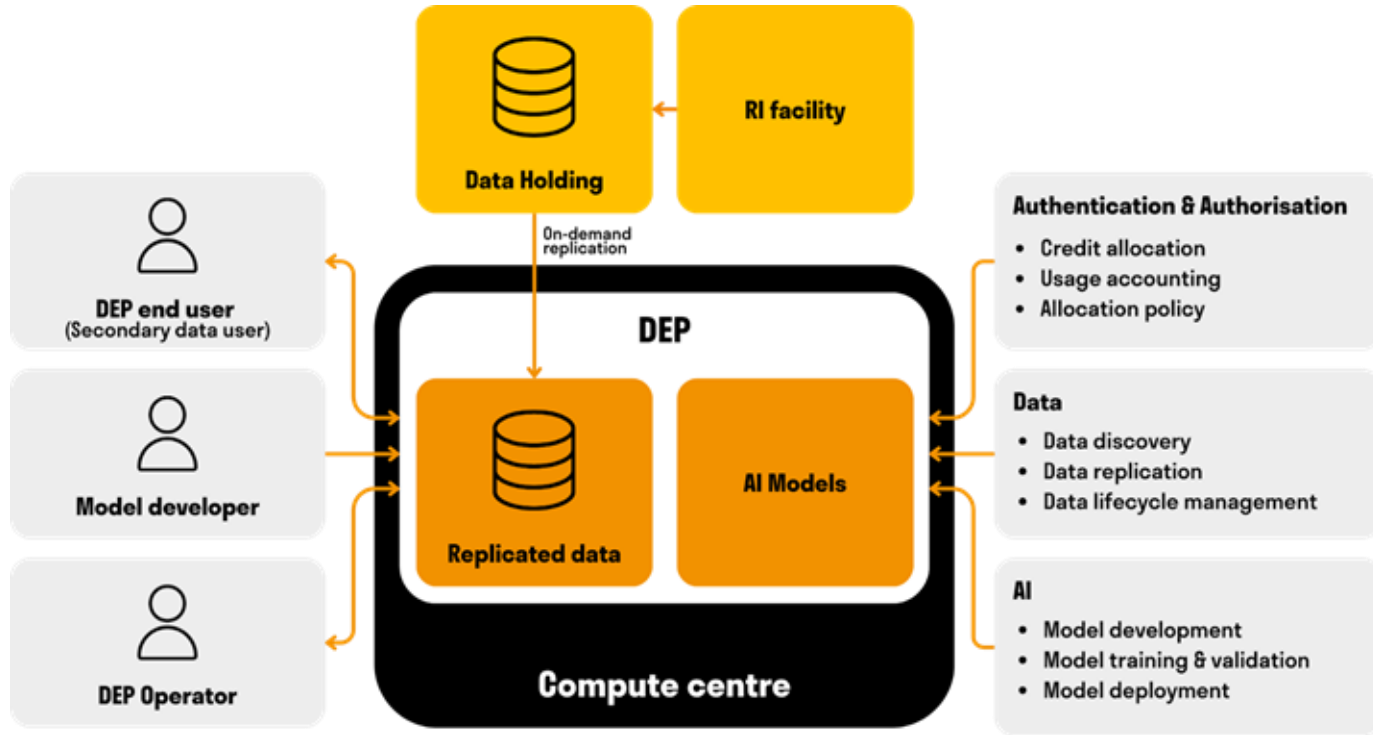
5. The Experiment - Shared Data & Final Analysis



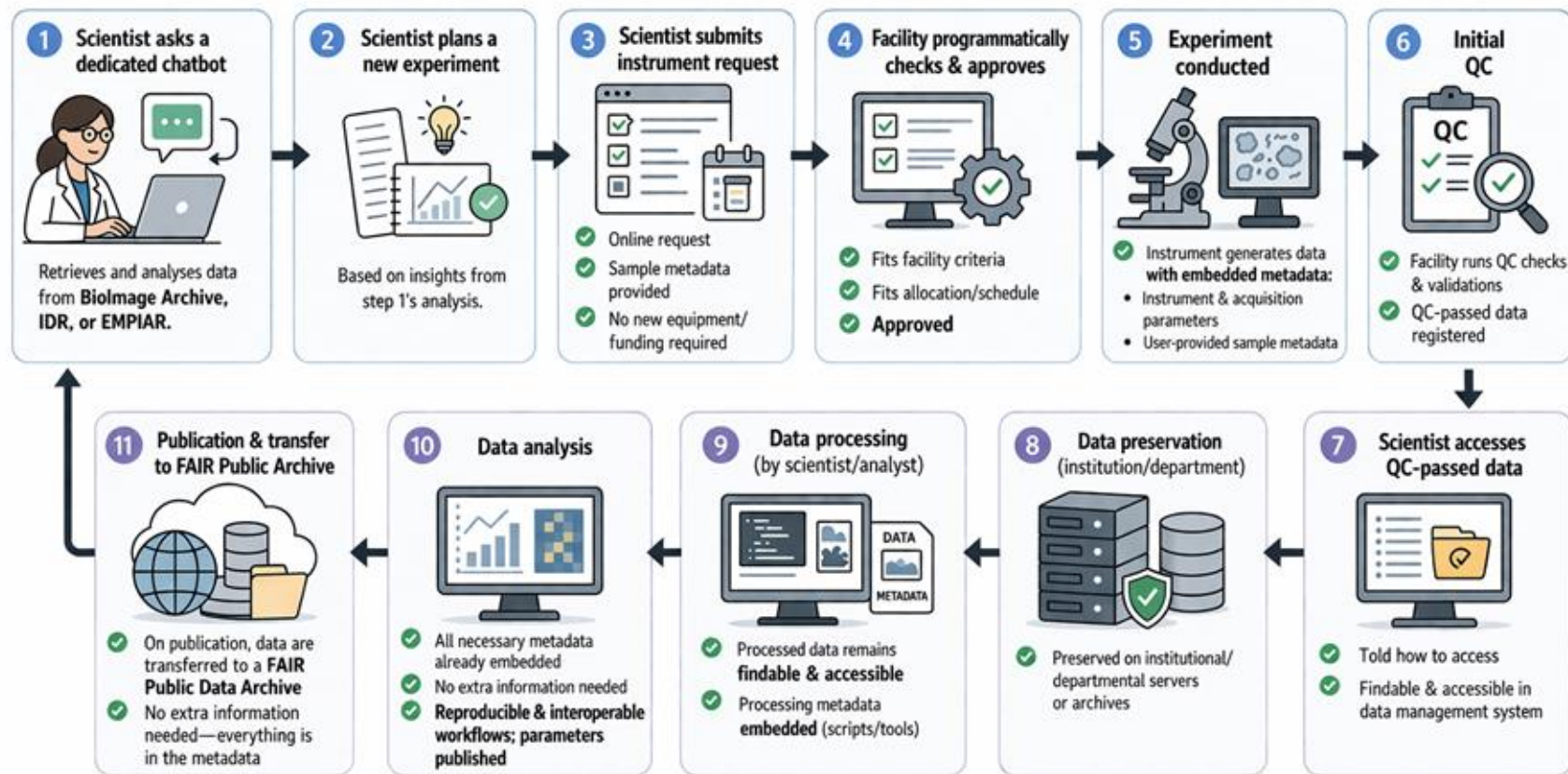
6. Dissemination and Future-Proofing



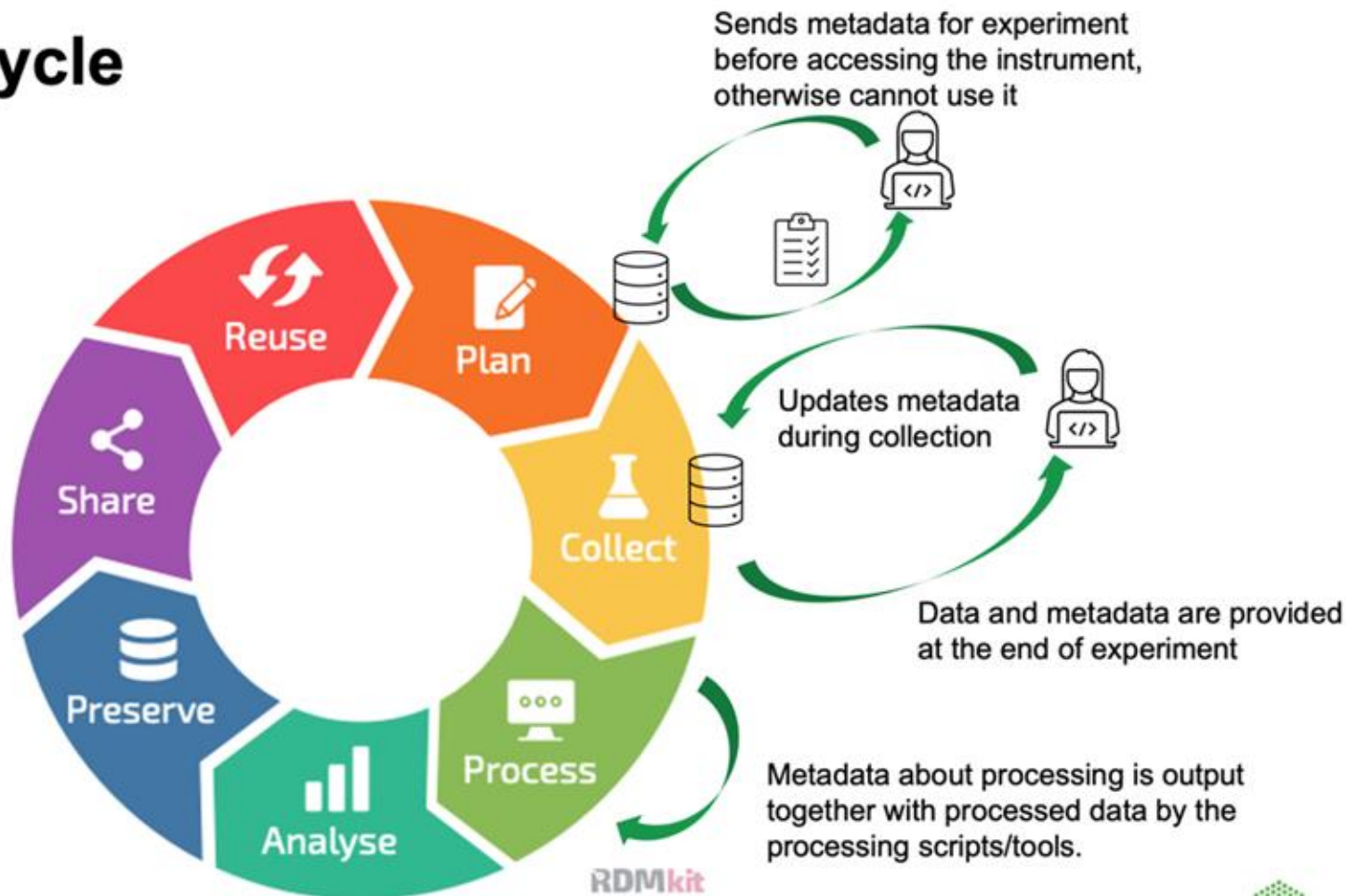
A scalable, open platform for scientific data valorisation



Ideal image data life cycle/management scenario?



Data Life Cycle



Group Discussion Exercise (20 minutes)

Describe the data life cycle journey of your imaging project



Write/Think about the journey of your image data

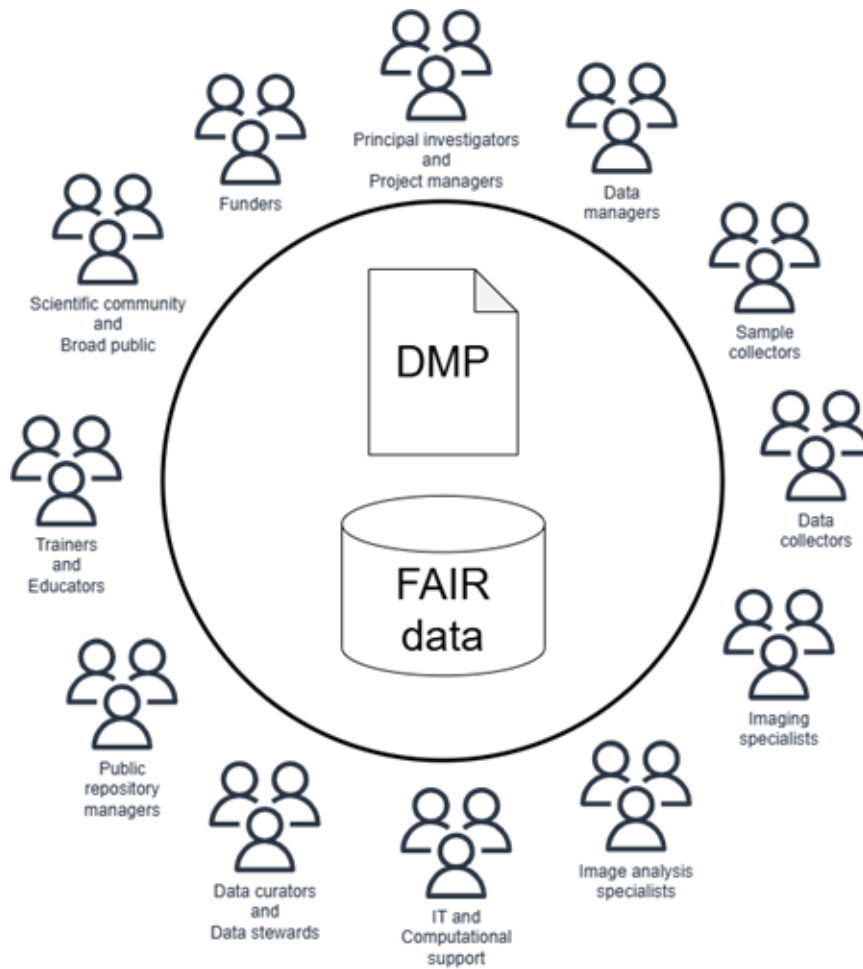
1. Which of the points in the ideal scenario works well in your setting and how do you do it?
1. What can be done to achieve the ideal scenario?

Google doc for note taking: <https://bit.ly/4vJyhqr>

Data Management Plans

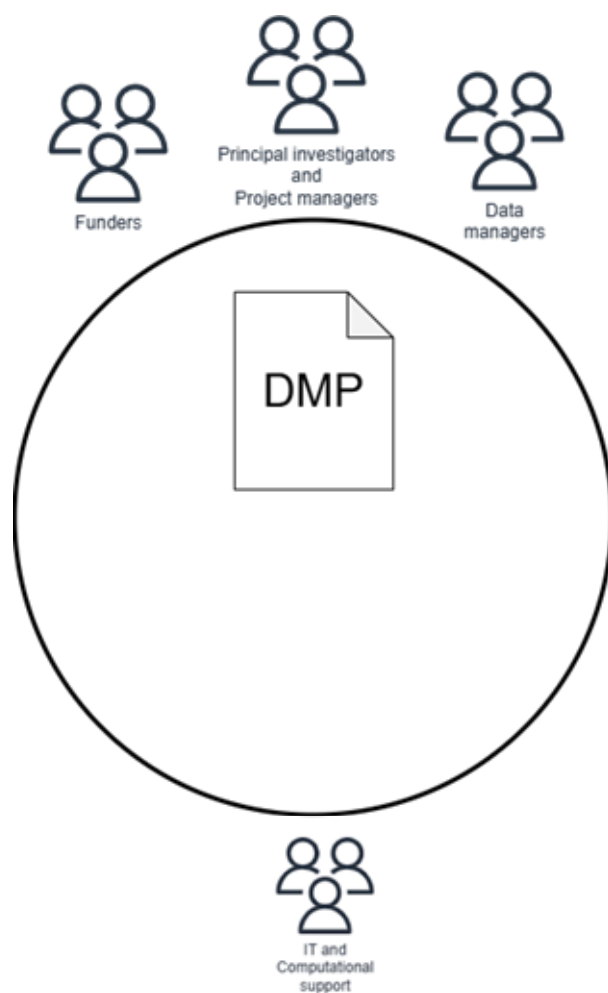
Data Management Plan

- **Strategy:** Define the strategy for managing the data and documents in your project
- **Cost:** You can avoid unexpected situations and costs See https://rdmkit.elixir-europe.org/costs_data_management
- **Why:**
 - Requirements from funders: Guidance to follow
 - It can help to determine potential risks and ways to mitigate them
 - It is a way to be more FAIR as it makes it easier to archive, share and reuse data
- **When:**
 - In early stages of a project, but it is a “**living document**” so it should be constantly revisited



Ensure that all stakeholders

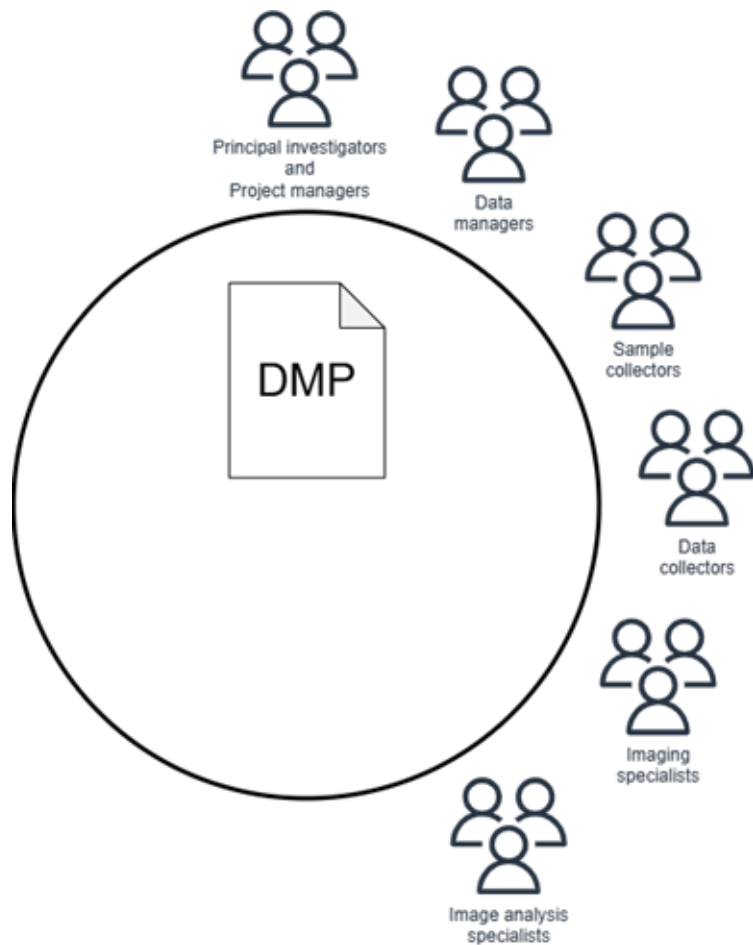
- understand data management practices
- know how to apply them,
- are aware of where to access training if they need additional knowledge or skills.



Principal Investigators and **Project Managers** bring **Data Managers** to draft the first version of the DMP in line with funder requirements.

Essential for Data Managers to understand the types of data that will be generated, review relevant ethical considerations, and identify suitable public repositories for data submission.

Who is missing?



Congratulations!

At kickoff, **schedule a data management planning session**

Data Manager to lead.

Data Stewardship Wizard

Open-source data management tool for writing and managing a DMP in a collaborative format

<https://ds-wizard.org/>

It is distributed as a series of Docker images

You set up SMART questionnaires that guide you through building a DMP

DMPonline

Online tool to create DMPs

<https://dmponline.dcc.ac.uk/>

You can find funder's templates

There are publicly shared user's DMPs

Example template from Euro-Biolmaging for Research Data Management plan

1-General Project Description

Project Name

Project ID ⓘ

Project Leader ⓘ

Project description ⓘ

Project Start Date

Project End Date

Funding ⓘ

<https://doi.org/10.5281/zenodo.11473803>

See also <https://gerbi-gmb.de/i3dbio/i3dbio-rdm/dmp/>

Data Management Plan (tools)

- Use the tool suggested by your funding agency or institution.
- Choose one of the following online DMP tools (ordered alphabetically).
 - [Argos](#) : the joint effort of OpenAIRE and EUDAT to deliver an open platform for Data Management Planning.
 - [Data Stewardship Wizard](#) : publicly available open-source tool to collaboratively compose data management plans through smart and customisable questionnaires with FAIRness evaluation.
 - [DAMAP](#) : tool for machine actionable Data Management Plans.
 - [DMPonline](#) : tool widely used in Europe and many universities or institutes provide a DMPonline instance to researchers.
 - [DMPTool](#) : widely used tool and many universities or institutes provide a DMPTool instance to researchers.
 - [DMPRoadmap](#) : DMP Roadmap is a Data Management Planning tool. Management and development of DMP Roadmap is jointly provided by the Digital Curation Centre (DCC), <http://www.dcc.ac.uk/>, and the University of California Curation Center (UC3), <http://www.cdlib.org/services/uc3/>. The DMPTool and DMPonline sites are both now running from the joint DMPRoadmap codebase.
 - [easy.DMP](#) : tool provided by the pan-European network EUDAT.
- Examples of useful resources for writing and implementing a DMP.
 - [FAIR Implementation Profile](#) and [FIP Wizard](#) are effective instruments for clearly defining and explaining the particular implementation choices required to effectively enact FAIR principles during and after the course of a research project.
 - [Research Data Management Organiser](#) : tool that supports the systematic planning, organisation and implementation of research data management throughout the course of a project.
 - Webinars explaining what a data management plan is and when you might need one, such as the [Research Management Plan](#) webinar produced in collaboration with Dr. Rob Hoof, Technical Coordinator of the Dutch TechCentre for Life Sciences, could be useful instruments to learn more about the topic.

Day-to-day data management

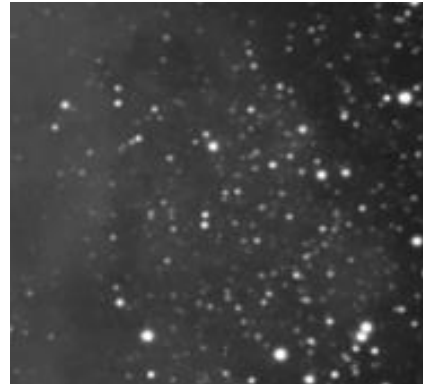
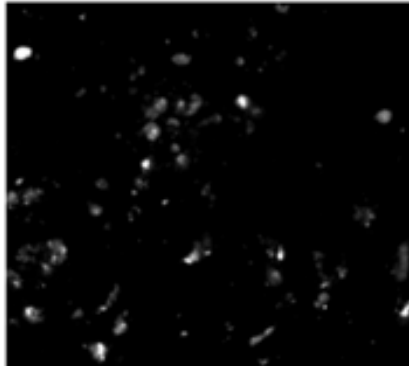
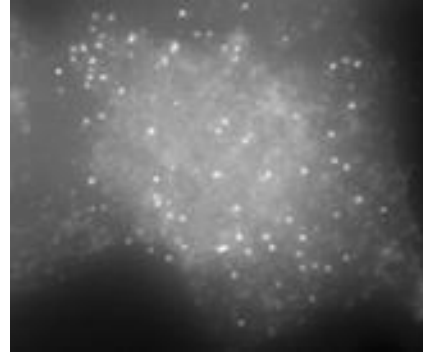
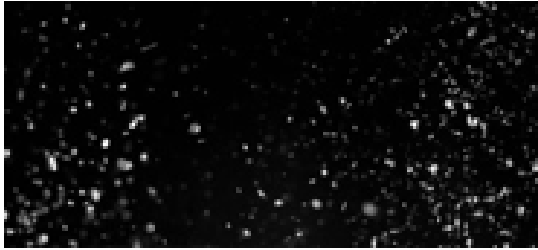
guidance on practical dos and don'ts

<https://www.ebi.ac.uk/training/events/data-management-bioimage-informatics-data-flow/>

<https://www.ebi.ac.uk/training/events/principles-research-data-management>

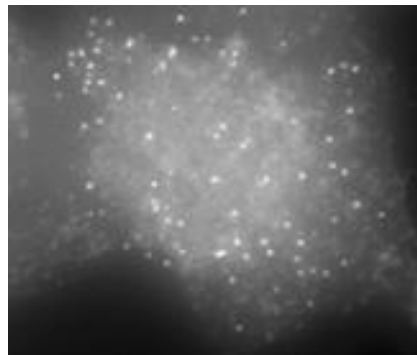
<https://gerbi-gmb.de/i3dbio/i3dbio-rdm/data-management-platforms/>

Metadata

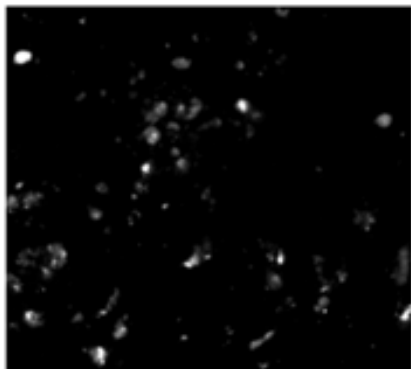




B&W photo of dust



smFISH on mouse cells

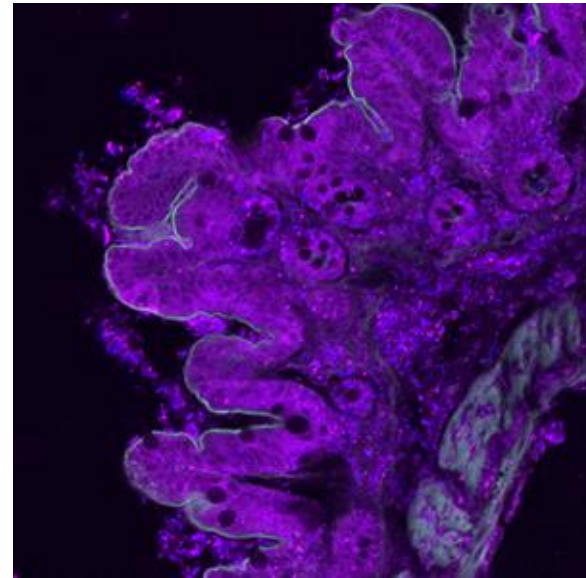
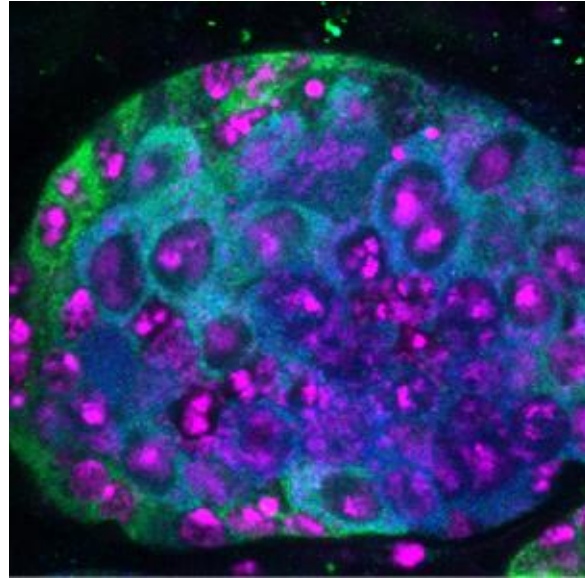
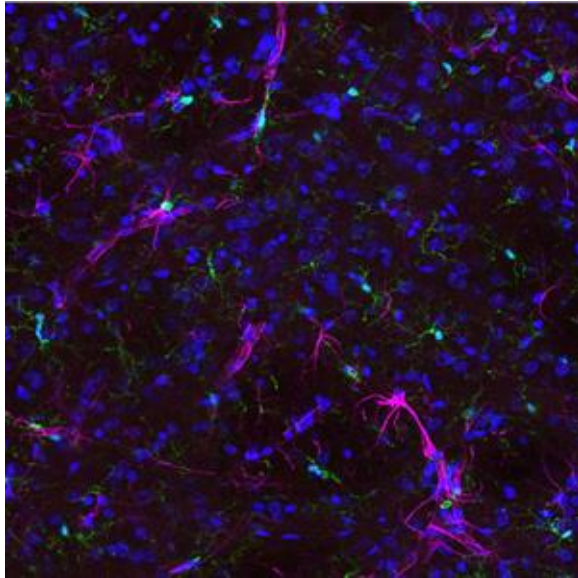


Fluorescence microscopy of mice tissue

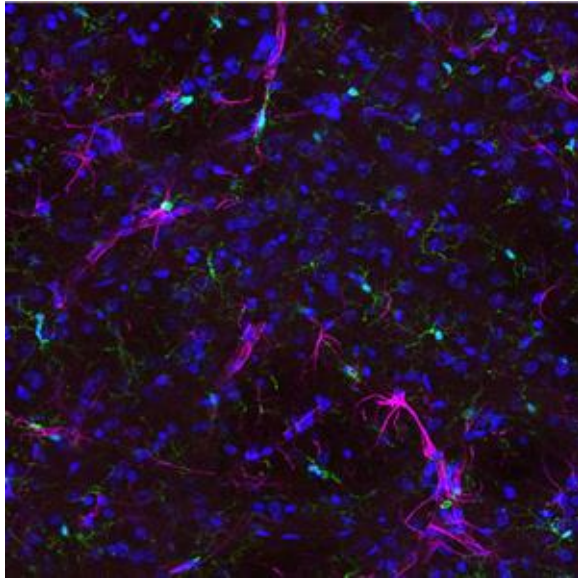


A section of M8 (The Lagoon) Nebula
(stars and interstellar dust and gas)

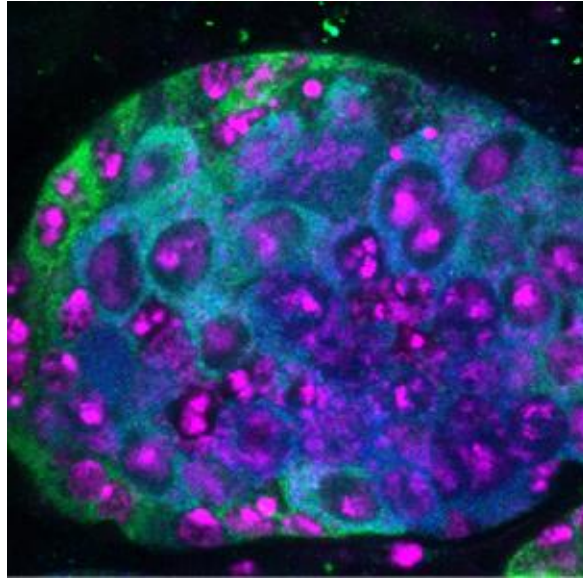
Which one of these spatial experiments was performed in a human sample?



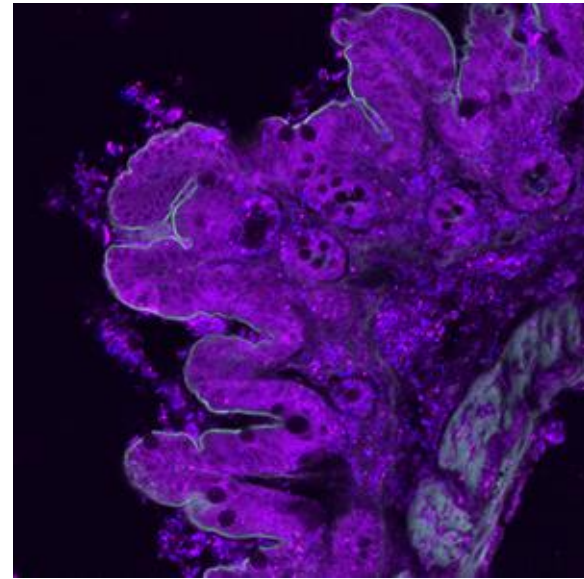
None! Experimental data without context is not useful



Rat (S-BIAD1280)



Fruit fly (S-BIAD1009)



Mouse (S-BIAD922)

Metadata

It's data about our data



It helps to contextualise the data

What?

How?

When?

Who?

Exercise (5 minutes)

Asses the following metadata recording

Can you re-use this or replicate this experiment? What information is missing?

<https://zenodo.org/records/16618585>

<https://zenodo.org/records/5140334>

Exercise (5 minutes)

What metadata needs to be there?

1. What metadata should be there for you to be able to do your responsibilities?
1. What metadata should be there if one needs to repeat the experiment?
1. What metadata should be there if it was someone else's data you'd like to re-use?



menti.com
3770 5336

<https://www.menti.com/alhcwchdasgg>

Exercise (for you to use for training)

The Case of the Missing Metadata

The Scenario:

Dr. X submitted the following dataset to the BIA:

Study Title: "Mitochondrial dynamics in neurons"

Description: We imaged neurons to look at mitochondria. The cells were stained and imaged on the confocal microscope. Images were processed in Fiji.

Files submitted: image_001.tif, image_002.tif, image_003.tif, processed_data.tif

Organism: Mouse (*mus musculus*)

Imaging Method: Confocal microscopy



Exercise

The Case of the Missing Metadata

What's Missing:

1. **Cell type specificity** - What type of neurons? (cortical, hippocampal, motor neurons?)
2. **Mouse strain, age, sex** - Critical for reproducibility
3. **Treatment/conditions** - Control vs experimental? What were they testing?
4. **Sample preparation** - Fixed or live? If fixed, what fixation method?
5. **Microscope details** - Which microscope? Which objective (magnification, NA)?
6. **Fluorophores/stains** - What was used to label mitochondria? (MitoTracker? Immunofluorescence? Genetically encoded marker?)
7. **Acquisition settings** - Laser power, detector gain, pixel size, z-step size
8. **Channel information** - How many channels? What does each show?
9. **Time information** - Single timepoint or time-lapse? If time-lapse, what intervals?
10. **File naming convention** - Which image is which condition/replicate/field of view? (though this should be in the metadata!)
11. **Image dimensions** - What are the physical dimensions (μm)? XY pixel size? Z-spacing?
12. **Raw vs processed** - Which files are raw data? What's in processed_data.zip?
13. **Analysis Information**: What specific steps in Fiji? (background subtraction, deconvolution, segmentation?)
14. **Software versions** - Which version of Fiji? Any plugins used? Is that enough?
15. **Analysis parameters** - Thresholds, filters applied?
16. **Biological replicates** - How many? Which images belong to which replicate?
17. **Technical replicates** - Multiple fields of view per sample

Metadata through the data life cycle

What to collect?

- Study-related:
 - Funder
 - Grant id
 - Researchers
 - Experimental design
- **Data-related:**
 - Sample and preparation information
 - Image acquisition protocols and parameters
 - Processing and analysis protocols

When to collect?

- Initially define in the Data Management Plan
- Before you even begin to collect any data
- During the whole data life cycle

How to collect?

- See next sections on metadata standardisation, data management tools and data standardisation

Points to consider

Metadata throughout the data life cycle



- What metadata to collect when during the data life cycle?
- What metadata is usually missing in your facilities?
- How could we prevent this at the point of data acquisition?

Do's and Don'ts for Metadata

Do's:

- Collect the metadata at the beginning, during and after the study is ended
- Use metadata schemas and standard vocabulary to describe your (meta)data
- It should make the data clear for another user or even for your future self
- Keep the data and metadata together

Don'ts:

- Don't leave the metadata gathering to when you start are writing the manuscript

Metadata standardisation: schemas, ontologies and controlled vocabularies

Metadata schema

- A structured way to describe a study
- It has specific components and attributes
- It will provide guidelines on how to organise, and format your data
- It helps with programmatic use

OME DATA MODEL

```
<?xml version="1.0" encoding="UTF-8"?>
<OME xmlns="http://www.openmicroscopy.org/Schemas/OME/2016-06"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://www.openmicroscopy.org/Schemas/OME/2016-06
    http://www.openmicroscopy.org/Schemas/OME/2016-06/ome.xsd">
  <Image ID="Image:0" Name="6x6x1x8-swath.tif">
    <AcquisitionDate>2010-02-23T12:51:30</AcquisitionDate>
    <Pixels DimensionOrder="XYZCT" ID="Pixels:0:0" PhysicalSizeX="10000.0"
      PhysicalSizeY="10000.0" Type="uint8" SizeC="1" SizeT="1" SizeX="6"
      SizeY="4" SizeZ="1">...</Pixels>
    <AnnotationRef ID="Annotation:1"/>
    <AnnotationRef ID="Annotation:2"/>
  </Image>
  <StructuredAnnotations>
    <!-- First Tag -->
    <MapAnnotation ID="Annotation:1">
      <Description>This is the description of the sample map A</Description>
      <Value>
        <M K="SampleKeyA">SampleValueA</M>
      </Value>
    </MapAnnotation>
    <!-- Second Tag -->
    <MapAnnotation ID="Annotation:2">
      <Description>This is the description of the sample map B</Description>
      <Value>
        <M K="SampleKeyB-1">SampleValueB-1</M>
        <M K="SampleKeyB-2">SampleValueB-2</M>
      </Value>
    </MapAnnotation>
  </StructuredAnnotations>
</OME>
```

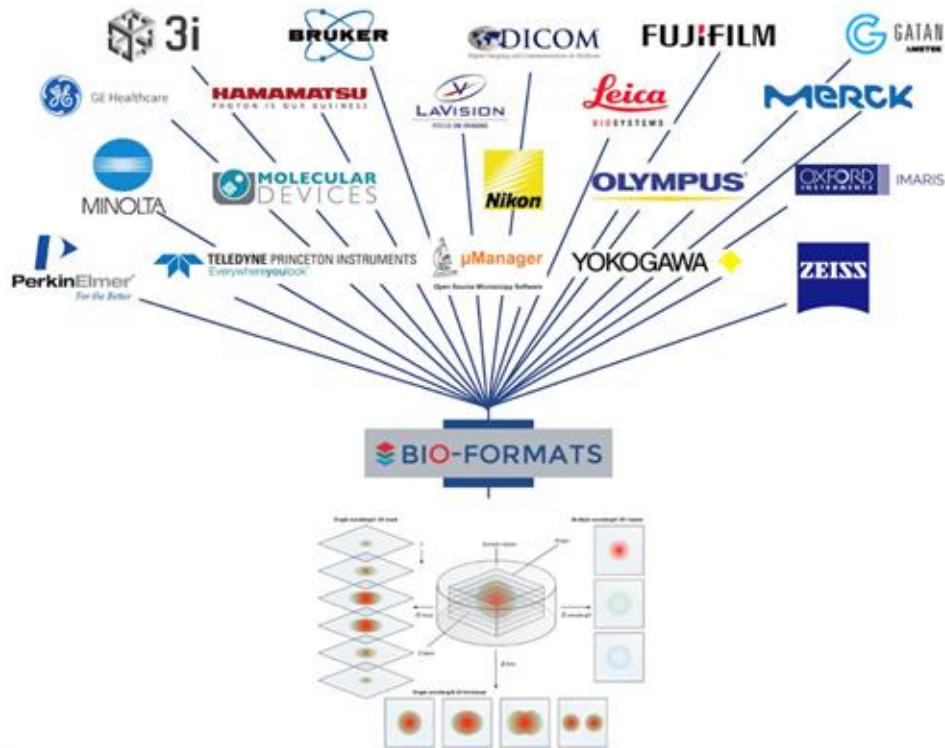
Goldberg, et al. (2005) *The Open Microscopy Environment (OME) Data Model and XML File: Open Tools for Informatics and Quantitative Analysis in Biological Imaging*. *Genome Biol.* 6:R47.
DOI: [10.1186/gb-2005-6-5-r47](https://doi.org/10.1186/gb-2005-6-5-r47)

<https://bit.ly/citing-ome>



Image	Describes the actual image and its metadata.
AcquisitionDate	The acquisition date of the image.
Description	A multi-line description for the image.
Name	A short description for the image. This would be used to, for example, select the image from a list.
ImagingEnvironment	Describes the environment that the biological sample was in during the experiment.
AirPressure	Air pressure in millibars [mbar].
CO2Percent	%CO2 as a percent-fraction from 0.0 to 1.0 [%].
Humidity	Humidity as a percent-fraction from 0.0 to 1.0 [%].
Temperature	Temperature [degrees Celsius].
ObjectiveSettings	Describes any settings on or around the objective.
CorrectionColor	An adjustable ring on the objective that corrects for changes in immersion medium refractive index. Arbitrary scale and unitless.
Medium	A description of a Medium used for the lens. e.g., Oil, Water, WaterDipping, Air, Multi, Glycerol, Other.
RefractiveIndex	Refractive index is that of the immersion medium.
Pixels	Defines the location and parameter of the Pixels, the actual binary image data.
DimensionOrder	The order in which the individual planes of data are interleaved. e.g., XYZCT, XYCTZ, XYCTZ, XYCTZ, XYCTZ, XYCTZ.
PhysicalSizeX	Physical size in x, y, z of a pixel in microns [um]
PhysicalSizeY	
PhysicalSizeZ	
SizeC	Dimensional size x, y, z, c, t of pixel data array
SizeT	
SizeX	
SizeY	
SizeZ	
TimeIncrement	Used for time series that have a global timing specification instead of per-timestep timing info, e.g., a video stream, [s].
Type	The variable type used to represent each pixel in the image. e.g., int8, int16, int32, uint8, uint16, uint32, float, bit, double, complex, double complex.

BIO-FORMATS



<https://www.openmicroscopy.org/bio-formats/>

Bio-Formats

1. **Core metadata** includes:
 - a. image resolution; number of focal planes, time points, channels, and other dimensional axes; byte order; dimension order;
 - b. color arrangement (RGB, indexed color or separate channels); and thumbnail resolution.
2. **Original metadata** is information specific to a particular file format. Nomenclature often differs between formats, as each vendor is free to use their own terminology.

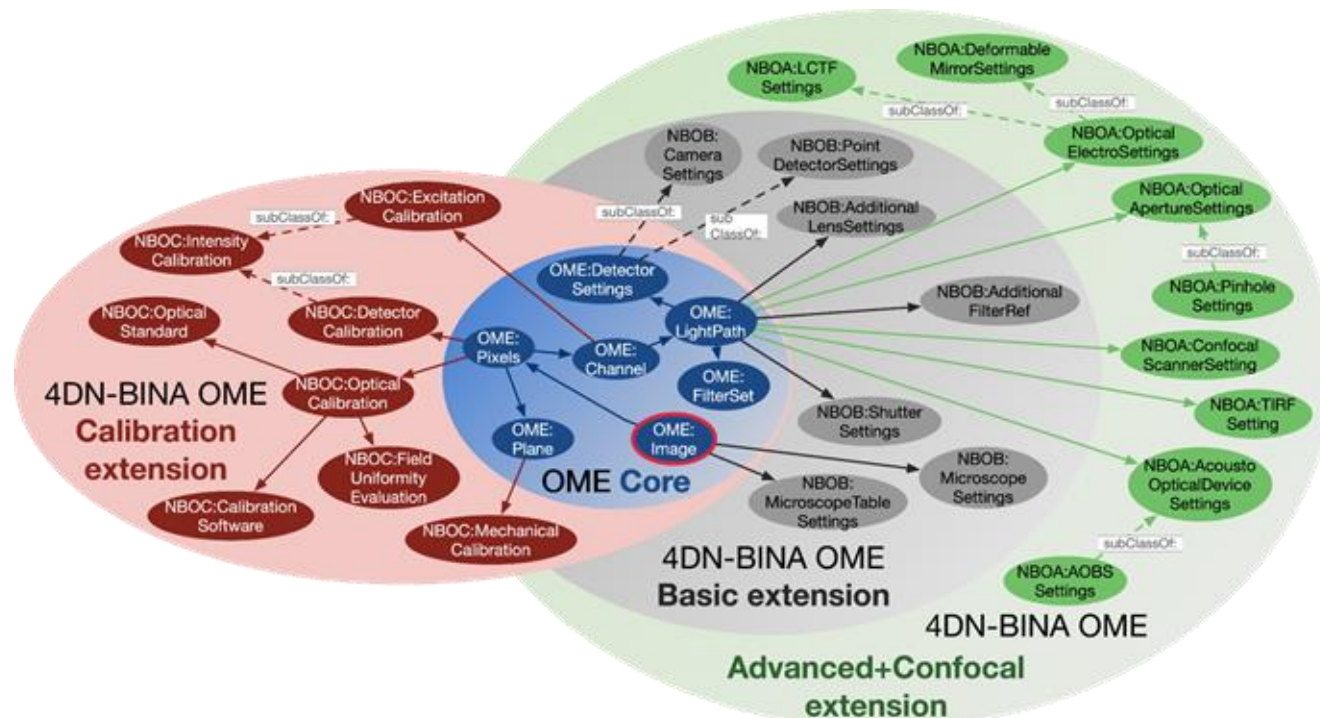
OME metadata is information from #1 and #2 converted by Bio-Formats into the OME data model.

<https://bio-formats.readthedocs.io/en/stable/about/index.html>

Examples tools

- Read data using Bio-Formats
 - <https://github.com/BioImagingUK/Training/blob/main/notebooks/BioFormats.ipynb>
- Read data using Python libraries
 - https://github.com/BioImagingUK/Training/blob/main/notebooks/Read_Image_Using_Python.ipynb

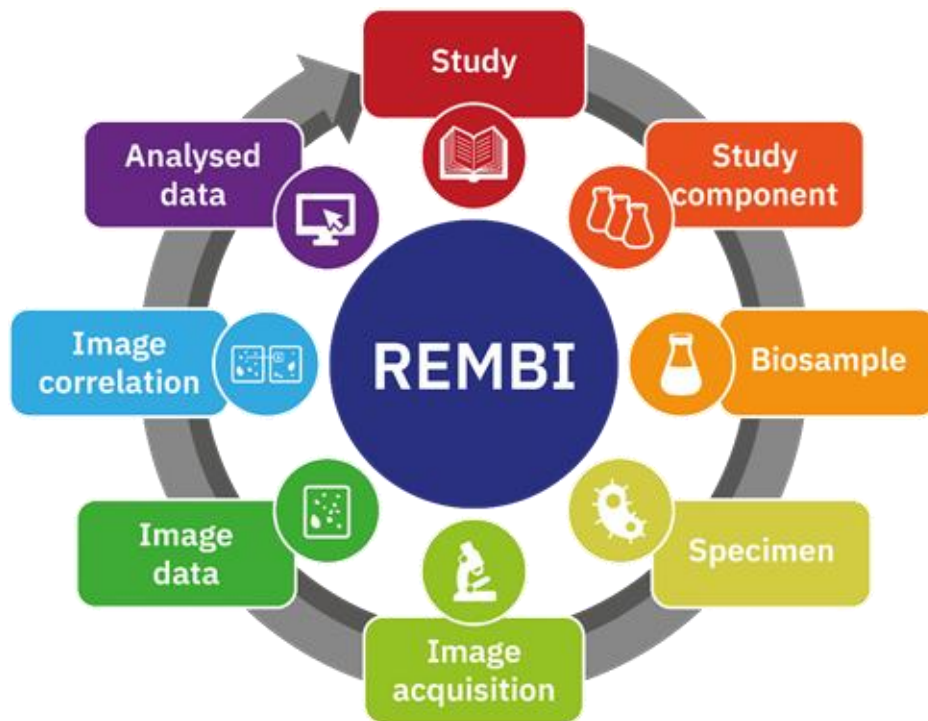
Extending a Model??



4D-Nucleome / BINA / OME / QUAREP-LiMi (2018-2019)

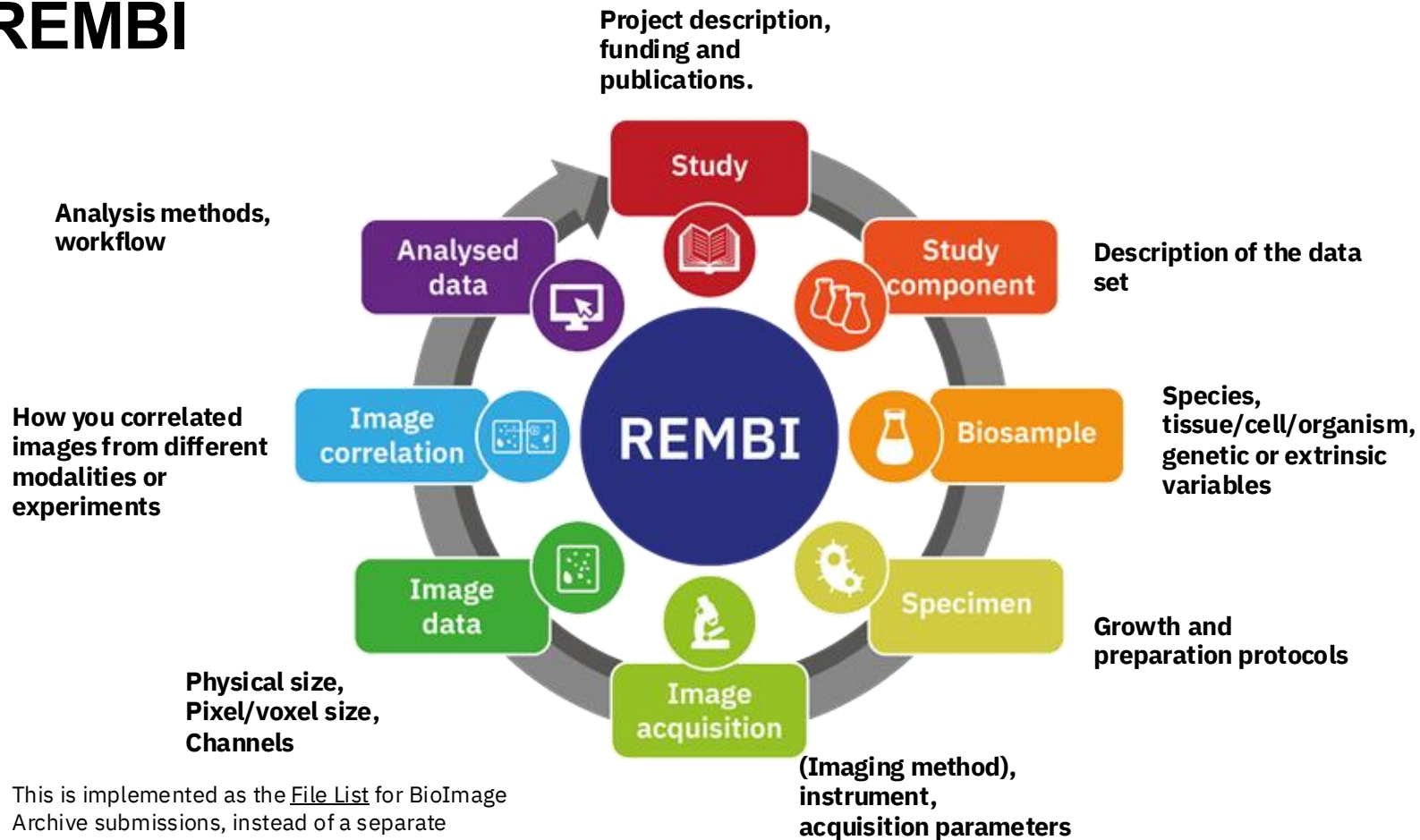
<https://github.com/WU-BIMAC/NBOMicroscopyMetadataSpecs>

REMBI: REcommended Metadata for Biological Images



Sarkans, U., Chiu, W., Collinson, L. et al. REMBI: Recommended Metadata for Biological Images—enabling reuse of microscopy data in biology. *Nat Methods* 18, 1418–1422 (2021). <https://doi.org/10.1038/s41592-021-01166-8>

REMBI



This is implemented as the [File List](#) for BioImage Archive submissions, instead of a separate component in the submission form.

<https://www.ebi.ac.uk/bioimage-archive/rembi-model-reference/>

Seeing REMBI on a BIA submission

<https://beta.bioimagearchive.org/bioimage-archive/study/S-BIAD1392>

MIFA: Metadata, Incentives, Formats and Accessibility

Standardise metadata



Annotators
Annotation type
Annotation method
Annotation coverage
Annotation criteria
Association with source image
Confidence level
Transformations
Spatial information



MIFA guidelines are meant to improve the reuse of bioimages and annotations for AI applications.

Zulueta-Coarasa, T., Jug, F., Mathur, A. *et al.* MIFA: Metadata, Incentives, Formats and Accessibility guidelines to improve the reuse of AI datasets for bioimage analysis. *Nat Methods* 22, 2245–2252 (2025). <https://doi.org/10.1038/s41592-025-02835-8>

Example of BIA submission with annotations

BETA

S-BIAD1392

PlantSeg core datasets: Ovules of *Arabidopsis thaliana*

Released: 2024-10-01

Rachele Tofanelli¹, Athul Vijayan¹, Kay Schneitz¹, Adrian Wolny², Anna Kreshuk²,
Teresa Zulueta-Coarasa¹

¹School of Life Sciences Weihenstephan, Technical University of Munich, Freising, Germany ²EMBL,
Heidelberg, Germany ³European Bioinformatics Institute

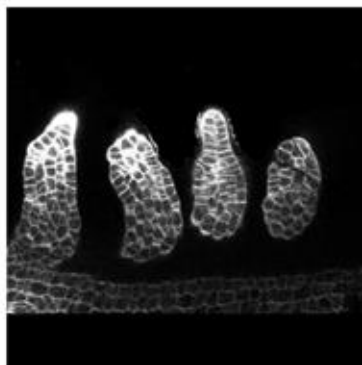
On this page

Curated Study Information

Models Related To This Work

Annotated Datasets

Viewable Annotated Images



Example image for this dataset



Example annotation for this dataset

View in BioStudies

Browse all files

Controlled vocabularies

- Gives clear definitions for terms
- Avoids issues like different spelling or even different languages

Standard vocabulary:

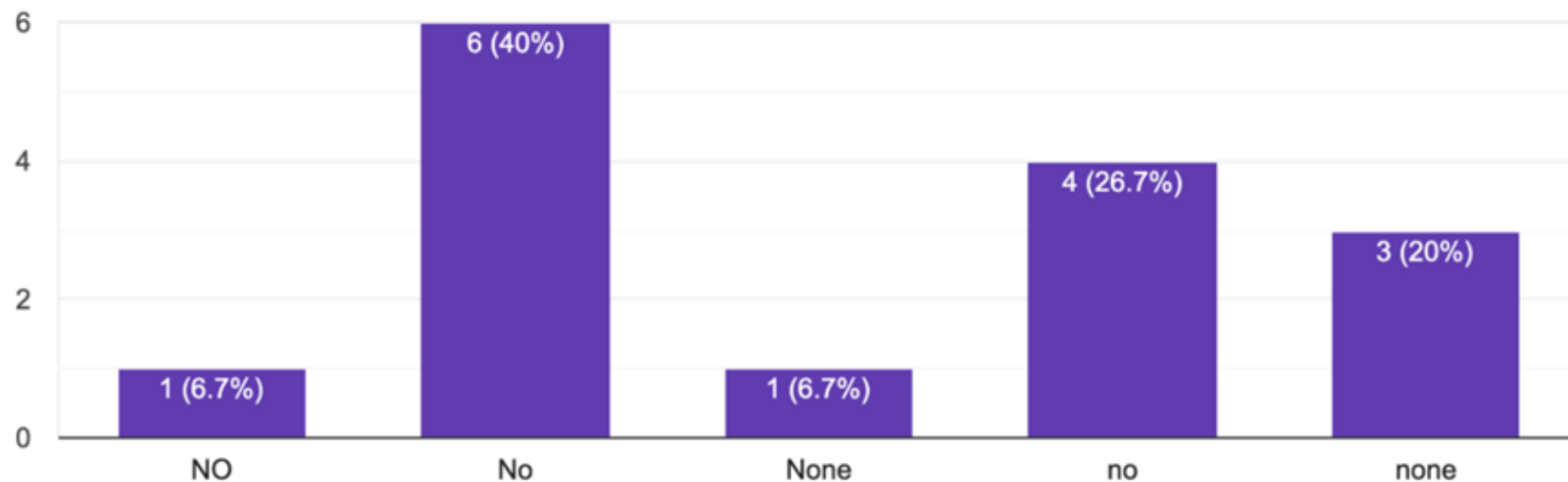
- List of terms
- Gives a more precise way of entering data

Taxonomy:

- List of terms but with a hierarchy
 - E.g. NCBI Taxonomy

Do you have any accessibility requirements?

15 responses



Ontologies

- A taxonomy but with added relationships between the terms
- It allows to make inferences
- Can describe properties
- Terms have a persistent identifier
- E.g. CHEBI (Chemical Entities of Biological Interest)

Ontologies

Ontologies > CHEBI > Classes > CHEBI:52735  Copy

azurite

 http://purl.obolibrary.org/obo/CHEBI_52735  Copy

Definition:

A mineral-based fluorescent blue dye.

Also appears in SLM

Related Synonyms Azurite blue 

Search CHEBI...

☐ Exact match ☐ Include obsolete terms ☒ Include imp



Tree



Graph

- chemical entity (203,368)
 - chemical substance (1,248)
 - mineral (193)
 - azurite**

Relationship Types:

☒ has role (1)

☒ subClassOf (1)



Exercise

Define ROI

Region of Interest or Return-on-investment

Adding a controlled vocabulary or an ontology term would avoid confusion

obo/NCIT_C85402

Definition:

A specific area of interest defined by a sequence of image overlays or a sequence of contours described as a single point (for a point ROI) or more than one point (representing and open or closed polygon).

Where to look for Ontologies

- Ontology Lookup Service (OLS) <https://www.ebi.ac.uk/ols4/ontologies>
- Ontobee <https://ontobee.org/>

Where to check if an ontology is maintained or how widely it's been adopted



The screenshot shows the 'GENERAL INFORMATION' section for the 'EDAM Bioimaging Ontology (EDAM-BIOIMAGING)'. It is identified as a 'Terminology Artifact' with the FAIRsharing ID '10.25504/FAIRsharing.g593w1'. The page lists the following details:

- Type:** Terminology artifact
- Registry:** Standard
- Description:** Bioimaging operations, data types, formats, identifiers and topics. EDAM Bioimaging is an extension to the EDAM ontology (edamontology.org) for bioimaging, developed in collaboration with partners from NEUBIAS (neubias.org).
- Homepage:** <https://github.com/edamontology/edam-bioimaging>

On the right side of the page, there are five circular icons representing different aspects of the ontology: a green circle with a white 'R' (likely for 'Registered'), a person icon, a document icon, a speech bubble icon, and a video camera icon.

<https://fairsharing.org/>

Example from the BIA



The screenshot shows the BioImage Archive website. The header includes the BioImage Archive logo and a search bar with the query 'arabidopsis AND root'. Below the header, there are navigation links: Home, Browse, Submit, Galleries, Help, Metadata Help, Policies, and About us. The main content area displays search results for 'arabidopsis AND root'. On the left, there are filters for Collection (BioImages - Core, EMPIAR) and Released (2025, 2024, 2023, 2022, 2021, 2020, 2017). The search results list two items:

- Item 1:** S-BIAD2226 • 7 August 2025 • 1 link • 42 files. Title: **Tonoplast organization in the rhizodermis of the Arabidopsis thaliana root tip: effect of root position on the surface vs. inside agar medium.** Description: ... line (Segami et al., 2014) to the Col-8 background. Primary root tip, rhizodermis 5 days seedlings with root growing inside the agar medium. BS-top Arabidopsis thaliana (thale cress) Arabidopsis thaliana line expressing plant vacuolar H⁺ pyrophosphatase 1 fused with monomeric GFP (VHP1:mGFP), obtained...
- Item 2:** S-BIAD968 • 15 December 2023 • 230 files • 193 views. Title: **An AGO10:miR165/6 module regulates meristem activity and xylem development in the Arabidopsis root** Description: ... This study represents data related to the role of AGO10 in arabidopsis root apical meristem. All samples used for this study are from Arabidopsis and occasionally Benthamiana Arabidopsis thaliana seedlings Arabidopsis thaliana seedlings ...

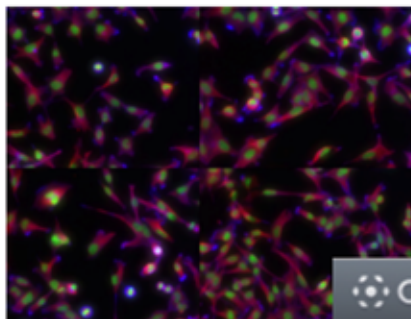
Experimental Factor Ontology (EFO)

Provides a systematic description of many experimental variables available in EBI databases

It combines several ontologies like CHEBI, FBBI, NCBI taxon



Linked Metadata



OMERO Studies Genes Phenotypes

Public

Type Gene Symbol...

Gene 1

- ASH2L (326) 6
 - idr0022-koedoot-cellmigration/screenA (288) 8
 - idr0006-fong-nuclearbodies/screenA (16) 1
 - idr0009-simpson-secretion/screenA (12) 6
 - idr0013-neumann-mitochek/screenA (6) 3
 - idr0012-fuchs-cellmorph/screenA (2) 1
 - HT28 2
 - HT28 [Well G13, Field 1]
 - HT28 [Well G13, Field 2]
 - idr0010-doll-dnadamage/screenA (2) 1

OMERO Studies Genes Phenotypes Cell Lines

Public Public data

Type Phenotype...

Match case ?

Phenotype 1

- CMPO_0000077 (20872) 8
 - idr0028-pascualvargas-rhogtpases/screenD (3920) 4
 - idr0028-pascualvargas-rhogtpases/screenA (3808) 4
 - idr0028-pascualvargas-rhogtpases/screenC (3220) 4
 - idr0033-rohban-pathways/screenA (2916) 12
 - idr0028-pascualvargas-rhogtpases/screenB (2856) 4
 - idr0001-graml-sysgro/screenA (2788) 122
 - idr0008-rohn-actinome/screenB (1098) 10
 - idr0012-fuchs-cellmorph/screenA (266) 55

Attributes 8

Cell Lines

Added by: Public data

Cell Line HeLa

Gene

9070 [icon]

ASH2L

elongated cells

elongated cell phenotype

CMPO_0000077 [icon]

Search OxO by term id

1 2 3

Examples...

HGNC:5009

OxO summary view

Jump directly to a datasource

Jump to a datasource ▾

←→↻

ebi.ac.uk/spot/oxo/

☆

📄

📁

📄

📄

📄

New Chrome available

OXO

ONTOLOGY MAPPING

Home

Documentation

About

Welcome to the EMBL-EBI Ontology Mapping Service

OxO is a service for finding mappings (or cross-references) including the [Ontology Lookup Service](#) and [feedback](#).

Search OxO by term id

Choose a target (optional)

Mapping Distance

1

2

3

Enter list of identifiers: [Examples...](#)

HGNC:5009

←→↻

ebi.ac.uk/spot/oxo/search

☆

📄

📁

📄

📄

📄

New Chrome available

Mapping results

Download as

Select a term to see more information. The evidence column tells you how many times we have seen this mapping and the distance is a how many hops across other mappings you need to go to find this mapping. Distance 1 is a direct mapping, the greater the distance the less likely it is that a mapping holds true. Max distance is set to 2.

Showing 1 to 5 of 5 entries

Search:

Input	Mapped id	Id source	Evidence	Distance
HGNC:5009	Genatlas:HGMA2	Genatlas	1	2
HGNC:5009	Reactome:P62926	Reactome	1	2
HGNC:5009	OMIM:600698	OMIM	1	2
HGNC:5009	Ensembl:ENSG00000149948	Ensembl	1	2
HGNC:5009	Orphanet:248487 (high mobility group AT-hook 2)	Orphanet	1	1

Show

50

entries

Previous

1

Next

EMBL-EBI, Wellcome Trust Genome Campus, Hinxton, Cambridgeshire, CB10 1SD, UK +44 (0)1223 49 44 44

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https://github.com/BiolMagingUK/Training/blob/main/notebooks/OXO_IDR.ipynb

EMBL-EBI

FoundingGIDE: Founding a Global Image Data Ecosystem

Bringing together image data resource owners and research infrastructures from Europe, Australia and Japan to develop the basis of **interoperable image data repositories**.

- common ontologies and metadata frameworks
- data interoperability

Biological Imaging

- [D2.1 – Landscape analysis of existing ontologies and recommendations on a set of imaging ontologies](#)
- [D6.1 – Report in metadata model overlap and gaps](#)

Preclinical Imaging

- [D4.1 – Report on the recommended ontologies for preclinical image datasets and technologies](#)

Recommended ontologies by FoundingGIDE

Ontologies used by BioImage Archive, IDR and SSBD

Category	Ontology/Controlled vocabulary	Acronym
Imaging Method	Biological Imaging Methods Ontology	FBbi
Taxonomy	NCBI Organismal Classification	NCBITaxon
Cell Line	Cell Line Ontology	CLO
Cell type	Cell Ontology	CL
Gene	Ensembl	ENSG
	Entrez Gene	NCBI_Gene
Protein	Universal Protein Resource	UniProt
Chemical Compound	PubChem	PubChem
	Chemical Entities of Biological Interest Ontology	ChEBI
Experimental Condition	Experimental Factor Ontology	EFO
	Ontology for Biomedical Investigation	OBI
Unit	Units of Measurement Ontology	UO
Disease	Systematized Nomenclature of Medicine-Clinical Terms	SNOMED CT
Cellular Phenotype	Cellular Microscopy Phenotype Ontology	CMPO
Anatomy	Uberon Multi-species Anatomy Ontology	UBERON
Biological Process	Gene Ontology	GO
Cellular Component		
Molecular Function		
Controlled Vocabulary	Medical Subject Headings	MeSH

Imaging technique/method

We recommend use of either the Biological Imaging Methods Ontology (FBbi) or EDAM ontologies.

If some subject domain is not captured in this list, we recommend looking at the ontologies in use in EFO.

Recommended ontologies by FoundingGIDE

- Using ontologies and taxons are better than not
- Some archives have specific recommendations

Recommendations for imaging facilities, data stewards, and biologists capturing imaging data.

Capture ontology IDs rather than (or as well as) text description or names. Many ontologies will have guidelines about how to reference their terms: e.g. NCBI taxon recommends NCBI:txid9606 for Homo Sapiens, and OBO ontologies recommend e.g. UBERON:0000948 when it is possible to provide context for the UBERON prefix. Preferably use ontologies that correspond to those in use of any database that may eventually store the data, but do not make that a limiting factor. It is preferable to have some IDs of a different ontology than none at all. If ontologies are missing terms, contact their maintainers to get them added.

Data management tools

Data Management Basis

Spreadsheets

- Easiest to implement
- Lightweight
- Get complicated to handle the bigger they get
- Unit of data is the cell

Electronic Lab Notebook (ELN)

- Electronic equivalent of a paper notebook
- Easier way of sharing
- Versioning
- can't truly delete anything

Database & File Storage (e.g. OMERO)

- Better suited for more complex data
- Requires IT knowledge to set up and maintain
- Easily scalable
- Can have data models and relationships
- Can provide provenance and versioning

Packaged Files (e.g. RO-Crate)

- Metadata and data is packaged together
- Easily scalable and shareable
- Can have (relational) data models
- Can have versioning and provenance

Which one to use

- Is it for one individual only or do you want others to be able to view and edit?
- Do you want to impose a structure?
- Do you want to impose value restrictions?
- Does it need controlled access?

A spreadsheet may be fine when:

- one person is exploring a small dataset
- the work is temporary
- the data is not yet final
- the consequences of errors are low

A managed system becomes important when:

- many people depend on the data
- the data must be reused later
- permissions matter
- data quality needs to be checked
- the data links to samples, files or workflows

What you still need for image data management

Option	What it gives you	What you still need to build
Relational database	Strong structured data management	(Image viewer, file handling), metadata model, user interface, permissions, import tools
Document database	Flexible metadata storage	(Image viewer, file management), validation, search interface, permissions
Graph database	Tracks relationships and provenance	(Image viewer, file management), user interface, data entry tools
OMERO	Image repository, web viewer, metadata, annotations, sharing, permissions, image import, APIs	Some custom metadata templates, integrations

Which one to use

- Individual researcher or small group: OMERO or data management servers or ELN if you have it setup at your institution. If not, spreadsheets
- Big research groups or labs: OMERO or data management servers if you have it setup at your institution. If not, electronic lab notebooks
- Imaging centres or institutes: OMERO or custom data management servers

→ RO-crates for data packaging and sharing.

In some cases can also be used for metadata management in combination with versioning tools (e.g. EMO BON uses ro-crates and Github for data management)

Electronic Lab Notebooks

Pros

- An almost electronic equivalent to a paper notebook but we added functionality
- Many commercial options (LabArchives, RSpace) that provide both free and paid versions
- Paid versions can be at an institutional level and have things like admins
- Easy way to transfer ownership of (meta)data when someone leaves the lab/institution

Cons

- Limited features on the free version
- You need an electronic equipment and internet connection to use them
- Trickier to use in labs with clean policies

Electronic Lab Review Notebooks

← → ↻ gerbi-gmb.de/i3dbio/i3dbio-rdm/electronic-lab-notebooks/ ☆ 📄 📁 🔍 New Chrome available

Regulatory Compliance: ELNs help ensure compliance with standards.

The ELN market is diverse, with a wide range of options available. Here's a brief overview of some popular ELN's in the field of Life Sciences (as of early 2025):

	LabArchives	Benchling	SciNote	elabFTW	RSpace	openBIS
Costs	subscription	subscription	subscription	Free (self hosting)	subscription	Free (self hosting)
Open Source Code	No	No	No	Yes	No	Yes
Link	Website	Website	Website	Website	Website	Website

When using both an ELN and OMERO, scientists should be able to conveniently annotate the required metadata in either their ELN or within OMERO, without the need to duplicate annotation efforts. For this purpose, the I3D:bio team is working on an ELN-integration layer that allows bridging data entries within an electronic lab notebook and OMERO. The tool is [available for testing on GitHub](#), but still considered under development.

Try out the OMERO-ELN-DataBridge-Tool developed by I3D:bio.

Exercise (5 minutes)

What can you do with this data? What needs to change?

File Name	Imaging Modality	Organism	Channel 1	Ch2	Resolution (um/pixel)	Experiment Date	Treatment
image_1.tif	Confocal	fly	actin-GFP	myosin:mCherry	0.05	2023/01/15	control
sample_B.tif	Microscopy	Drosophila	actin:GFP	myosin:mCherry	512x512	02-15-2023	drug1
sample_3.tif		mouse	actin	myosin:mCherry	0.1	Jan 25th, 2023	drug2

Answers

- File name: Inconsistent naming of images and image formats
- Imaging modality: which microscopy? Empty values
- Organism: inconsistent naming; both scientific name and common name
- Channel 1: inconsistent formatting (actin-GFP, actin:GFP), is actin same as actin-GFP?
- Resolution: wrong values; 512x512 is not resolution but likely the image pixel size
- Experiment date: inconsistent format
- Treatment: what are drug1 and drug2?
- General: better to avoid space characters in column headers and file names, lowercase, uppercase inconsistency, inconsistent naming of column headers

Tips for spreadsheets

Do's

Use first row as header

Use atomic values

One table per sheet

Impose formats to each cell where possible

Add one tab to work as a README

Define how you will code for missing values

Don'ts

Don't convey information using different formats

Don't spread headings across multiple columns

Data model and metadata schema

What are the high level concepts to record, and how are they related?

For each high level concept, what information must we record, in what format, using what allowed values?

- **In a spreadsheet:** the column names, allowed values and rules.
- **In an ELN:** templates, required fields, sample links and metadata sections.
- **In a relational database:** tables, fields and relationships.
- **In a document database:** expected structure of each document, often with flexible optional fields.
- **In a graph database:** nodes and relationships

← → ↻ bff.allencell.org/app?c=index%3A0.25%2Cimage_webclient_url%3A0.25%2CFile+Path%3A0.25%2CThumbnail%3A0.25... ☆ 📄 📁 📌 New Chrome available

BioFile Finder

Powered by BigFile Finder

Open-source datasets · Learn · Help

+ ADD

New Query

Data source +
idr0168-zhang-mllocalizatio...

Group by +

Filter +

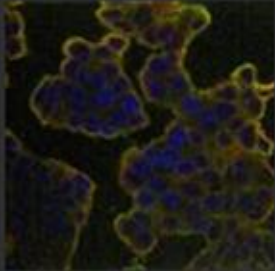
Sort +

index	image_webclient_url	File Path	Thumbnail
0	https://idr.openmi...	https://idr.openmi...	https://idr.openmi...
1	https://idr.openmi...	https://idr.openmi...	https://idr.openmi...
2	https://idr.openmi...	https://idr.openmi...	https://idr.openmi...
3	https://idr.openmi...	https://idr.openmi...	https://idr.openmi...
4	https://idr.openmi...	https://idr.openmi...	https://idr.openmi...
5	https://idr.openmi...	https://idr.openmi...	https://idr.openmi...
6	https://idr.openmi...	https://idr.openmi...	https://idr.openmi...
7	https://idr.openmi...	https://idr.openmi...	https://idr.openmi...
8	https://idr.openmi...	https://idr.openmi...	https://idr.openmi...
9	https://idr.openmi...	https://idr.openmi...	https://idr.openmi...
10	https://idr.openmi...	https://idr.openmi...	https://idr.openmi...
11	https://idr.openmi...	https://idr.openmi...	https://idr.openmi...
12	https://idr.openmi...	https://idr.openmi...	https://idr.openmi...
13	https://idr.openmi...	https://idr.openmi...	https://idr.openmi...
14	https://idr.openmi...	https://idr.openmi...	https://idr.openmi...
15	https://idr.openmi...	https://idr.openmi...	https://idr.openmi...
16	https://idr.openmi...	https://idr.openmi...	https://idr.openmi...
17	https://idr.openmi...	https://idr.openmi...	https://idr.openmi...
18	https://idr.openmi...	https://idr.openmi...	https://idr.openmi...
19	https://idr.openmi...	https://idr.openmi...	https://idr.openmi...
20	https://idr.openmi...	https://idr.openmi...	https://idr.openmi...
21	https://idr.openmi...	https://idr.openmi...	https://idr.openmi...
22	https://idr.openmi...	https://idr.openmi...	https://idr.openmi...
23	https://idr.openmi...	https://idr.openmi...	https://idr.openmi...

1 Total Files Selected 1 Unique Files Selected 277 files

OPEN WITH

15180304



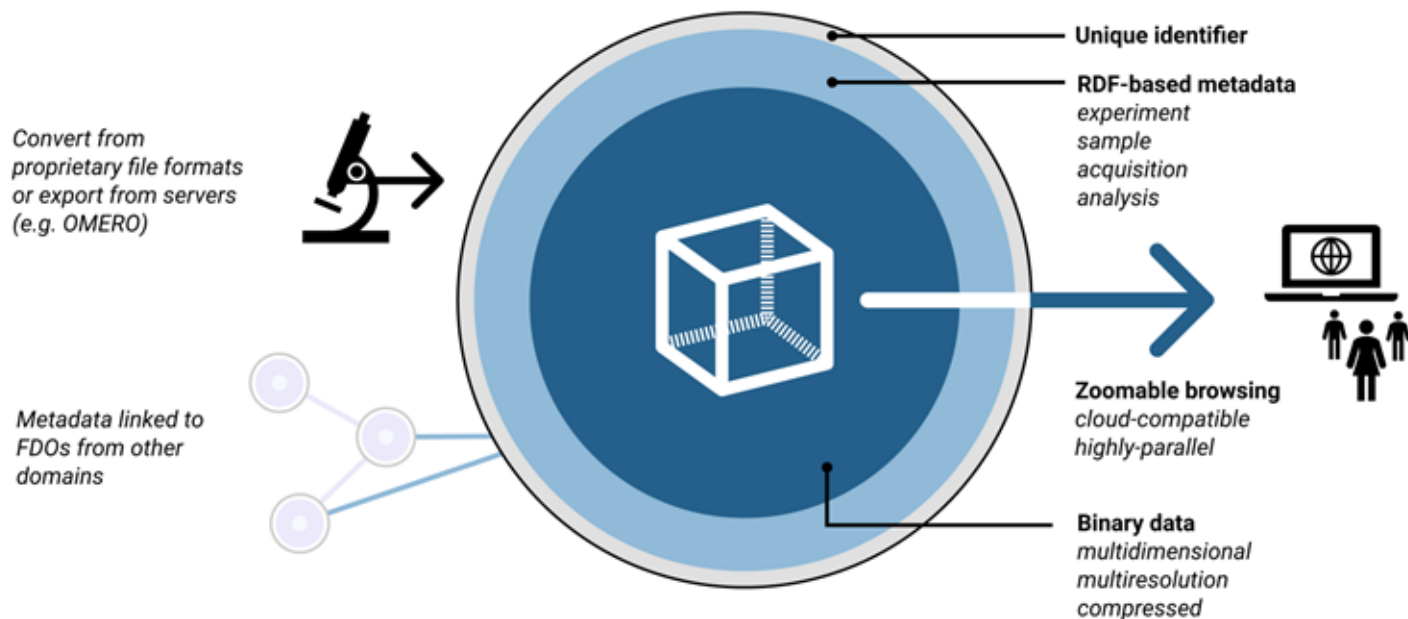
Metadata

Antibody Comment	Atlas Antibodies
Antibody Identifier	HPA028487
Antibody Identifier URL	https://v19.proteinatlas.org/search/HPA028487
Cell Line	MCF-7
data_source	idr
File Path	https://idr.openmicroscopy.org/webclient/img_detail/15180304/

<https://bff.allencell.org/>

FAIR-IO (2023-2028)

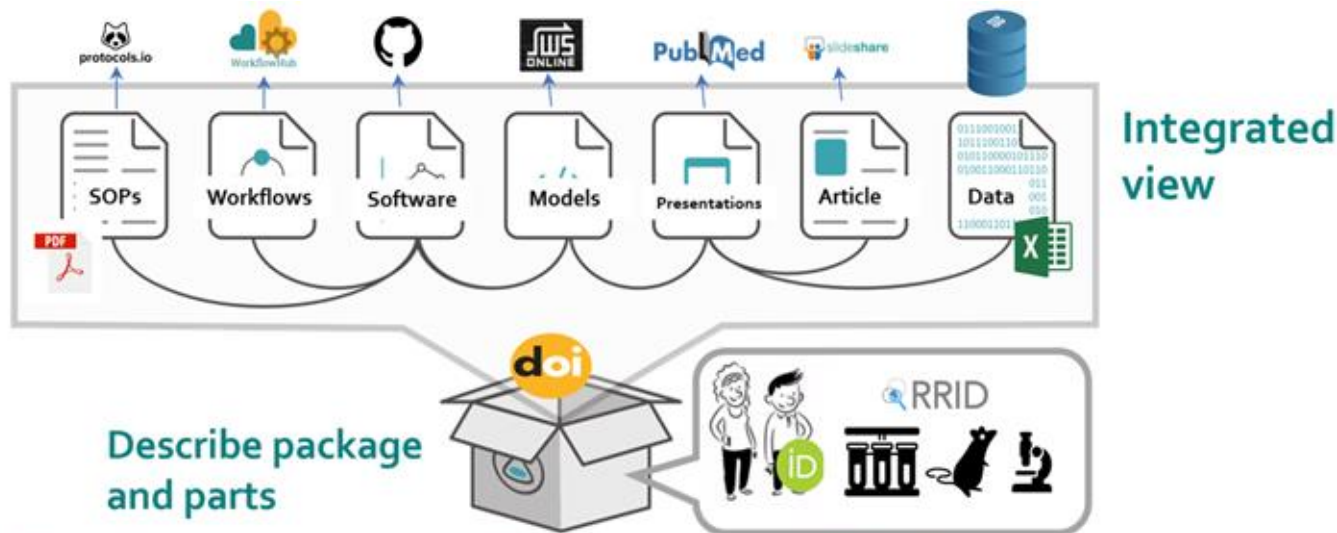
FAIR Image Objects (*FAIR-IO*), an *FDO*-subtype for bioimaging



<https://nfdi4bioimage.github.io/FAIR-IO/specification/index.html>

Research Object Crate (RO-Crate)

- A way to package data and metadata together:
 - Paper + authors + protocols + workflows + data
- It uses JSON-LD and [Schema.org](https://schema.org)
- It's a way to comply with the FAIR principles



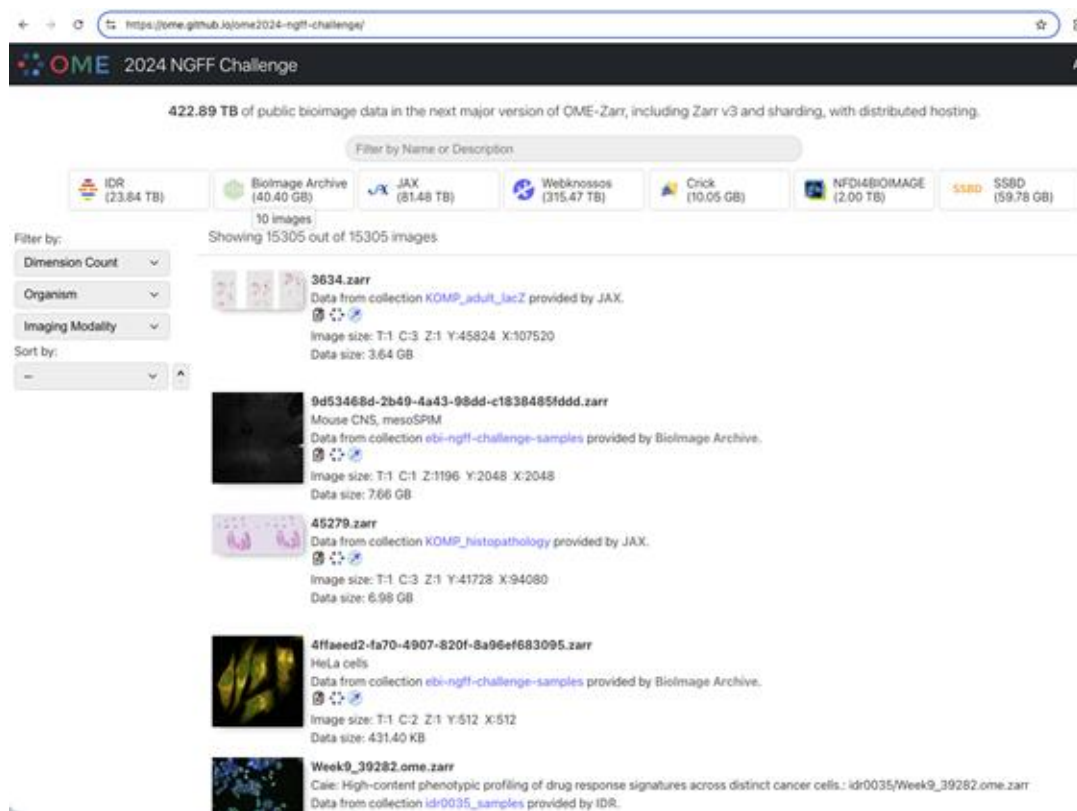
RO-Crates can link and be linked to research objects from many different sources using persistent identifiers like ORCIDs and DOIs.

Image credit: Goble, C. (2024, February 16). FAIR Digital Research Objects: Metadata Journeys. University of Auckland Seminar, Auckland. Zenodo. <https://doi.org/10.5281/zenodo.10710142>

Research Object Crate (RO-Crate)

Metadata and data sharing

- OME-NGFF 2024 challenge



The screenshot displays the OME 2024 NGFF Challenge website. At the top, it states "422.89 TB of public bioimage data in the next major version of OME-Zarr, including Zarr v3 and sharding, with distributed hosting." Below this, there's a search bar and a list of data sources: IDR (23.84 TB), BioImage Archive (40.40 GB), JAX (81.48 TB), Webknossos (315.47 TB), Crick (10.05 GB), NFDI4BioIMAGE (2.00 TB), and S5BD (59.78 GB). The page shows "10 images" and "Showing 15305 out of 15305 images". On the left, there are filters for Dimension Count, Organism, and Imaging Modality, and a sort by dropdown. The main content area lists several data crates with their IDs, descriptions, and metadata:

- 3634.zarr**: Data from collection *KOMP_adult_lacZ* provided by JAX. Image size: T:1 C:3 Z:1 Y:45824 X:107520. Data size: 3.64 GB.
- 9d53468d-2b49-4a43-98dd-c1838485fddd.zarr**: Mouse CNS, mesoSPIM. Data from collection *ebi-ngff-challenge-samples* provided by BioImage Archive. Image size: T:1 C:1 Z:1196 Y:2048 X:2048. Data size: 7.66 GB.
- 45279.zarr**: Data from collection *KOMP_histopathology* provided by JAX. Image size: T:1 C:3 Z:1 Y:41728 X:94080. Data size: 6.98 GB.
- 4ffaee2-fa70-4907-820f-8a96ef683095.zarr**: HeLa cells. Data from collection *ebi-ngff-challenge-samples* provided by BioImage Archive. Image size: T:1 C:2 Z:1 Y:512 X:512. Data size: 431.40 KB.
- Week9_39282.ome.zarr**: Caia: High-content phenotypic profiling of drug response signatures across distinct cancer cells: *idr0035/Week9_39282.ome.zarr*. Data from collection *idr0035_samples* provided by IDR.

Research Object Crate (RO-Crate)

Metadata and data sharing

- OME-NGFF 2024 challenge
- FoundingGIDE search
- BIA submission (in progress)

```
{
  "@context": [
    "https://w3id.org/ro/crate/1.1/context",
    {
      "organism_classification": "https://schema.org/taxonomicRange",
      "BioChemEntity": "https://schema.org/BioChemEntity",
      "channel": "https://www.openmicroscopy.org/Schemas/Documentation/Generated/OME-",
      "obo": "http://purl.obolibrary.org/obo/",
      "FBcv": "http://ontobee.org/ontology/FBcv/",
      "acquisition_method": {
        "@reverse": "https://schema.org/result",
        "@type": "@id"
      },
      "biological_entity": "https://schema.org/about",
      "biosample": "http://purl.obolibrary.org/obo/OBI_0002648",
      "preparation_method": "https://www.wikidata.org/wiki/Property:P1537",
      "specimen": "http://purl.obolibrary.org/obo/HSO_0000308"
    }
  ],
  "@graph": [
    {
      "@id": "./",
      "@type": "Dataset",
      "name": "141-Sato-CellMorphology/Fig3a_FIB-SEM_synapse",
      "description": "FIB-SEM image of a synapse in a 141-Sato cell morphology dataset."
    }
  ]
}
```

FoundingGIDE Search Portal

This search portal provides a single federated search interface over a number of biological imaging databases that contribute to gide. We currently display data from the [Image Data Resource \(IDR\)](#), [SSBD:repository](#) and [SSBD:database](#), and the [Bioline Archive \(BIA\)](#).

 foundingGIDE About

Search GIDE databases

Publisher

- ☐ BioImage-Archive (804)
- ☐ SSBD:repository (175)
- ☐ SSBD:database (158)
- ☐ IDR (138)

Organism

- ☐ Homo sapiens (519)
- ☐ Mus musculus (378)
- ☐ Arabidopsis thaliana (46)
- ☐ Drosophila melanogaster (43)
- ☐ Saccharomyces cerevisiae (32)

[Show 10 more](#)

Page Size:



[2D and 3D instance segmentation of nuclei from volume electron microscopy data](#)

Kedar Narayan

Published: 2026-02-03 Licence: [CC0](#)

2D and 3D instance segmentation of nuclei from array tomography (AT) and FIB-SEM datasets, respectively. In this deposition, 10 AT and 2 FIB-SEM datasets and paired annotations are included. Note: in several datasets, partial annotations of nuclei at image edges have been deleted. These are indicated in the annotation metadata.

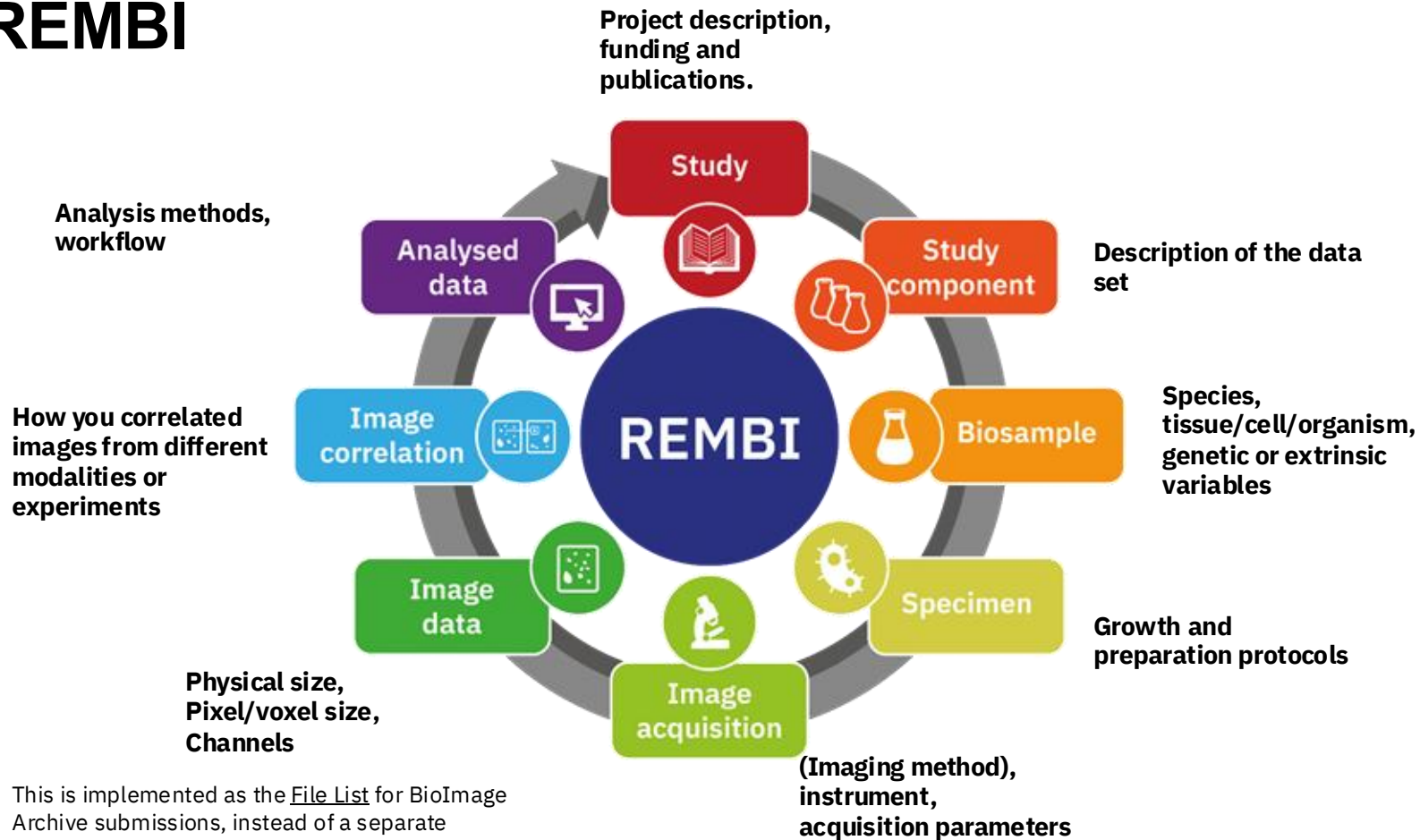
Imaging Methods: [scanning electron microscopy \(SEM\)](#), [focussed ion beam scanning electron microscopy \(FIB-SEM\)](#)

Organisms: [Homo sapiens](#), [Mus musculus](#)

<https://embl-ebi.cloud.panopto.eu/Panopto/Pages/Viewer.aspx?id=b2be292d-73d9-47e8-bbd2-b43200cfb63c>

Let's put things together: A possible solution

REMBI



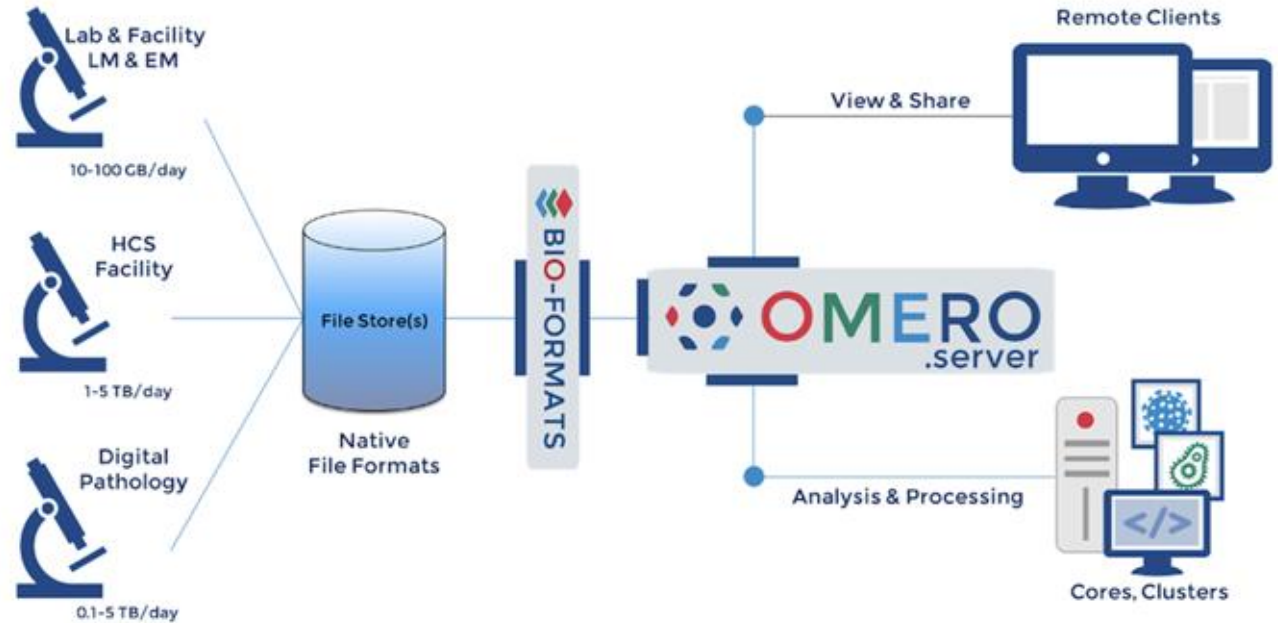
This is implemented as the [File List](#) for BioImage Archive submissions, instead of a separate component in the submission form.

<https://www.ebi.ac.uk/bioimage-archive/rembi-model-reference/>

OMERO

Benefit: Full power of Database

Challenge: Custom API, bottleneck



Allan et al (2012) Nature Methods



Export and Transfer

- [omero-cli-plugin](#): transfer object and annotations

Two objects to highlight:

- -barchive creates a package compliant with BIA submission standards
- -rocrate generates a RO-Crate compliant package with flat structure

Installation

- **Manually**
 - Instruction for [RHEL 9](#) and [Ubuntu 24.04](#)
- **Docker:** containerisation
 - <https://ome.github.io/training-docker/>
- **Ansible:** configuration management
 - <https://github.com/ome/prod-playbooks>

Server Specification

- It depends on usage and type of data
- Training server:
 - 4 CPU 40GB RAM
- Teaching server (Indonesia)
 - 16 CPU 100GB RAM
- Dundee server of Faculty of Life Sciences
 - 4 CPU 32GB RAM (server)
 - 2 CPU 15 GB RAM (web)

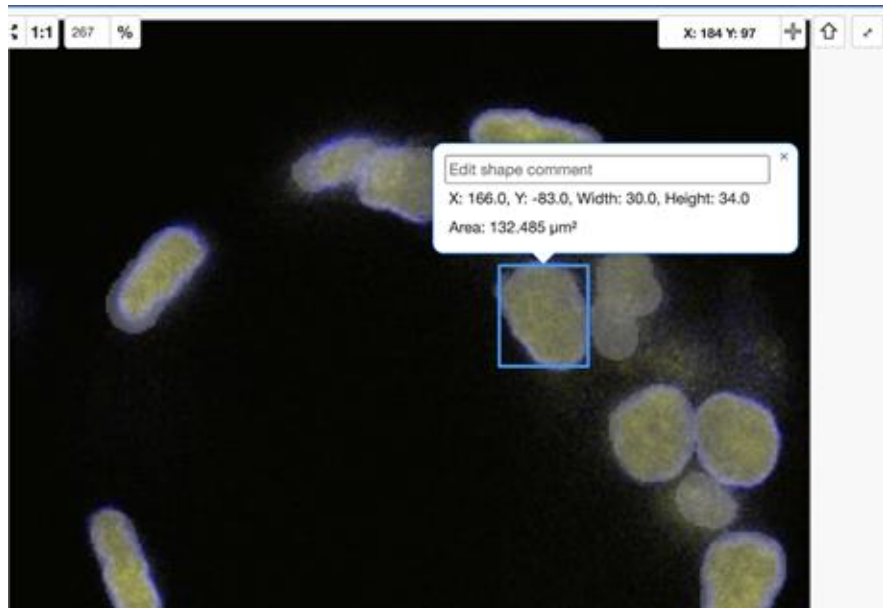
Exercise: Quick OMERO tour

Server: <https://workshop.openmicroscopy.org>

Users: user-1, user-2, etc.

Password: harvard

Guide: <https://omero-guides.readthedocs.io/en/latest/figure/docs/>



Z-sections/Timepoints: 236 x 1
 Channels: LaminB1, Dapi
 ROI Count: 61

Attributes 2

Info Settings ROIs [61]

Save Undo Redo ☐ Show Comments

Comment 10

Z 119 ☐ T 1 ☐

ROIs Planes

☒ Show Z T C Comment

- ☒ 110
- ☒ 111
- ☒ 112
- ☒ 113
- ☒ 114
- ☒ 115
- ☒ 116
- ☒ 117
- ☒ 118
- ☒ 119

Organism

Added by: Public data

Organism [Homo sapiens](#)

hers

Added by: Public data

Type microglia
 in Region caudate
 Disease State HC
 Patient ID HC10
 Sex female
 Age of Death 89
 I 13
 Channels Iba1; pS129

Added on Project [idr0173-breiter-alfasynuclein](#)

Sample Type tissue
 Organism [Homo sapiens](#)
 Study Title Early Parkinson's disease is defined by α -synuclein oligomer-driven aggregation susceptible cells
 Study Type protein localization
 Imaging Method spinning disk confocal microscopy

Publication Title α -synuclein-driven cell susceptibility in Parkinson's Disease
 Publication Authors Breiter JC, Beckwith JS, Brock EE, Lachica J, Toomey CE, Fu B, Weiss LE, Wood NW, Vendruscolo M, Gandhi S, Lee SF
 Release Date 2025-11-20
 License CC BY 4.0 <https://creativecommons.org/licenses/by/4.0/>

data

Image acquisition

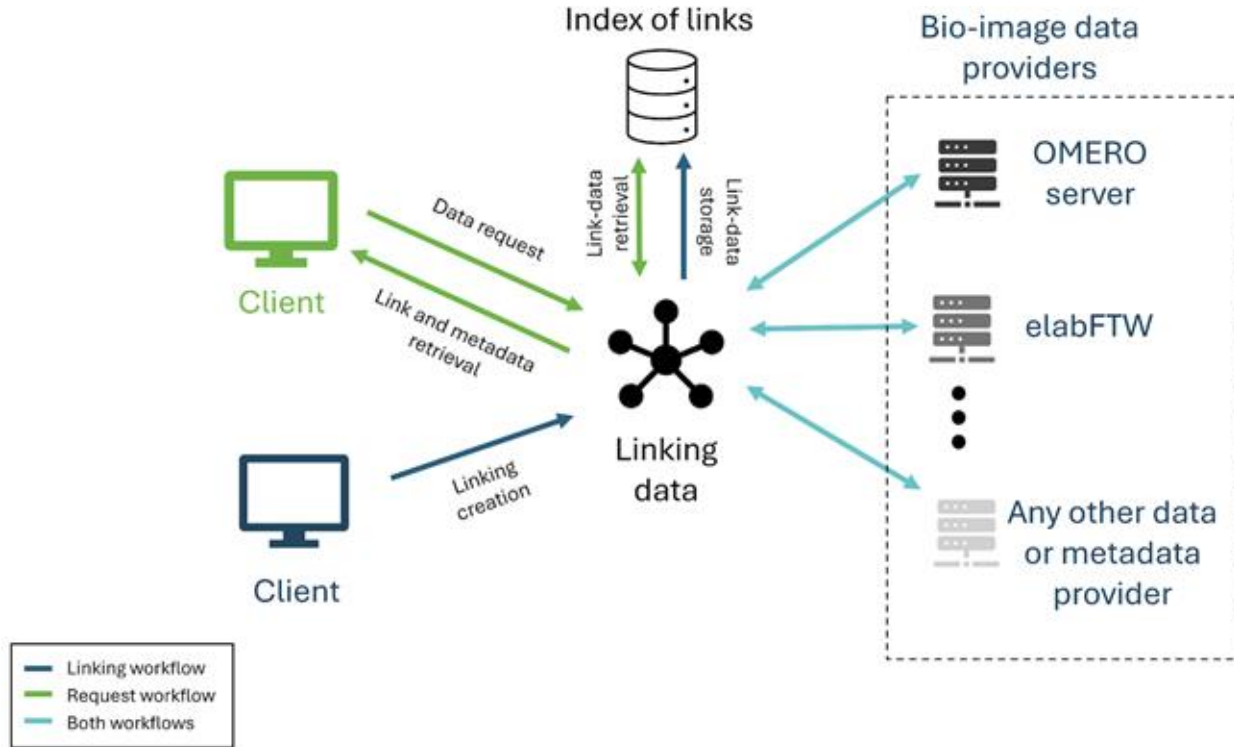
OMERO.guides and Videos

- OMERO.guides – including community workflow contributions

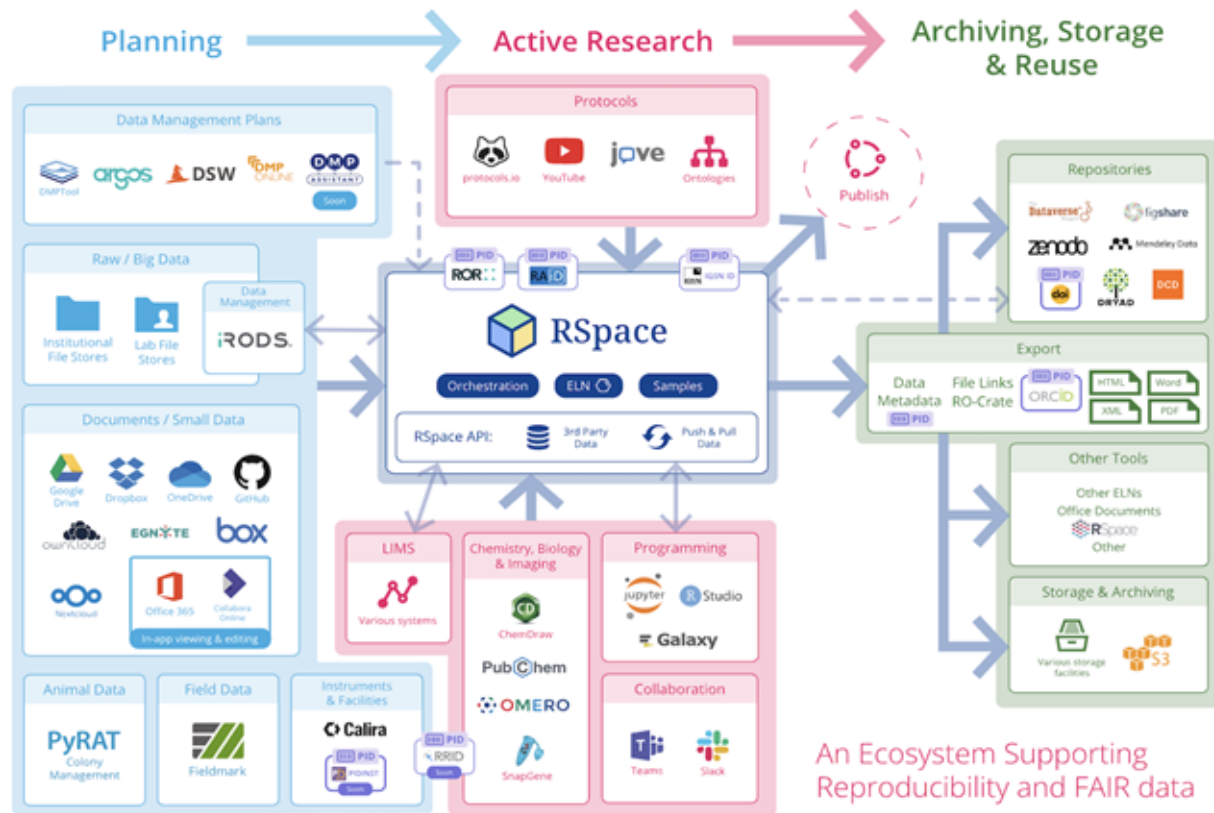


<https://omero-guides.readthedocs.io/en/latest/index.html>

OMERO & ELN



Rspace: Research Infrastructure Orchestrator



An Ecosystem Supporting
Reproducibility and FAIR data

Data management end goal: Successful real life examples

How to organise and share my imaging data?

Multimodal data management for marine biologists,
environmental scientists and imaging specialists

<https://www.ebi.ac.uk/training/events/how-organise-and-share-my-imaging-data/>

List of Webinars

1. Principles of research data management
2. Mastering data management: Go FAIR, Go Open, Go Smart — EMO BON and TREC showcases
3. Towards open and standardised imaging data: an introduction to Bio-Formats, OME-TIFF, and OME-Zarr
4. Data management in a bioimage informatics data flow
5. A FAIR and scalable workflow for plankton phenogenomics at single cell level
6. Open FAIR data: the role of public data archives
7. A journey to FAIR bioimage data

A journey to FAIR bioimage data

Stefanie Weidtkamp-Peters, Heinrich-Heine-Universität Düsseldorf

Tom Boissonnet, Heinrich-Heine-Universität Düsseldorf

Christian Schmidt, DKFZ Heidelberg



NFDI4
BIOIMAGE

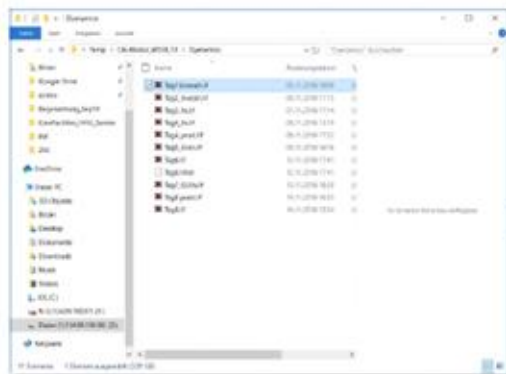
Info and recording: <https://www.ebi.ac.uk/training/events/journey-fair-bioimage-data>

Presentation slides: <https://zenodo.org/records/15796252>

What happens in between ...in reality?



NFDI4
BIOIMAGE

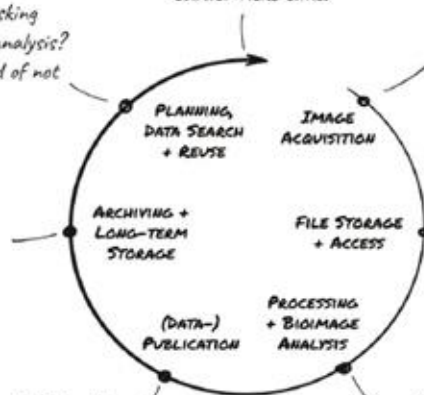


WHAT? They are asking
for my data for their analysis?
But the naming is kind of not
intuitive ...

Most of the data is still
on the microscope PC and I
have some on this stick and
there was an external
drive ...

SHIT, how did I do this
measurement?! I'll just write
'as previously done'.

Gosh, what a nightmare.
I really need to think about
data management MUCH
earlier next time!



I will remember the
laser power - I usually
take the same value
anyway.

I just leave my
images on the microscope
PC. Nobody would delete
anything on there,
right?

We always use this old
Python script this one postdoc
once wrote for analysis.



BEEN THERE, DONE THAT.

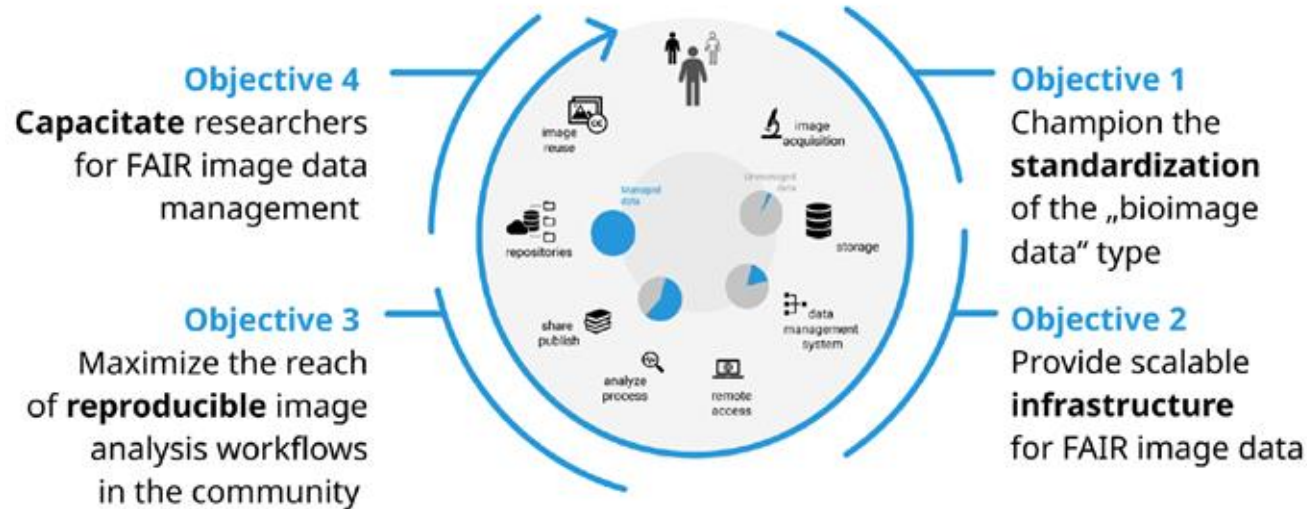
HENNING FALK

"[Imaging Data Lifecycle](#)", NFDI4BIOIMAGE Consortium (2024):
NFDI4BIOIMAGE data management illustrations by [Henning Falk](#),
Zenodo, <https://doi.org/10.5281/zenodo.14186100>, is used under
a [CC BY 4.0](#) license. Modifications to this illustration include cropping.

NFDI4BIOIMAGE - a consortium of the National Research Data Infrastructure



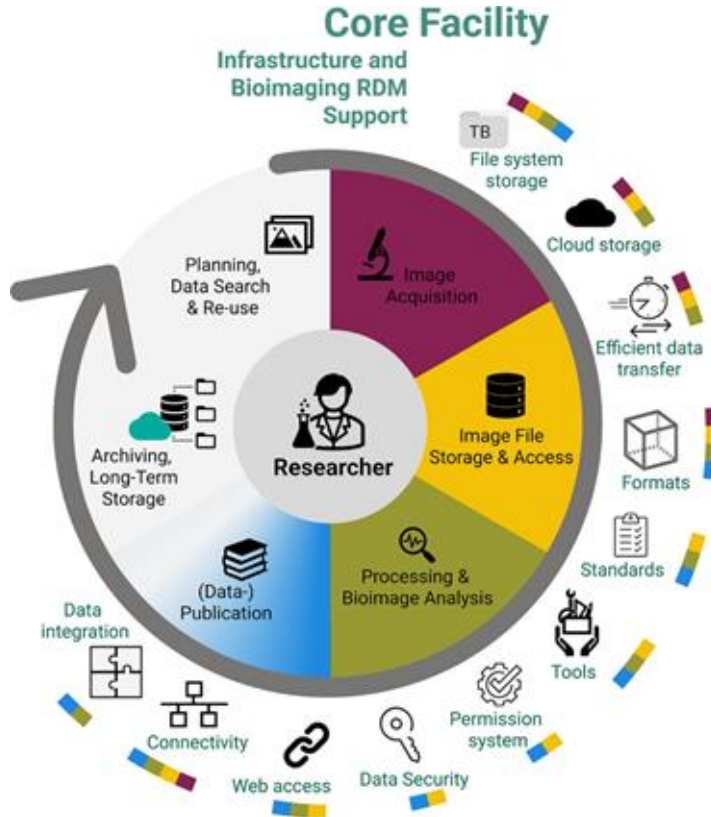
NFDI4BIOIMAGE - Main objectives



6 Task Area teams



A practical guide to bioimaging research data management in core facilities



Schmidt, C., Boissonnet, T., Dohle, J., Bernhardt, K., Ferrando-May, E., Wemet, T., Nitschke, R., Kunis, S., & Weidtkamp-Peters, S. (2024). A practical guide to bioimaging research data management in core facilities. *Journal of Microscopy*, 294, 350–371.

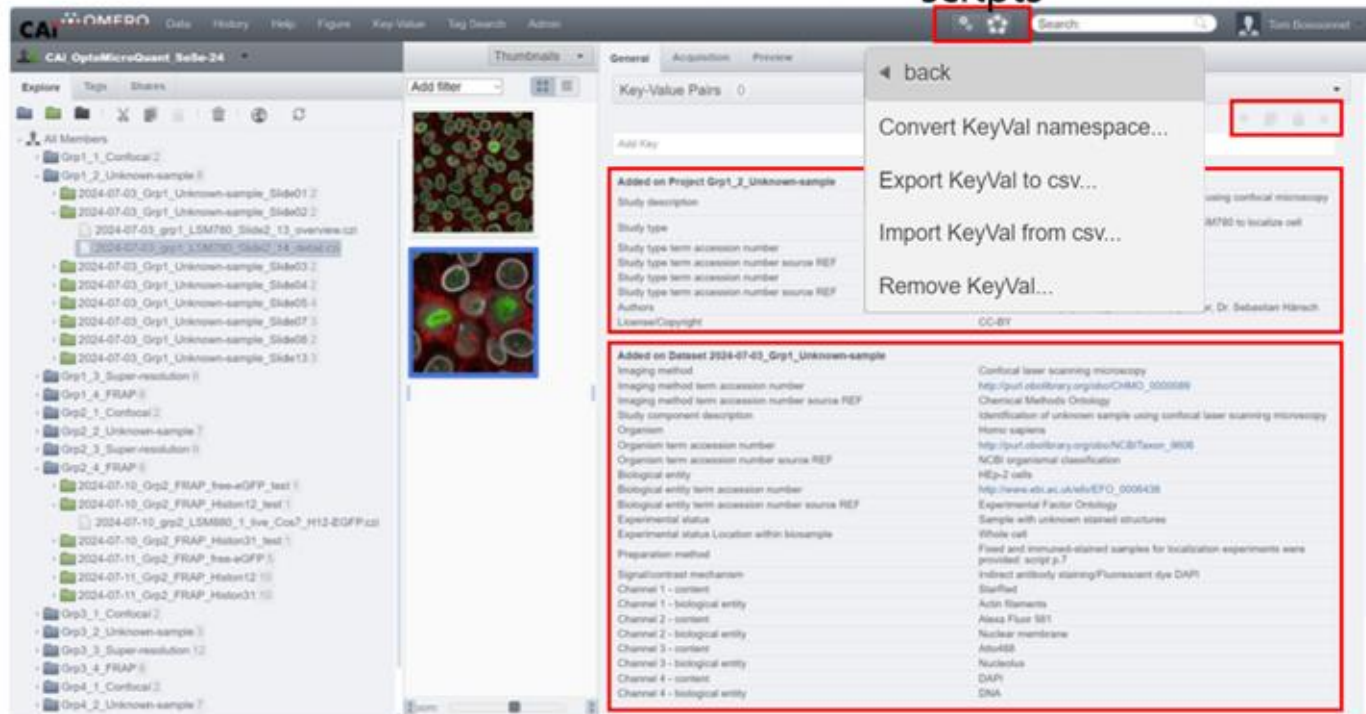
<https://doi.org/10.1111/jmi.13317>

A practical guide to bioimaging research data management in core facilities

TABLE 1 Common issues and solutions regarding bioimage data management planning.

Common issue	Proposed solution
Underestimating the complexity of bioimage data handling	Consulting and taking advice from core facility staff. Using DMP templates that have been approved or provided by core facilities.
Reusing existing data is not considered	Data search and review in public repositories and institutional archives can become part of experiment planning. Reusing data from repositories can help users familiarise themselves with the data types and test workflows before acquiring their own data. Suitable archives can be found on re3data.org or on FAIRsharing.org . Bioimaging-specific archives include, for example, the BioImage Archive (BIA) and the Image Data Resource (IDR). Institutional archives may be another data source.
Lack of adaptation strategies for changing demands throughout the experiment	Writing a data management plan and using it as a living document to adapt where required, and record changes.
Suboptimal instrument choice	User training and guidance on instrument choice and setup, as well as recommendations on sample preparation. Consulting the core facility and taking advice on instrument and software choice.
Image analysis and data processing are not planned prior to the acquisition	Consulting with core facility staff, image analysts and data stewards before starting a new project. Searching example data and protocols that have established the desired procedure previously to test image analysis a priori. Planning suitable data structures and metadata for analysis and running a pilot experiment before adjusting and scaling up.
The data volumes, storage space and data transfer requirements are not considered before the acquisition	Gathering comprehensive information about typical image sizes of the core facility's instruments. Highlight limits to memory load and data transfer with large files. Estimating expected data volumes and required storage and ensuring resource availability.
The importance of metadata collection is overlooked	Preparing a metadata annotation checklist for the data even before acquisition, for example, based on REMBI. Optionally, enforce annotation as part of a usage policy.
Lack of a strategy for the publication of image data	Considering data publication, candidate public repositories, and reviewing submission guidelines. Defining authorship and licensing criteria with collaboration partners early in the project.

Working with images — key-value pairs scripts



CAI OptoMicroQuest

Explore

Grp1_1_Confocal2

Grp1_2_Unknown-sample

2024-07-03_Grp1_Unknown-sample_Slide012

2024-07-03_Grp1_Unknown-sample_Slide022

2024-07-03_gpr1_LSM780_Slide2_13_overview.czi

2024-07-03_gpr1_LSM780_Slide2_24_detail.czi

2024-07-03_Grp1_Unknown-sample_Slide032

2024-07-03_Grp1_Unknown-sample_Slide042

2024-07-03_Grp1_Unknown-sample_Slide052

2024-07-03_Grp1_Unknown-sample_Slide062

2024-07-03_Grp1_Unknown-sample_Slide072

2024-07-03_Grp1_Unknown-sample_Slide082

2024-07-03_Grp1_Unknown-sample_Slide132

Grp1_3_Super-resolution

Grp1_4_FRAP

Grp2_1_Confocal2

Grp2_2_Unknown-sample

Grp2_3_Super-resolution

Grp2_4_FRAP

2024-07-10_Grp2_FRAP_free-eGFP_test

2024-07-10_Grp2_FRAP_Histon12_test1

2024-07-10_gpr2_LSM880_1_free_Cox7_H12-eGFP.czi

2024-07-10_Grp2_FRAP_Histon31_test1

2024-07-11_Grp2_FRAP_free-eGFP1

2024-07-11_Grp2_FRAP_Histon121

2024-07-11_Grp2_FRAP_Histon311

Grp3_1_Confocal2

Grp3_2_Unknown-sample

Grp3_3_Super-resolution12

Grp3_4_FRAP

Grp4_1_Confocal2

Grp4_2_Unknown-sample

Key-Value Pairs

Added on Project Grp1_2_Unknown-sample

Study description

Study type

Study type term accession number

Study type term accession number source REF

Study type term accession number

Study type term accession number source REF

Authors

License/Copyright

CC-BY

Added on Dataset 2024-07-03_Grp1_Unknown-sample

Imaging method

Imaging method term accession number

Imaging method term accession number source REF

Study component description

Organism

Organism term accession number

Organism term accession number source REF

Biological entity

Biological entity term accession number

Biological entity term accession number source REF

Experimental status

Experimental status Location within biosample

Preparation method

Signalcontrast mechanism

Channel 1 - content

Channel 1 - biological entity

Channel 2 - content

Channel 2 - biological entity

Channel 3 - content

Channel 3 - biological entity

Channel 4 - content

Channel 4 - biological entity

Confocal laser scanning microscopy

http://purl.obolibrary.org/obo/CHMO_0000089

Chemical Methods Ontology

Identification of unknown sample using confocal laser scanning microscopy

Homo sapiens

http://purl.obolibrary.org/obo/NCITaxon_3606

NCBI organismal classification

H2o-2 cells

http://www.ebi.ac.uk/efo/0006438

Experimental Factor Ontology

Sample with unknown stained structures

Whole cell

Fixed and immunolabeled samples for localization experiments were provided: script p.7

Indirect antibody staining/Fluorescent dye DAPI

StarRad

Actin filaments

Alexa Fluor 568

Nuclear membrane

Atto488

Nucleolus

DAPI

DNA

copy
paste

Recommended Metadata for Biological Images, Sarkans et al, Nature Methods 2021

A journey to FAIR bioimage data

27

Fluorescence lifetime separation of fluorophores in *C. elegans* embryos and hermaphrodites

Cornelia Wetzer ¹, Marcelo Leomil Zoccoler ², Svetlana Iarovenko ², Chukwuebuka William Okafornta ³, Anja Nobst ¹, Hella Hartmann ¹, Thomas Müller-Reichert ¹, Robert Haase ³, Gunar Fabig ¹

¹ Technische Universität Dresden ² Institute of Molecular Biotechnology of the Austrian Academy of Sciences, Vienna ³ Leipzig University

Accession S-BIAD1967

DOI [10.6019/S-BIAD1967](https://doi.org/10.6019/S-BIAD1967)

ReleaseDate 2025-06-18

Description Fluorescence lifetime imaging microscopy (FLIM) translates the duration of excited states of fluorophores into lifetime information as additional source of contrast in images of biological samples. This offers the possibility to separate fluorophores particularly beneficial in case of similar excitation spectra. Here, we demonstrate the distinction of fluorescent molecules based on FLIM phasor analysis, called lifetime separation, in live-cell imaging using open-source software for analysis. We showcase two applications using *Caenorhabditis elegans* as a model system. First, we separated the highly spectrally overlapping fluorophores mCherry and mKate2 to distinctively track tagged proteins in six-dimensional datasets to investigate cell division in the developing early embryo. Second, we separated fluorescence of tagged proteins of interest from masking natural autofluorescence in adult hermaphrodites. For FLIM data handling and workflow implementation, we developed the open-source plugin napari-FLIM-phasor-plotter to implement conversion, visualization, analysis and reuse of FLIM data of different formats. Our work thus advances technical applications and bioimage data management and analysis in FLIM microscopy for life science research.

Keywords FLIM, live-cell imaging, phasor, napari

Acknowledgements Imaging was performed at the Light Microscopy Facility at the Center for Molecular and Cellular Bioengineering (CMCB-LMF) of TU Dresden. We wish to acknowledge the support of the CMCB-LMF team, particularly Ellen Geibelt for technical support and maintenance of the microscope, and Silke Tulok (Core Facility Cellular Imaging at the Faculty of Medicine at TU Dresden) for support in bioimage data management using OMERO. We thank the Bio-Image Analysis Technology Development team of Physics of Life at TU Dresden (BiA-PoL) for provision of infrastructure and fruitful discussions, Tom Boissonnet (Heinrich Heine Universität Düsseldorf, I3D:bio) for support with metadata annotation and Josh Moore (German BioImaging - Gesellschaft für Mikroskopie und Bildanalyse e.V.) for feedback on the resulting manuscript.

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Funding statement C.W. is funded by the German consortium NFDI4BIOIMAGE (Deutsche Forschungsgemeinschaft, grant number NFDI 46/1, project number 501864659). Research in the Müller-Reichert lab is supported by grants

The screenshot shows a web-based interface for managing data files. At the top, there's a header 'Data files' with a search bar and a 'Show' dropdown menu set to 'entries'. Below this is a table with a single column 'Name'. The table contains four entries, each with a checkbox on the left and a file icon on the right. The entries are: 'TMR17_2_sptw_summed_intensity_with_segme', 'TMR17_3_sptw_summed_intensity_with_segme', 'TMR31_1_sptw_summed_intensity_with_segme', and 'TMR31_2_sptw_summed_intensity_with_segme'. The last entry is partially visible as 'TMR31_3_sptw_summed_intensity_with_segme'.

Fluorescence lifetime s hermaphrodites

Cornelia Wetzker ¹, Marcelo Leomil Zoccolero ², Svetlana Müller-Reichert ², Robert Haase ², Gunar Fabi ²

¹ Technische Universität Dresden ² Institute of Molecular Bio

Accession S-BIAD1967

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Funding statement C.W. is funded by the Germ grant number NFDI 46/1, project number 501

Interdisciplinary Study on Drug-Induced-Phospholipidosis of Repurposing Libraries through Machine Learning and Experimental Evaluation in Different Cell Lines

Maria Kuzikov ¹, Adelinn Kalman ², Jeanette Reinshagen ¹, Johanna Huchting ¹, Kun Qian ², Hanna Axelsson ², Marianna Tampere ², Päivi Östling ², Brinton Seashore-Ludlow ², Yojana Gadiya ², Philip Gribbon ¹, Andrea Zaliani ¹, Jens Wendt ¹

¹ Fraunhofer Institute for Translational Medicine and Pharmacology ² Department of Oncology-Pathology, Karolinska Institutet, Science for Life Laboratory, Solna, Sweden ³ Department of Medical Biochemistry and Biophysics, Karolinska Institutet, Science for Life Laboratory, Solna, Sweden ⁴ Bonn Aachen International Center for Information Technology ⁵ University of Münster ⁶ NFDI4Bioimage

Accession S-BIAD2282

DOI [10.6019/S-BIAD2282](https://doi.org/10.6019/S-BIAD2282)

ReleaseDate 2025-09-12

Description Phospholipidosis (PLD), a cellular adverse effect that is, among others, caused by numerous cationic amphiphilic drugs. Interest is raised within pharma discovery to predict this phenomenon, as it can impact the outcome of phenotypic cellular screens and significantly delay drug development processes. The development of accurate and validated machine learning models for predicting drug-induced PLD across different cell lines and research centers could provide a valuable early application tool for the pharmaceutical industry, potentially accelerating drug discovery and reducing the risk of late-stage failures. We report here the assembly, curation, testing and modeling of one of the largest datasets of repurposed drugs (5000+) tested for PLD induction on different cell lines. A machine-learning classification method was developed and validated to predict whether molecules are prone to induce PLD effects when applied in cell-based screens.

This constitutes the accompanying publication of the acquired image data.

Keywords Phospholipidosis, HCS, Machine learning, Drug repurposing

Acknowledgements We thank Jens Wendt (NFDI4BIOIMAGE, DFG project ID 501864659) for help with metadata annotation and upload to the BioImage Archive repository.

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Funding statement This work and the authors were primarily funded by the EU-Horizon HLTH Remedi4ALL (EU 101057442).

Funding Horizon Europe: [101057442](https://doi.org/10.101057442)

Vero-E6 Kidney cells from African green monkey

Data files

Show 5 entries Search:

☐	Name	Size	Section	source
☐	ESP0030315.zarr.zip	12.21 GB	Study Component	
☐	ESP0030316.zarr.zip	12.19 GB	Study Component	
☐	ESP0030317.zarr.zip	12.08 GB	Study Component	
☐	ESP0030460.zarr.zip	12.83 GB	Study Component	
☐	ESP0030461.zarr.zip	12.75 GB	Study Component	

Showing 26 to 30 of 32 entries

Previous 1 2 3 4 5 6 7 Next

Download all files

ORCID: Data claiming

You can sign-in to ORCID to claim your data

☐ Remember me on this computer

Similar Studies

- PolyA tail segmentation improves the stability of the template DNA and increases the translatability of in vitro transcribed mRNA. [S-BIAD2347]

EMO BON: European Marine Omics Biodiversity Observation Network

<https://www.ebi.ac.uk/training/events/mastering-data-management-go-fair-go-open-go-smart-emo-bon-and-trec-showcases/>

Centralised data management via GitHub

CENTRALISED data space to deal with DISTRIBUTED data

- Allowing each data team to use formats familiar to them
- **But adding interoperability to that via metadata, standards, templates, formatting**

Why GITHUB?

- GitHub offers version control, automated actions, mini-webpages, issue tracking, data (re)formatting, etc
- Workflows to reformat data as required for external publishers
- For large files: need to be archived somewhere else, but via LFS can be dealt with as if are on GitHub

DATA PACKAGING via Research Object Crate (RO-Crate)

- Packaging research data with their metadata
- Who, what, where, when, why, how and internal/external data linking



End-user best practices for data organisation

Folder structure tips

- Use folders to separate raw from processed/analysed data
- It is always useful to add a [README.md](#) file to guide people on the contents and structure of that folder and subfolder(s)
- Organize your files in a logical structure that reflects the relationships and hierarchy between these files
- Use file naming conventions that will make it easier to identify and retrieve the data

File naming tips

- Try and use meaningful names related to the data
- Provide a version of the file
 - Use a minimum of 2 digits
 - Avoid using `_final`, `_final_final`
 - Don't use initials to version a file
 - Use version control e.g. [GitHub](#) or [Gitlab](#)
- Try and avoid using personal names
 - e.g. `mut1_confocal_GFP_Melina`
- Add date at the beginning if you want to sort out chronologically
 - Follow the YYYY-MM-DD format
- Avoid using special characters or accents
 - e.g. '+' will cause problems in a BIA submission
- Try and avoid using white spaces to separate words.
 - Best to use underscores, or camelCase
- Avoid extra long names
- If there are known abbreviations in the field, use them

Good data management practices

- Folder and file structures are good for data organisation but it is not data management.
- Folder structures and file names are **NOT** your metadata storage.
 - Filenames have a character limit, you can't put all metadata in a filename.
 - Folder structures and filenames are not accountable and changes are not trackable. They don't provide provenance.
 - What happens if you or someone else move the data or accidentally rename the data?
 - What happens if you forget what the abbreviation you put in meant?

Storage and sharing strategies and tools

Hot, Warm or Cold Storages: Which one to choose?

- **Hot Storage:** Refers to data that is frequently used and accessed. The problem does not go away with Cloud storage.
- **Cold Storage:** is the opposite; this is data that you want to keep (probably off site and on slower equipment), but you rarely need to access. Less expensive to store than hot data. Cold storage could sit off-site on a service like [Amazon Glacier](#)
- **Warm Storage:** Something in the middle. Data stored close enough for easy access, but with fewer financial hits than your hot storage data e.g Network-attached Storage or a remote file server e.g. SharePoint.

	Hot	Warm	Cold
Access Speed	Fast	Slower	Slowest
Access Frequency	Regular	Irregular	Rare
Storage Media	Standard Hard drive, easy access cloud storage	Network Attached storage (NAS), remote file server	Archive Cloud storage (e.g. Amazon Glacier), Tape
Cost	Higher	Medium	low
Ideal User	Everyone	Users with lots of data	Users with lots of data

Aspects to consider for storage strategy

- **Scalability**
- **Data access frequency**
- **Data retention requirements**
- **Budget**
- **Data security**

SPEAK TO YOUR IT. DO NOT DO IT YOURSELF

Sharing data

- Public repositories e.g. BIA, IDR, EMPIAR. See https://rdmkit.elixir-europe.org/bioimaging_data
- Host your own OMERO instance and publish data using public account
- Share your data as OME-Zarr - best works with hot storage (fast cloud)

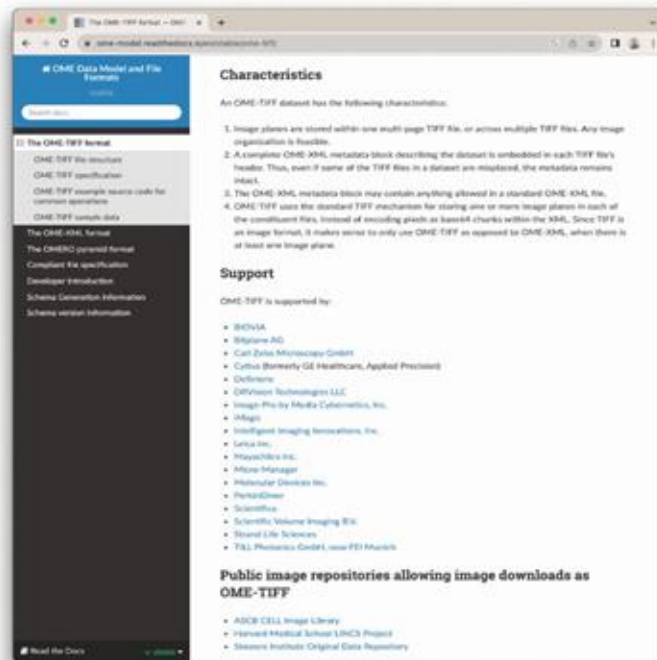
Sharing data - transfer methods and tools

- Ftp - command line/programmatic, or apps like FileZilla, Cyberduck
- Globus - basic local transfers via free personal account; server/HPC integration and managed sharing require institutional setup
- Tools requiring institutional setup and payment, such as Aspera
- Hot storage/cloud s3
 - Http - you don't need to download or transfer the data to access and analyse or view it.
 - AWS Console
 - command line tools (aws s3, rclone etc.)
 - Applications like CyberDuck

Data formats - Standardisation and interoperability



<https://ome-model.readthedocs.io/en/stable/ome-tiff/specification.html>



OME-TIFF (Pyramidal) Common Exchange Format

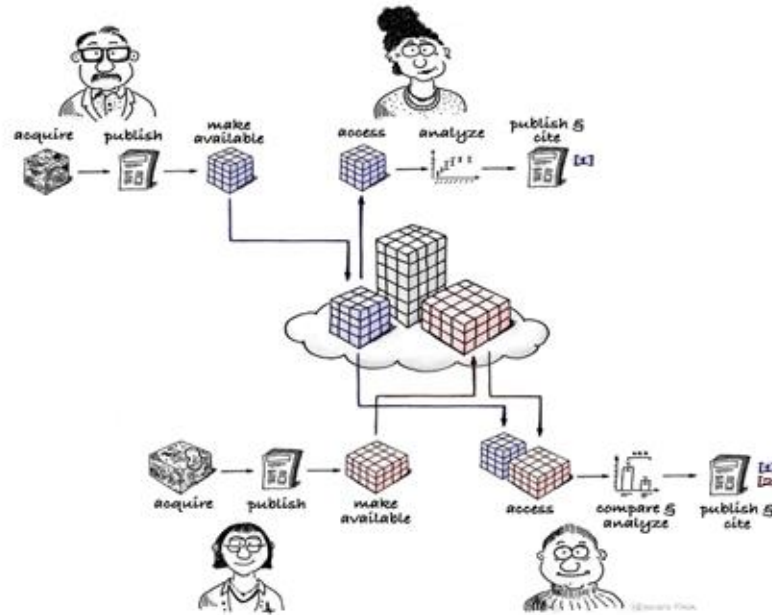
Benefit: Pre-calculated multi-resolution

Challenge: Limited annotation options

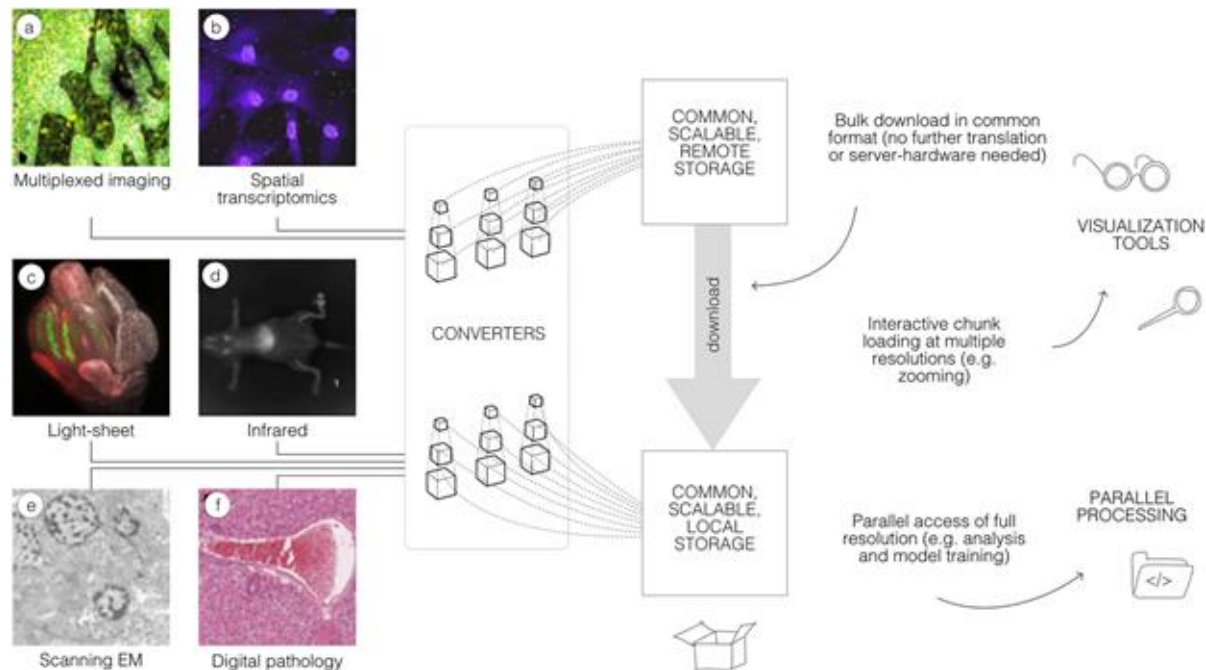
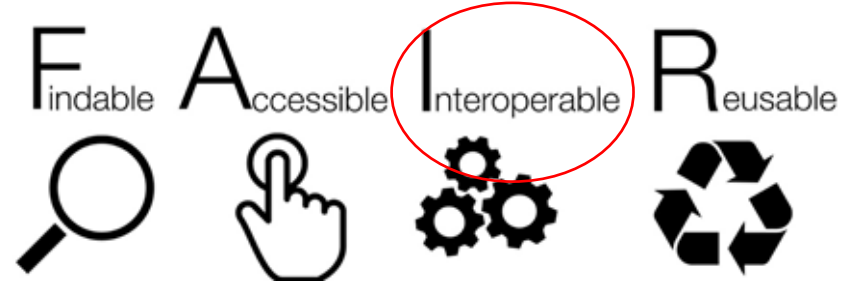


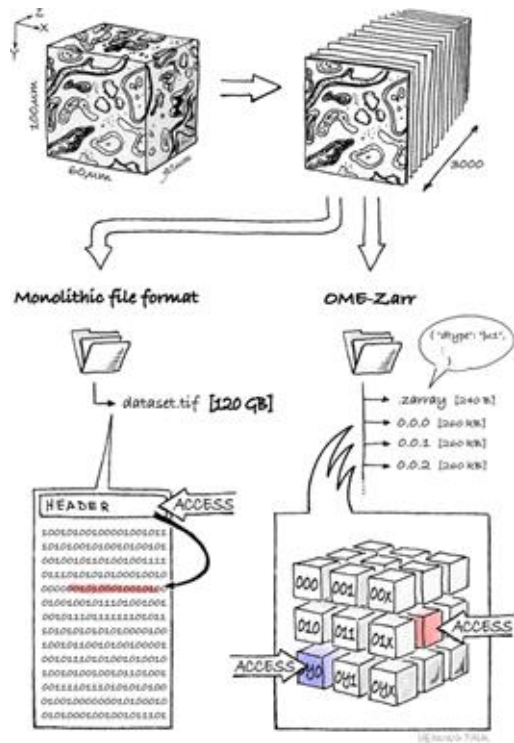
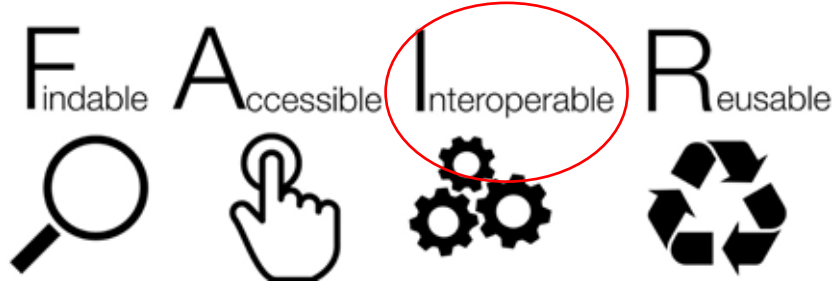
<https://www.openmicroscopy.org/2018/11/29/ometiffpyramid.html>

Challenge: FAIR principles even for largest images



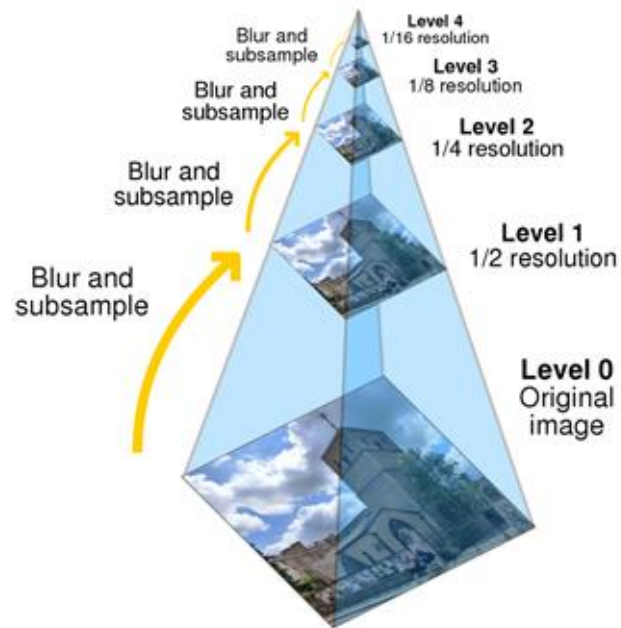
"FAIR re-use" by [@DrHenningFalk](#)
©2022 [@NumFOCUS](#) is used under a CC-BY 4.0 license.





Next generation file
format:

chunking + pyramids



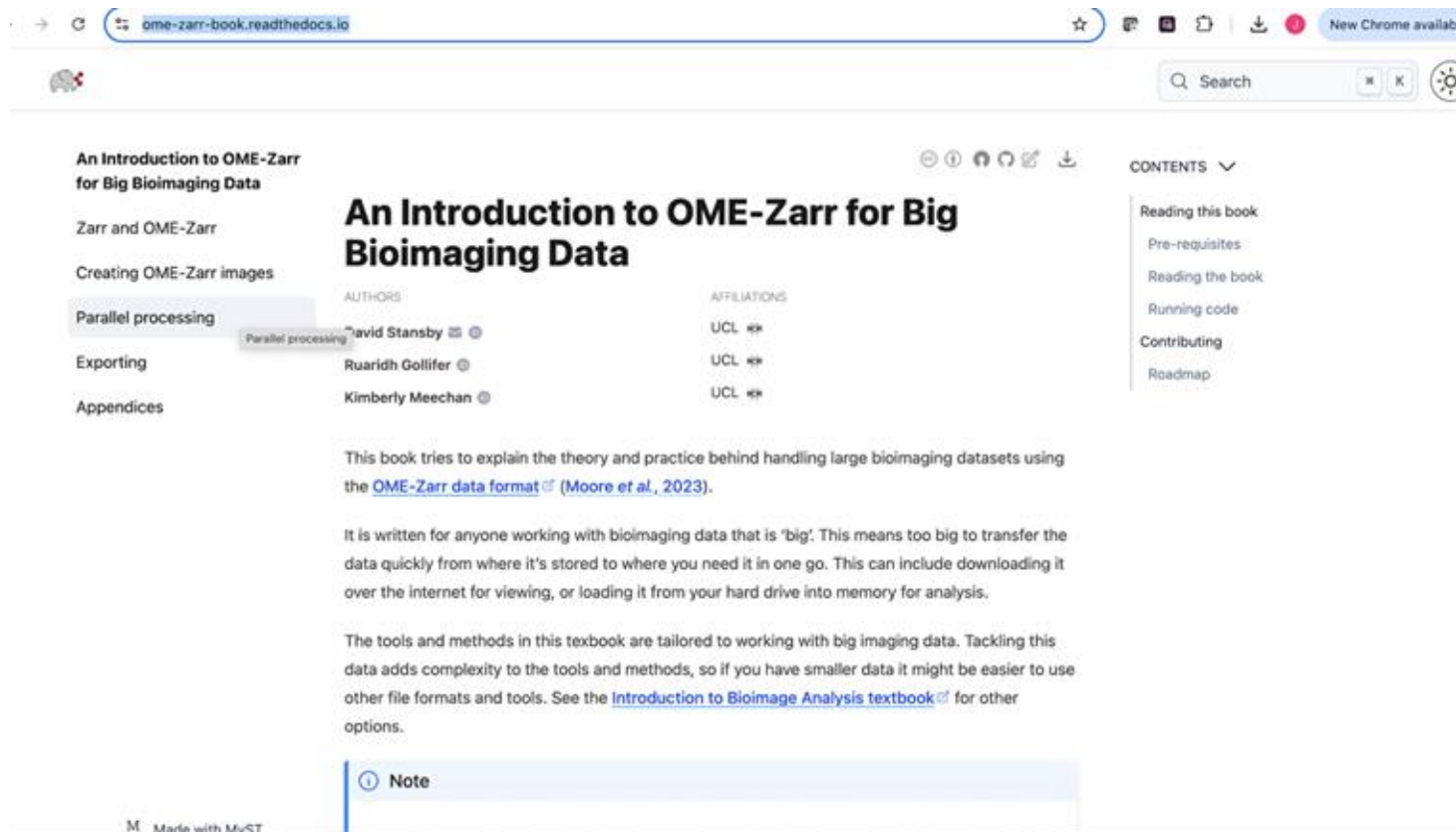
Why would I use OME-Zarr?

1. **Chunking:** “Zarr” files are stored in independently-accessible blocks.
1. **OME-Zarr:** “Zarr” with embedded standardized metadata.

When should I not use OME-Zarr?

1. Files are not big and you are working locally.
1. If data is lossy compressed e.g. SVS => Huge Increase in the number of files.

When do know more about it?



The screenshot shows a web browser displaying the OME-Zarr book website. The browser's address bar shows the URL `ome-zarr-book.readthedocs.io`. The website has a dark theme. On the left, a sidebar lists the book's sections: "An Introduction to OME-Zarr for Big Bioimaging Data", "Zarr and OME-Zarr", "Creating OME-Zarr images", "Parallel processing" (which is highlighted), "Exporting", and "Appendices". The main content area features the title "An Introduction to OME-Zarr for Big Bioimaging Data" in large, bold letters. Below the title, it lists the authors: David Stansby, Ruairidh Gollifer, and Kimberly Meechan, along with their affiliations (UCL). The text explains that the book covers the theory and practice of handling large bioimaging datasets using the OME-Zarr data format, citing Moore et al., 2023. It also mentions that the book is written for anyone working with 'big' data, which is too large to transfer quickly. The tools and methods are tailored for working with big imaging data. At the bottom of the main content area, there is a blue box with a note icon and the word "Note". On the right side, there is a "CONTENTS" dropdown menu with links to "Reading this book", "Pre-requisites", "Reading the book", "Running code", "Contributing", and "Roadmap". The footer of the website says "M. Made with MkDocs".

ome-zarr-book.readthedocs.io

New Chrome available

Search

An Introduction to OME-Zarr for Big Bioimaging Data

Zarr and OME-Zarr

Creating OME-Zarr images

Parallel processing

Exporting

Appendices

Parallel processing

An Introduction to OME-Zarr for Big Bioimaging Data

AUTHORS

David Stansby

Ruairidh Gollifer

Kimberly Meechan

AFFILIATIONS

UCL

UCL

UCL

CONTENTS

Reading this book

Pre-requisites

Reading the book

Running code

Contributing

Roadmap

This book tries to explain the theory and practice behind handling large bioimaging datasets using the [OME-Zarr data format](#) (Moore et al., 2023).

It is written for anyone working with bioimaging data that is 'big'. This means too big to transfer the data quickly from where it's stored to where you need it in one go. This can include downloading it over the internet for viewing, or loading it from your hard drive into memory for analysis.

The tools and methods in this textbook are tailored to working with big imaging data. Tackling this data adds complexity to the tools and methods, so if you have smaller data it might be easier to use other file formats and tools. See the [Introduction to Bioimage Analysis textbook](#) for other options.

Note

M. Made with MkDocs

<https://ome-zarr-book.readthedocs.io/>

How does it look?

ome.github.io/ome-ngff-validator/?source=https://livingobjects.ebi.ac.uk/idr/zarr/v0.5/idr0062A/6001240_labels.zarr

OME NGFF Validator

: 6001240_labels.zarr

Validating: [/6001240_labels.zarr.json](#)
Using version 0.5. Using schema: [/v0.5/schemas/image.schema](#)

✓

Open with:

```
{
  "attributes": {
    "ome": {
      "version": "0.5",
      "_creator": {
        "name": "omero-zarr"
      },
      "multiscales": [
        {
          "axes": [...], // 4 items
          "datasets": [...], // 3 items
        }
      ]
    }
  }
}
```

Multiscale 0
3 Datasets checked ✓

Path: [Object.json](#)

	Array	Chunk	Shard
Bytes	70.35 MB	131.07 KB	5.24 MB
Shape	[2,236,275,271]	[1,1,256,256]	[1,10,512,512]
Counts	2,236,2,2	1888 Chunks	48 Shards
Type	uint16	numpy.ndarray	

Load chunk: 0 0 0 0

https://ome.github.io/ome-ngff-validator/?source=https://livingobjects.ebi.ac.uk/idr/zarr/v0.5/idr0062A/6001240_labels.zarr

Zarr converter with UI

Name	Link	Description
NGFF- Converter		A desktop application for conversion of bioimage formats into OME-Zarr or OME-TIFF.

<https://ngff.openmicroscopy.org/resources/tools/index.html#zarr-converters-with-a-ui>

Zarr converters

Name	Link	Description
BatchConvert		A Nextflow based command-line tool that wraps bioformats2raw for parallelised conversion of image data collections to OME-Zarr.
bioformats2raw		Java application to convert image file formats, including .mrxs, to an intermediate Zarr structure compatible with the OME-Zarr specification.
BioIO Conversion		CLI & scripting tool for easily converting images to OME-Zarr. Requires bioio-ome-zarr
EuBI-Bridge		A tool for distributed conversion of microscopic image collections into the OME-Zarr format.
ITKIOOMEZarrNGFF		An ITK external module for IO of images stored in OME-Zarr file format.
multiscale-spatial-image		Generate a multiscale, chunked, multi-dimensional spatial image data structure that can be serialized to OME-Zarr.
Nextflow (nf-omezarr)		A Nextflow pipeline for converting directories of images to OME-Zarr using bioformats2raw
stack-to-chunk		A Python library to convert stacks of 2D images to OME-Zarr with minimal memory use and maximum concurrency.
stack_to_multiscale_ngff		A tool for converting multi-terabyte stacks of images into a multiscale OME-Zarr.

<https://ngff.openmicroscopy.org/resources/tools/index.html#zarr-converters>

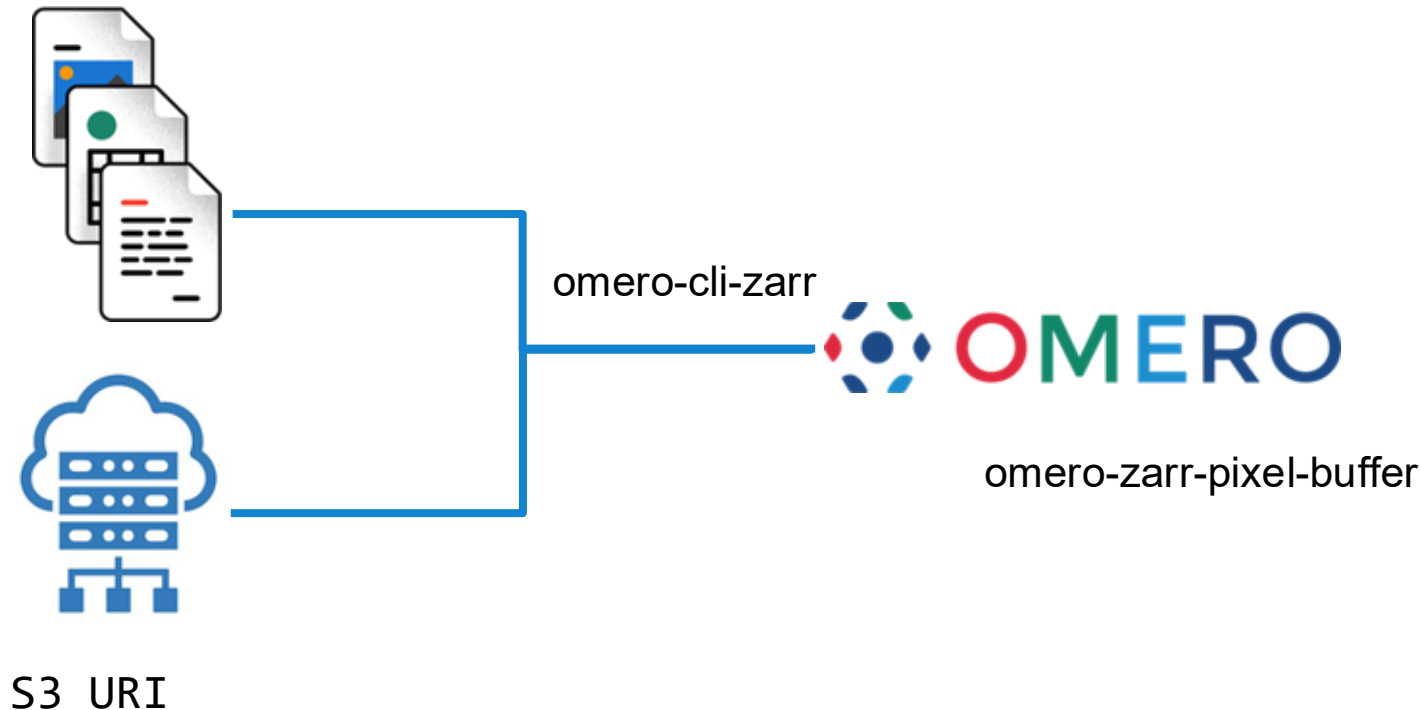
Zarr Viewers

Name	Link	Description
AGAVE		Desktop application for viewing multichannel volume data powered by your GPU
FIJI (MoBIE / BigDataViewer)		FIJI plug-in for exploring and sharing big multi-modal image and associated tabular data
FIJI (n5-ij)		FIJI plug-in for loading and saving image data to OME-Zarr and other formats supported by the N5 API
ITKWidgets		Python tool for interactively viewing images (ex. in Jupyter)
Microscopy Nodes		Blender add-on for visualizing high-dimensional microscopy data
napari		napari plug-in for viewing Zarr
Neuroglancer		A browser-based volume viewer
Viv (Aviator, Vizarr, Vitessce, ...)		A toolkit for interactive visualization of high-resolution, multiplexed bioimaging datasets. The viv toolkit is used by the Aviator , Vizarr and Vitessce image viewers, among others
Vol-E		A browser-based volume viewer
WEBKNOSSOS		An open-source tool for annotating and exploring large 3D image datasets

<https://ngff.openmicroscopy.org/resources/tools/index.html#zarr-viewers>

OMERO & OME-Zarr: "in-place" import

/data/CMU-1.ome.zarr



Example: Access possibilities of images

image from idr0044, McDole et al.	Download	IDR API Access	OME-Zarr Access via S3
Load image subregion, e.g., single chunk or tile	No , only per file	Yes	Yes
Lazy loading	No	No	Yes . Use Dask collections: <code>da.from_zarr(endpoint_url)</code>
Easily analyze in parallel	No , depends on file format which may require a translation library.	Difficult due to the transfer protocol used (<code>zeroc-ice</code>)	Yes . Use Dask schedulers: <code>dask.delayed(analyze)(t, c, z)</code>

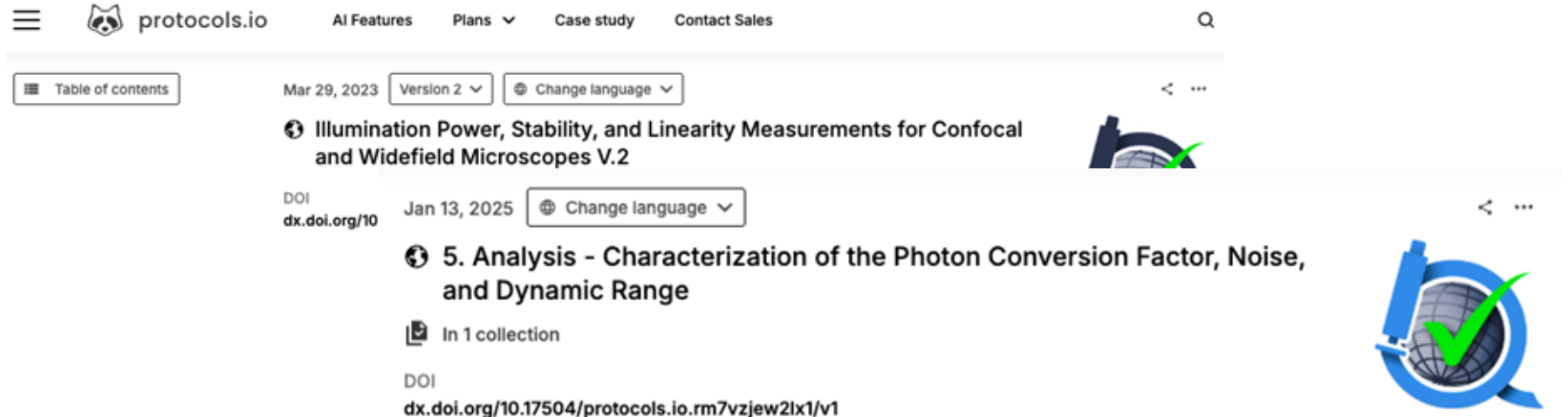
Exercises

- Read Data in IDR using Python API
https://github.com/BioImagingUK/Training/blob/main/notebooks/Read_Image_From_IDR.ipynb
- Explore the multi-resolution
https://github.com/BioImagingUK/Training/blob/main/notebooks/Multi_Resolution.ipynb
- Read OME-Zarr data using Python
https://github.com/BioImagingUK/Training/blob/main/notebooks/Read_OME_Zarr_Image.ipynb

Data preservation: what to preserve?

Quality Control checks

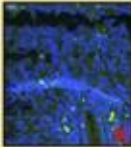
- QUAREP-LiMi: Quality Assessment and Reproducibility for Instruments & Images in Light Microscopy
<https://quarep.org/>
 - comprehensive set of community-agreed guidelines, protocols, automation procedures and other resources to improve quality control, quality assurance, and instrument/method calibration.

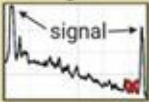


The screenshot displays the protocols.io website interface. At the top, the navigation bar includes a hamburger menu, the protocols.io logo, and links for 'AI Features', 'Plans', 'Case study', and 'Contact Sales'. A search icon is located on the right. Below the navigation bar, a 'Table of contents' button is visible on the left. The main content area shows a protocol titled 'Illumination Power, Stability, and Linearity Measurements for Confocal and Widefield Microscopes V.2' with a version of 'Version 2' and a 'Change language' dropdown. The protocol is dated 'Mar 29, 2023'. Below this, another protocol titled '5. Analysis - Characterization of the Photon Conversion Factor, Noise, and Dynamic Range' is shown, dated 'Jan 13, 2025', with a 'Change language' dropdown. This protocol has a DOI of 'dx.doi.org/10.17504/protocols.io.rm7vzjew2lx1/v1' and is noted as being 'In 1 collection'. A large blue circular logo with a green checkmark and a globe is positioned in the bottom right corner of the screenshot.

Quality Control checks

- Corrupt data (machine read)
 - Intensity distribution, e.g. percentage of saturated pixels, percentage of blank pixels
 - Focus
 - md5 checks after transfers or use transfer protocols that support data integrity
- Automated checks at the instrument or primary storage.

Artifact removal Errors • Tissue folds or tears, deposition errors • Cell loss • Dust speckles / dye aggregates • Focus issues • Uneven staining	Quality control metric • Visual inspection for manual ROI annotation • Intensity distributions (eg. percentage of saturated pixels) • Sharpness	Extra tissue deposition 	Oversaturation 	Tissue fold 
---	--	--	---	---

Illumination correction and background removal				
Errors • Uneven illumination • Remaining background signal • Removal of true signal	Quality control metric • Visual inspection • Signal to noise ratio • Variance of background intensity	Background signal 	Uneven illumination 	Homogenous background signal 

Stitching & Registration				
Errors • Gaps or overlap between adjacent tiles • FOV shifts between cycles	Quality control metric • Visual inspection • Normalized cross correlation • Mutual information	Cell duplication 	Misaligned 	Correct 

Segmentation				
Errors • Merged cells • Incorrect splits • Partial overlaps	Quality control metric • Visual inspection • Average Precision • F1 score • Jaccard index • Accuracy • Cell size	Merged cells 	Split cells 	Correct 

Gating and cell type annotation				
Errors • Subjective gating • Wrong annotations • Missed annotations	Quality control metric • Expert annotation • Jaccard index • Davies-Bouldin index • Silhouette index • Intensity oversaturation • Cell marker impurity	Cell type merging 	Correct 	

Vierdag, W. A. M., & Saka, S. K. (2024). A perspective on FAIR quality control in multiplexed imaging data processing. *Frontiers in bioinformatics*, 4, 1336257.
<https://doi.org/10.3389/fbinf.2024.1336257>

What to preserve

- Preserve raw data (keep it untouched)
- Can preserve processed data (or raw data plus processing and analysis workflows)
- Analysis workflows - [workflowhub.eu](https://www.workflowhub.eu)

Data Retention Policies and Data Embargoes

Data retention policies vary depending on a funder or repository perspective.

Funder (UKRI):

- Different policies depending on the research council.
- BBSRC and MRC say primary data should be retained for 10 years after the project ended.
- BBSRC say data should be shared as soon as possible after publication, and preferably no more than 3 years after the data was generated
- MRC also should be retained for at least 20 years when it's population health and clinical data
- EPSRC also mention 10 years, but metadata should be published after 12 months of the data being generated

Public Image Repository:

- Data can be embargoed for a period after submission.

Licensing and Data Retention

Licenses used by the archives

Creative Commons (CC)

<https://creativecommons.org/share-your-work/cclicenses/>

CC 0: You can reuse the data with no restrictions

CC BY: You can reuse the data with no restrictions other than give proper credit to the creator

There are other more restrictive licenses

Open-source Software Licenses

Do what you want, as long as you include attribution/copyright notice

Examples:

MIT: Very short, minimal requirements. Just include the license and attribution.

Apache 2.0: Like MIT, but also provides an explicit **patent grant**

BSD (2-Clause, 3-Clause): Similar to MIT, with slight differences in attribution requirements.

What do public repositories offer

The Open Knowledge Foundation: Open Data Means Better Science

Jennifer C. Molloy 

Published: December 6, 2011 • <https://doi.org/10.1371/journal.pbio.1001195>

Article

Authors

Metrics

Comments

Media Coverage

Patterns of database citation in articles and patents indicate long-term scientific and industry value of biological data resources

David Bousfield,^{1,2} Johanna McEntyre,³ Sameer Velankar,³ George Papadatos,³ Alex Bateman,³ Guy Cochrane,³ Jee-Hyub Kim,³ Florian Graef,³ Vid Vartak,³ Blaise Alako,³ and Niklas Blomberg^{0,1}

[Author information](#) ▶ [Article notes](#) ▶ [Copyright and License Information](#) ▶

Increased visibility, citations and collaboration

> [PLoS One](#). 2007 Mar 21;2(3):e308. doi: 10.1371/journal.pone.0000308.

Sharing detailed research data is associated with increased citation rate

Heather A Piwowar¹, Roger S Day, Douglas B Fridsma

[Published: 12 March 2015](#)

The open access advantage considering citation, article usage and social media attention

[Xianwen Wang](#) , [Chen Liu](#), [Wenli Mao](#) & [Zhichao Fang](#)

[Scientometrics](#) **103**, 555–564 (2015) | [Cite this article](#)

5672 Accesses | **119** Citations | **368** Altmeteric | [Metrics](#)

[nature](#) > [career feature](#) > article

CAREER FEATURE | 13 May 2019

Data sharing and how it can benefit your scientific career

Open science can lead to greater collaboration, increased confidence in findings and goodwill between researchers.

Funder and Journal requirement

What to include in your application

Contents

- Introduction
- Case for support
- **Data management plan**
- Diagrammatic work plan
- Justification of resources



- methods for data sharing – planned mechanisms for making these data available, for example through deposition in existing public databases or on request, including access mechanisms

Data availability

Please provide a Data Availability statement in the Methods section under “Data Availability”; detailed guidance can be found in our [data availability and data citations policy](#). Certain data types must be deposited in an appropriate public structured data depository (details are available [here](#)), and the accession number(s) provided in the manuscript. Full access is required at publication. Should full access to data be required for peer review, authors must provide it.

Editorial | Published: 17 June 2025

Light microscopy reporting for reproducibility

Nature Cell Biology 27, 877 (2025) | [Cite this article](#)

9656 Accesses | 1 Citations | 20 Altmetric | [Metrics](#)

We announce a cross-journal pilot at Nature Portfolio journals with a goal of implementing standardized light and fluorescence microscopy reporting to improve methodological description and aid in reproducibility efforts.

In the interest of fostering reuse of light and fluorescence microscopy data, we also encourage authors to deposit their microscopy datasets in public repositories, as is routinely expected in other fields, such as genomics and proteomics, which has expanded both the community and the value of data. For deposition, we recommend the [Image Data Resource](#), the [BioStudies](#) database, the [BioImage Archive](#) or figshare, which is [integrated in our submission system](#).

Long term data management is hard



“Kit’s deluge” by [Henning Falk](#), ©2022 [NumFOCUS](#), is used under a [CC BY 4.0](#) license.

Lots of data!

When you're looking for a dataset, use BIA, IDR and EMPIAR first, instead of google or sometimes even papers.

BioImage Archive: <https://www.ebi.ac.uk/bioimage-archive/>
<https://beta.bioimagearchive.org/>

IDR: <https://idr.openmicroscopy.org/>

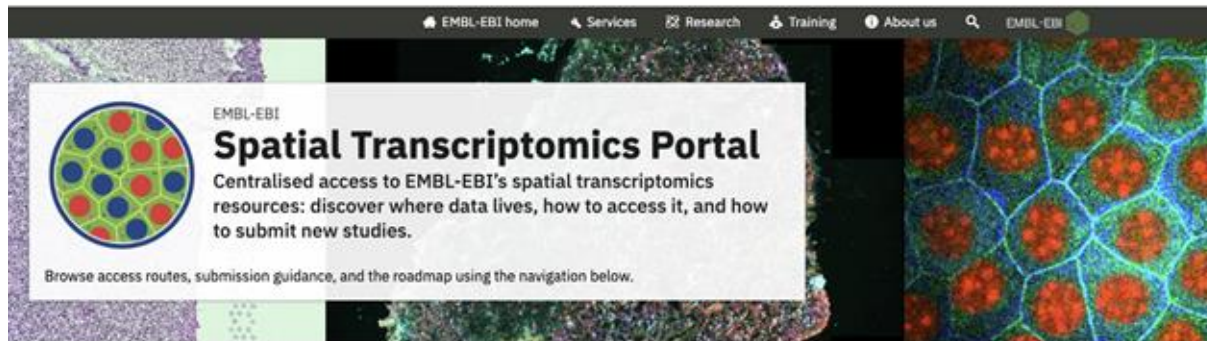
EMPIAR: <https://www.ebi.ac.uk/empiar/>

Exercise (5 minutes)

Find data on your research topic

- <https://beta.bioimagearchive.org/>
- <https://idr.openmicroscopy.org/>

Spatial Transcriptomics Portal



Spatial transcriptomics (ST) brings gene expression into its tissue context. EMBL-EBI hosts both sequencing-based and imaging-based ST data across established resources. We are working towards a unified platform for spatial transcriptomics. In the interim, this hub directs you to the relevant sequencing and imaging archives for retrieving data or submitting new studies.

Access →

Find where sequencing and imaging data live today.

Submit →

Choose the right deposition route for your modality.

Metadata →

Our recommendations for the metadata to include in your submissions.

Roadmap →

See what we are building next. We are currently in Phase 1.

FAQ →

Answers to common questions.

Contact →

Get in touch with the data teams.

About →

Why this site exists and how to use it.

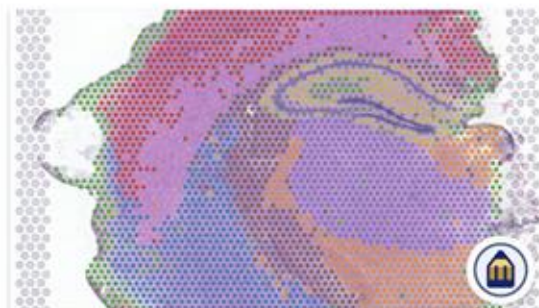
<https://www.ebi.ac.uk/spatial-transcriptomics/>

Spatial Transcriptomics Portal



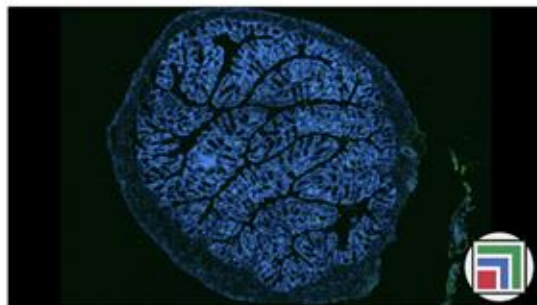
Submit spatial transcriptomics data

The submission path depends on the primary modality. If your submission is related to a sequencing-based experiment, all data, including images, should be deposited to ArrayExpress. If you are submitting an imaging-based experiment, please deposit all data types to the BioImage Archive. Follow the route below and include clear [spatial metadata](#) and sample descriptions.



Sequencing-based ST ➔

e.g. GeoMx or Visium technologies



Imaging-based ST ➔

e.g. CosMx, MERSCOPE, or Xenium technologies

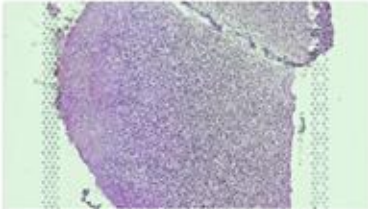
Spatial Transcriptomics Portal



Access spatial transcriptomics data

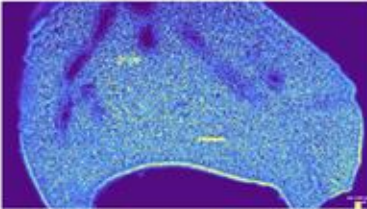
ST data is discoverable in existing EMBL-EBI resources. Choose the route that matches your modality and analysis needs. Each resource provides its own filters, metadata schema, and download options. Use their in-site documentation to refine search and access strategies.

Access routes



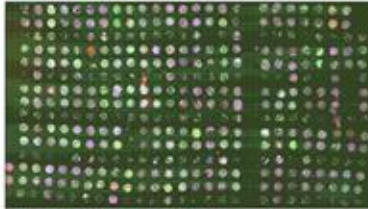
ArrayExpress Collection ➔

Sequencing-based spatial transcriptomics studies and metadata.



BioImage Archive ➔

Imaging-based spatial molecular data and associated files.



Spatial Omics Gallery ➔

Curated spatial omics datasets with imaging and metadata context.

Spatial Transcriptomics Portal

Metadata

While guidance on how to submit through Annotare or the BioImage Archive is available in the respective documentation pages, we are working to establish a shared metadata standard for spatial transcriptomics experiments.

Below are our recommendations for the metadata to include in your submissions in order to facilitate interpretation and reuse of your data.

Sample

Specimen

Experimental

Analytical

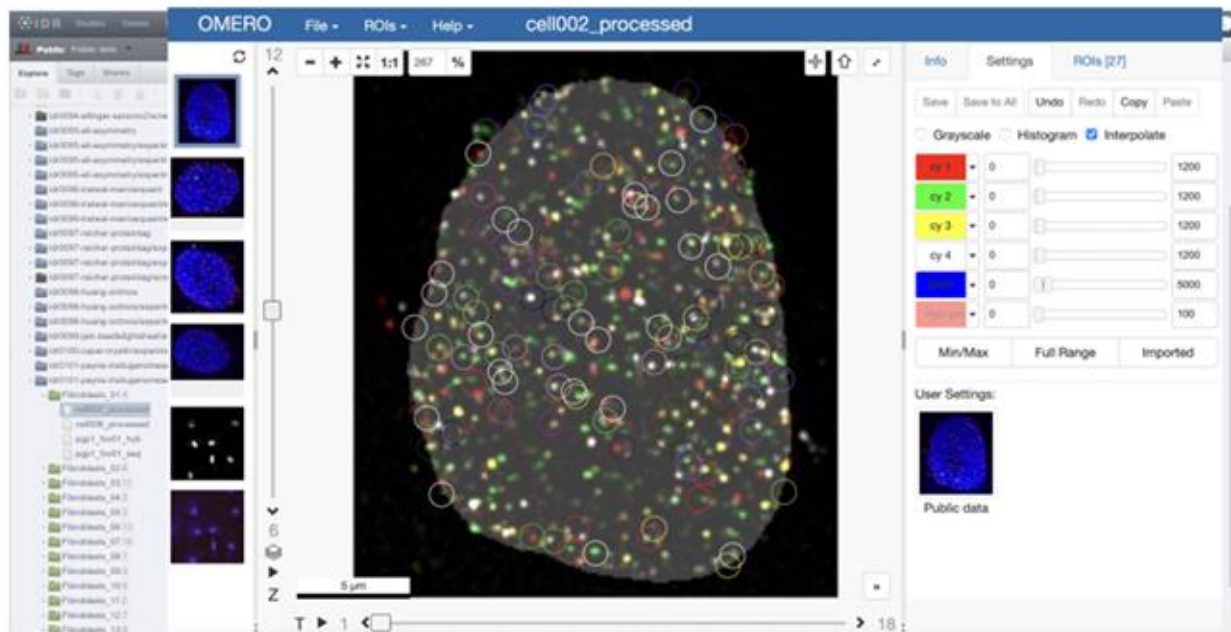
File types imaging based

File types sequencing based

Sample

Field	Definition	Example	Imaging-based ST	Sequencing-based ST
Sample ID	A unique identifier assigned to a biological specimen used in the imaging experiment	liver_01	required	required
Slide ID	A unique identifier for the glass slide or cartridge on which the tissue section was mounted and processed	BSMC_01	required	required
Position on slide	Describe where the capture area is located on the slide. This is required if the slide contains multiple capture areas, for example for visium	Area A1; number 1	optional	optional
Sample preservation method	The method used to preserve the tissue prior to processing (e.g., fresh frozen, FFPE)	fresh frozen	required	required
Organism name	The scientific name of the source organism (e.g., Homo sapiens, Mus musculus)	Homo sapiens (NCBITaxon:9606)	required	required
Time from collection to preservation	Elapsed time between tissue excision/collection and preservation, reported in hours	1h	optional	optional
Section	The thickness of the tissue section	5 µm	required	required

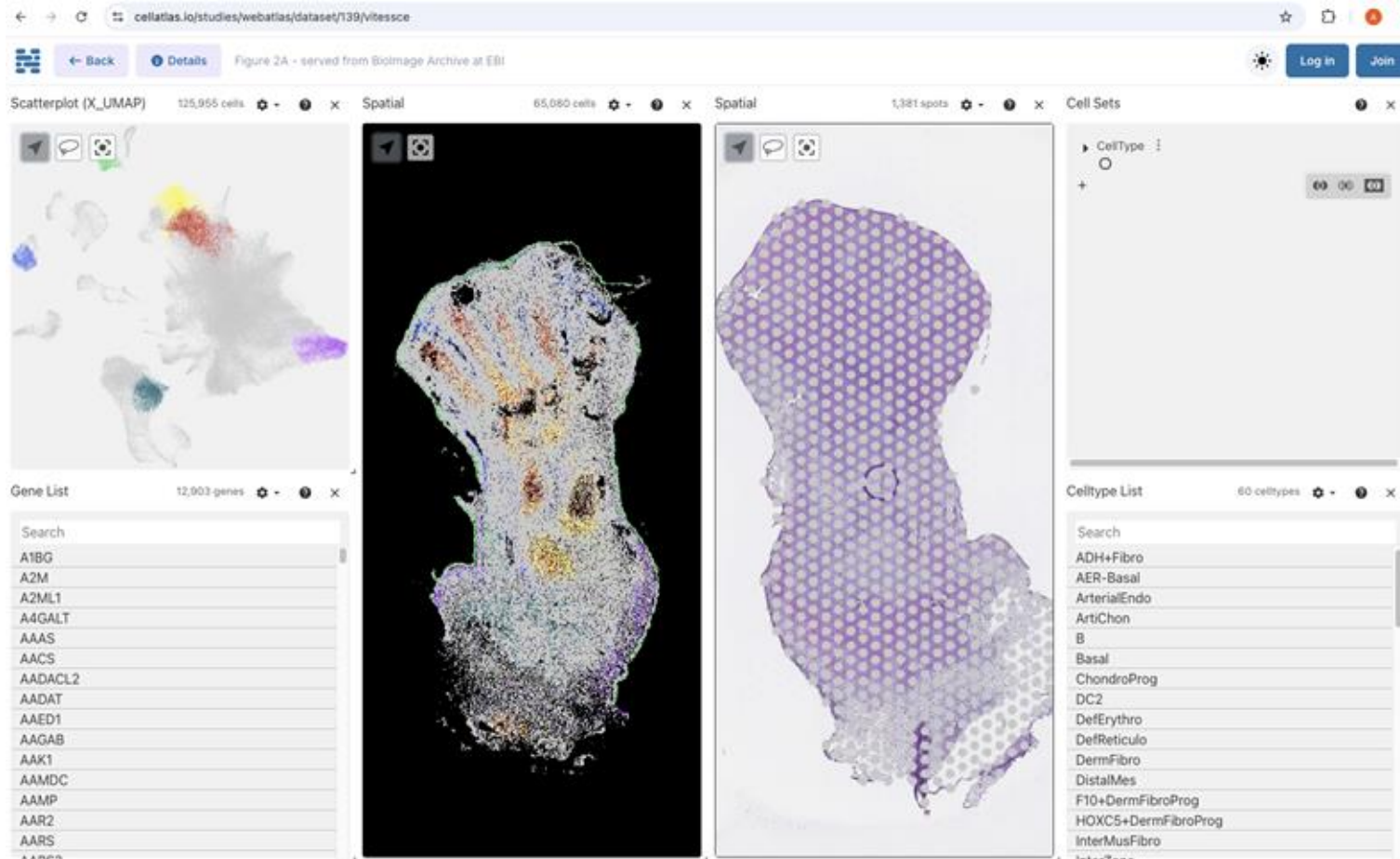
In Situ Sequencing - Linking Multimodal Metadata



idr0101, Payne et al *Science* DOI:10.17867/10000169

Connect Data stored in different resources using the Nucleome Browser
<https://www.nature.com/articles/s41592-022-01559-3>

WebAtlas integrated scRNA-seq. ISS and Visium dataset of the human lower limb



Li, T., Horsfall, D., Basurto-Lozada, D. et al. WebAtlas pipeline for integrated single-cell and spatial transcriptomic data. *Nat Methods* 22, 3–5

(2025). <https://doi.org/10.1038/s41592-024-02371-x> S-BIAD822, S-BIAD887

Re-use cases

Unsupervised and Source-Free Ranking of Biomedical Segmentation Models

Talks et al (2024). <https://arxiv.org/pdf/2503.00450>

S-BIAD895

ZeroCostDL4Mic - Stardist example training and test dataset

Johanna Jukkala ¹, Craig Russell ²

¹ ² European Molecular Biology Laboratory, European Bioinformatics Institute

Accession **S-BIAD895**

<https://www.ebi.ac.uk/biostudies/BioImages/studies/S-BIAD895?query=s-biad895>

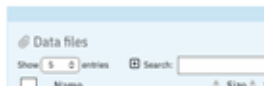
S-BIAD1410

ReSCU-Nets: recurrent U-Nets for segmentation of multidimensional microscopy data

Raymond Hawkins ¹, Negar Balaghi Ly ¹, Kathryn Rothenberg ¹, Michelle Ly ¹, Rodrigo Fernandez-Gonzalez ¹

¹ University of Toronto

Accession **S-BIAD1410**



<https://www.ebi.ac.uk/biostudies/bioimages/studies/S-BIAD1410?query=s-biad1410>

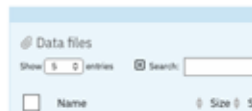
S-BIAD634

An annotated fluorescence image dataset for training nuclear segmentation methods

Sabine Taschner-Mandl ¹, Inge M. Ambros ¹, Peter F. Ambros ¹, Klaus Beiske ², Allan Hanbury ³, Wolfgang Doern ⁴, Tamara Weiss ⁵, Maria Berneder ¹, Magdalena Ambros ¹, Eva Bozsaky ¹

¹ Children's Cancer Research Institute ² Oslo University Hospital ³ TU Wien ⁴ TRAB-Applied and Translational Radiobiology, Department of Radiation Oncology, Medical University of Vienna ⁵ European Bioinformatics Institute (EMBL-EBI)

Accession **S-BIAD634**



Used to train instance segmentation datasets

Used as an instance target dataset

<https://www.ebi.ac.uk/biostudies/bioimages/studies/S-BIAD634>

Re-use cases

Unsupervised and Source-Free Ranking of Biomedical Segmentation Models

Talks et al (2024). <https://arxiv.org/pdf/2503.00450>

S-BIAD895

ZeroCostDL4Mic - Stardist

Johanna Jukkala¹, Craig Russell²

¹ ² European Molecular Biology Laboratory, European Bioinformatics

Accession **S-BIAD895**

<https://www.ebi.ac.uk/biostudies>

S-BIAD1410

ReSCU-Nets: recurrent U-Nets for s data

Raymond Hawkins¹, Nigar Balaghi¹, Kathryn Rothenberg¹, Michelle Ly¹, Rodrigo Fernandez-Gonzalez¹

¹ University of Toronto

Accession **S-BIAD1410**

<https://www.ebi.ac.uk/biostudies/bioimages/studies/S-BIAD1410?query=s-biad1410>

S-BIAD634

An annotated fluorescence image dataset for training nuclear segmentation methods

Sabine Taschner-Mandl¹, Inge M. Ambros¹, Peter F. Ambros¹, Klaus Beiske², Allan Hanbury³, Wolfgang Doern⁴,
Tamara Weiss¹, Maria Berneder¹, Magdalena Ambros¹, Eva Bozsaky¹

¹ Children's Cancer Research Institute ² Oslo University Hospital ³ TU Wien ⁴ TRAB-Applied and Translational Radiobiology, Department of Radiation
Oncology, Medical University of Vienna ⁵ European Bioinformatics Institute (EMBL-EBI)

Accession **S-BIAD634**

<https://www.ebi.ac.uk/biostudies/bioimages/studies/S-BIAD634>

Take home message

- The **original authors did not perform the reuse** to train instance segmentation datasets
- It illustrates the **diverse reuse potential** of annotated biomedical image datasets

Used as an
instance target
dataset

Re-use cases

Reanalysis of cryo-EM data reveals ALK-cytokine assemblies with both 2:1 and 2:2 stoichiometries

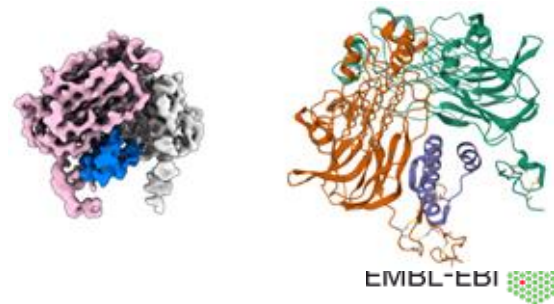
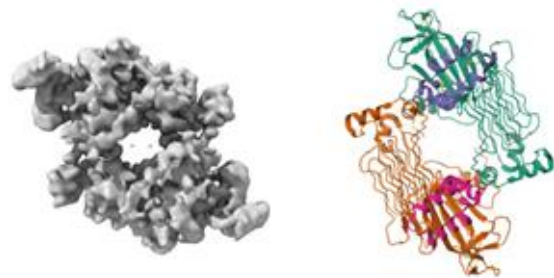
Felix et al., *PLoS Biology* (2025) <https://pubmed.ncbi.nlm.nih.gov/40208865/>

Independent re-processing of raw EMPIAR data

- Used the **uncleaned particle stack** from [EMPIAR-10930](#)
- Avoided particle binning used in the original workflow

The re-analysis revealed that:

- ~53% of particles in the original dataset correspond to a **previously unresolved 2:1 ALK:ALKAL2 stoichiometry**
- Preferred particle orientations had obscured this assembly in the original analysis
- A **new 3.2 Å cryo-EM structure** of the **2:1 ALK^{TG}-EGFL-ALKAL2 complex** was reconstructed



Re-use cases

Reanalysis of cryo-EM data reveals ALK-cytokine assemblies with both 2:1 and 2:2 stoichiometries

Felix et al., *PLoS Biology*

Independent re-proc

- Used the **unc**
- Avoided parti

The re-analysis reveal

Take home message

- The **original authors did not perform the reuse**
- The **same raw dataset** supported **multiple, biologically distinct interpretations**
- Methodological advances made *years later* unlocked new structural information
- Without public deposition of raw data, this discovery would not have been possible

- ~53% of particles in the original dataset correspond to a **previously unresolved 2:1 ALK:ALKAL2 stoichiometry**
- Preferred particle orientations had obscured this assembly in the original analysis
- A **new 3.2 Å cryo-EM structure** of the **2:1 ALK^{TG}-EGFL-ALKAL2 complex** was reconstructed



Submission to the public image repositories



BioImage Archive

A free, publicly available resource for life sciences images from any imaging modality, at any scale



Data are either **associated with a peer-reviewed publication, or of value beyond** a single experiment

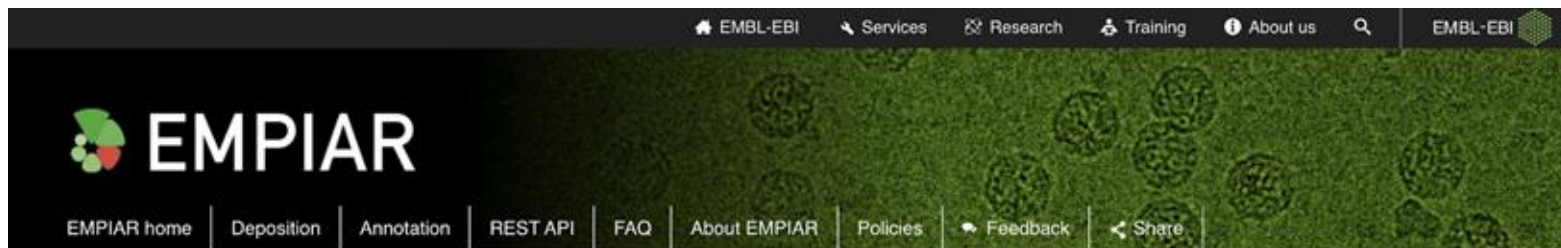
Out of scope:

- Datasets linked to an identifiable individual
- Datasets with a primary use in medical treatment (e.g. diagnostic imaging)

<https://www.ebi.ac.uk/bioimage-archive>

New website (beta version): <https://beta.bioimagearchive.org/>

EMPIAR - Electron Microscopy Public Image Archive



EMPIAR, the Electron Microscopy Public Image Archive, is a public resource for raw images underpinning 3D cryo-EM maps and tomograms (themselves archived in [EMDB](#)). EMPIAR also accommodates 3D datasets obtained with volume EM techniques and soft and hard X-ray tomography. [More ...](#)

As of **2021-11-23**, EMPIAR contains **727** entries, taking up **1.39 PB** of storage.

Browse and download EMPIAR datasets using the table below. The EMDB search system also supports [searches of all EMPIAR entries](#) ([video tutorial](#)).

EMPIAR News

- This recorded talk from the M&M2021 conference brings you up-to-date about the status and plans of EMPIAR and the Biolmage Archive in under 20 minutes.
- EMPIAR has recently moved to a new location. [What do I need to know?](#)

Show entries

Search:

License:



Quick links

Talks and Tutorials

EMPIAR Quick tour

Statistics

Re-use case study

EMPIAR in the news

Contact us

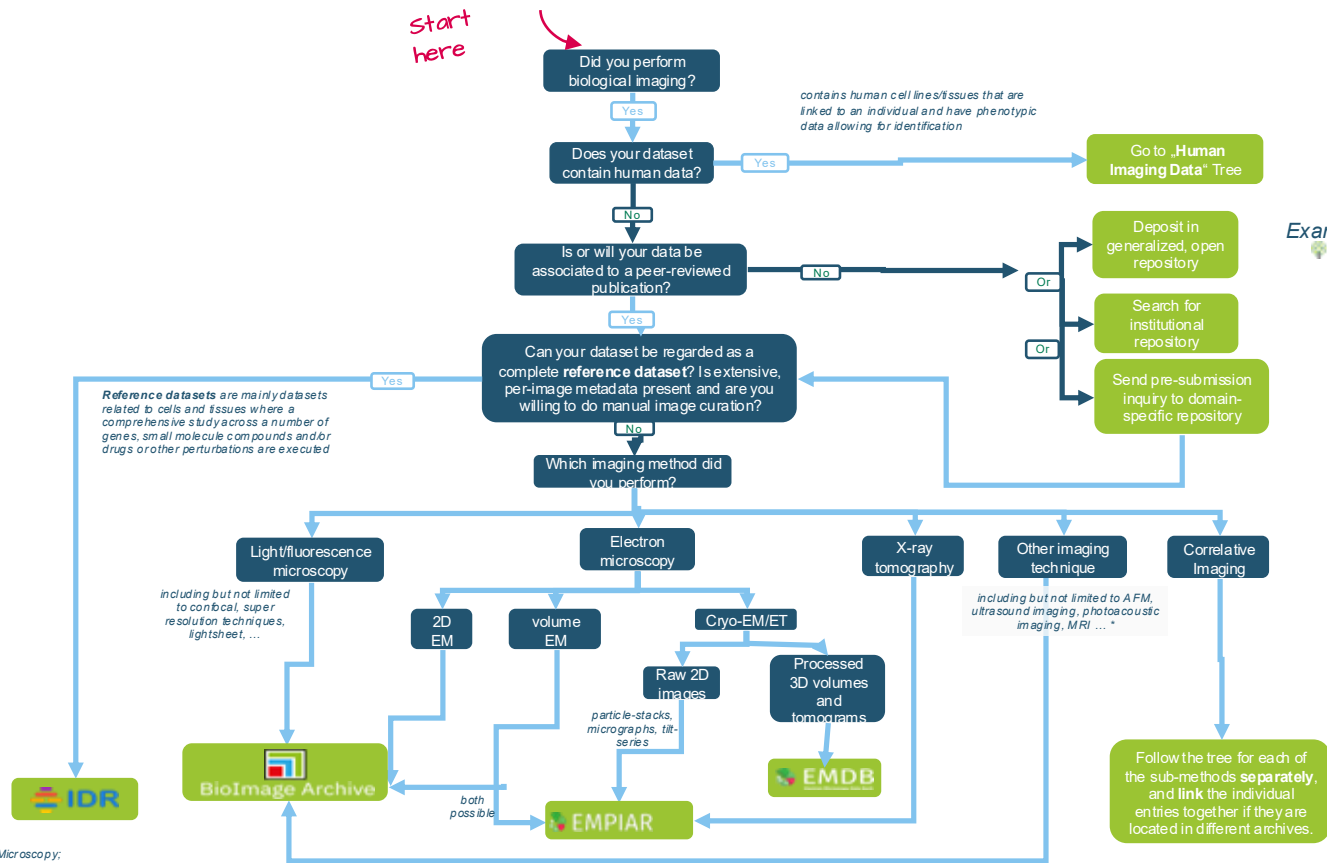
EMDB

PDBe

Biolmage Archive

Where do I deposit my open bioimaging data?

start
here



Examples:

DRYAD

zenodo

BioStudies

figshare

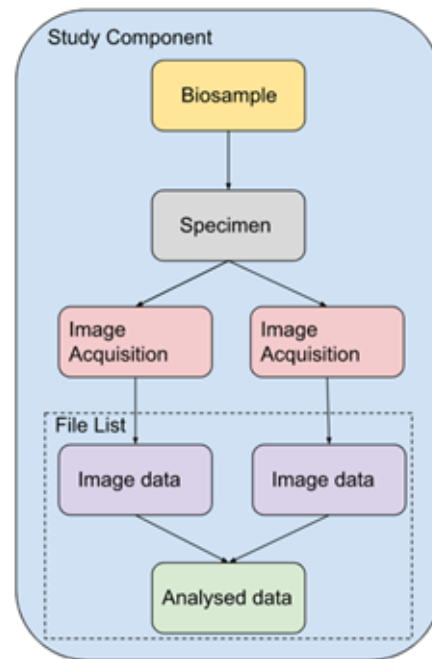
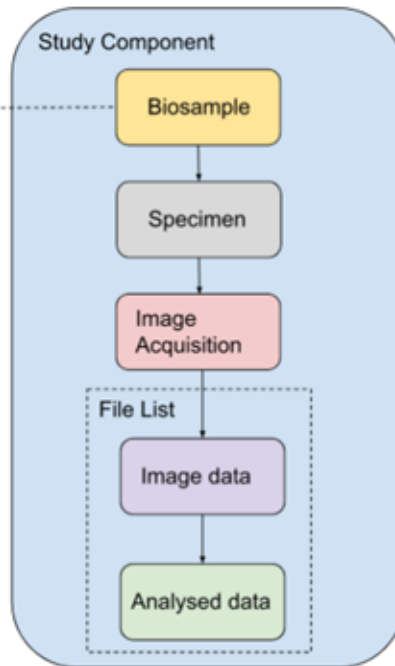
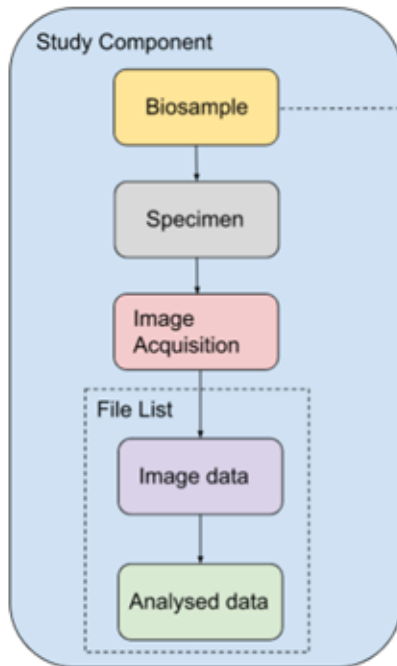
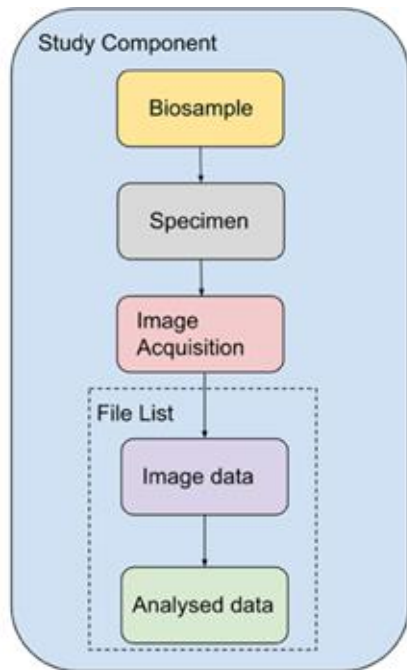
* AFM: Atomic Force Microscopy;
MRI: Magnetic Resonance Imaging

Submitting a study to the BIA

1. Create an account in Biostudies (<https://www.ebi.ac.uk/biostudies/submissions/signin>)
2. Upload your data: web interface, FTP, Aspera, (Globus)
3. Create a draft and fill all the metadata fields that are mandatory or necessary for your submission
4. See how best to organize your submission:
 - a. [One study component or more?](#)
 - b. Do you have annotations?
 - c. [Spatial-transcriptomics](#)
1. You will need one file list per study or annotations component
2. Submit when you are ready
3. Release date is your choice

More [detailed information on the BIA website](#) (with a short 2 min video) and the [BIA online tutorial](#)

REMBI - Study Component Examples

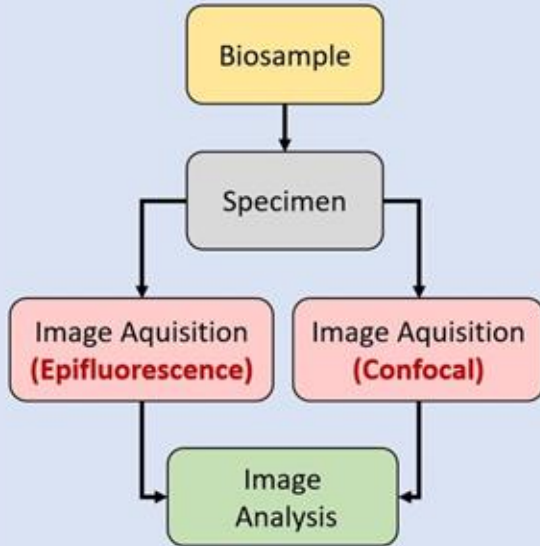


REMBI - Real Life Examples of Study Components

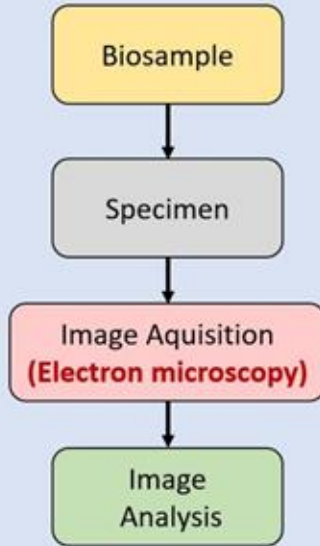
Bronchial Epithelia
no genetic alteration

Bronchial Epithelia
CRISPR-knockout of interferon-genes

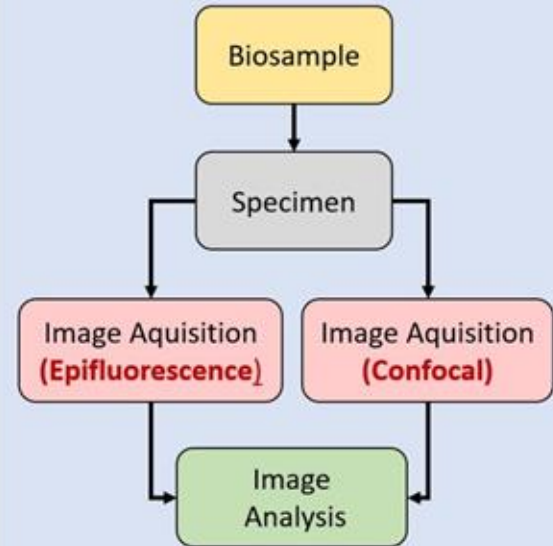
Study Component 1



Study Component 2



Study Component 3



Isabel Kemmer

REMBI - Real Life Examples of Study Components



The screenshot displays the BioImage Archive website interface. The top navigation bar includes links for Home, Browse, Submit, Galleries, Help, Metadata Help, Policies, and About us. The main content area shows a study titled "Bronchial epithelia from adults and children: SARS-CoV-2 sp formation and type III interferon infectivity restriction" by Nicolas Landrein, Harald Wodrich, Guillaume Beucher, Thomas Trian, Marie-Line Andreola, and Isabel Kemmer. The study is associated with the accession number S-BIAD677. The right sidebar lists study components, each with a "Show" and "Download" button. Three red arrows point to these buttons for the first three components.

BioImage Archive
Search BioImage
Examples: brain, retina

Home | Browse | Submit | Galleries | Help | Metadata Help | Policies | About us

BIOSTUDIES / STUDIES / S-BIAD677

Release Date: 2 May 2023 • Modified: 2 May 2023 • Views: 568

Bronchial epithelia from adults and children: SARS-CoV-2 sp formation and type III interferon infectivity restriction

Nicolas Landrein ¹, Harald Wodrich ¹, Guillaume Beucher ¹, Thomas Trian ¹, Marie-Line Andreola ¹, Isabel Kemmer ²

¹ University of Bordeaux ² Euro-BioImaging ERIC

Accession S-BIAD677

Study Component 531 files [Show](#) [Download](#)

Name Infected Multiciliated Cells Form Syncytia with Basal Cells at the Apical Side of the BE

Description we used human reconstituted bronchial epithelia (BE) to investigate the onset of infection and replication of SARS-CoV-2 with widefield and confocal microscopy. BE are an important tissue because subsequent infection of the bronchial tissue determines whether a SARS-CoV-2 infection results in severe or mild respiratory illness, by controlling the spread into the lower respiratory tract. These features differentiate our approach from similar studies using primary respiratory cells from either upper airway (nasal, tracheal)

File List file_list_sc1.json [JSON](#) [XML](#) [PageTab](#)

- ▶ Biosample
- ▶ Specimen
- ▶ Image Acquisition
- ▶ Image Acquisition
- ▶ Image Analysis

Study Component 2 files [Show](#) [Download](#)

Name Electron microscopy of BE after SARS-CoV-2 infection

Description Study of the ultra structure of the BE during an SARS-CoV-2 infection.

File List file_list_sc2.json [JSON](#) [XML](#) [PageTab](#)

- ▶ Biosample
- ▶ Specimen
- ▶ Image Acquisition

Study Component 57 files [Show](#) [Download](#)

Name An Accelerated Type III IFN Response Protects BE from SARSCoV-2 Infection

Description We study the effect of Type III IFN by knocking out the IFN (CRISPR/Cas9) and by treating the BE with Type III IFN

File List file_list_sc3.json [JSON](#) [XML](#) [PageTab](#)

- ▶ Biosample
- ▶ Specimen
- ▶ Image Acquisition
- ▶ Image Acquisition
- ▶ Image Analysis

<https://www.ebi.ac.uk/biostudies/bioimages/studies/S-BIAD677>

Transfer methods available at the BIA

Submission size	Recommended upload method
If less than 50GB total size and Less than 20GB per individual file	<u>Submission tool upload</u>
If up to 1TB total size	FTP
If larger than 1TB	

Please note that we cannot accept single files larger than 2TB.

We also offer Globus by request

Submission Tool

Upload your data via the web interface or get the credentials for FTP/Aspera



BioStudies.
SUBMISSION TOOL

Submissions Drafts Files Help

Home

Upload File Upload Folder FTP/Aspera

Name

Size (in bytes) Actions

FTP/Aspera Upload

You can upload directories to your BioStudies file space via FTP or Aspera. Please click on the relevant method below. Your secret directory is /

Upload using FTP

You can use any FTP client to upload your data, such as [FileZilla](#).

The FTP credentials are:

- Host: `ftp-private.ebi.ac.uk`
- Username: `bs-upload`
- Password:
- Secret directory:

Please use the credentials above to connect to the FTP server, and then upload your submission files into your secret directory.

Upload using Aspera

Submission Tool

Click here to create a draft



Submissions

Drafts

Files

Help

+

New submission

Please fill in the form below. The Validation tab on the right-hand side lists those fields still incomplete or incorrect. Use the **Add or** + buttons to quickly add new rows or tables.

Please note: all fields with * are required

Submit



BioImage Archive

TMP_2025-04-29T14:46:16.756678558Z

Submission Tool - Biostudies

[Submit](#)

New submission

Please select a BioStudies collection that you are submitting to.
Imaging datasets must be submitted using the "BioImages" template. Submissions that do not do so may have their release delayed. For further details please see the [EMBL-EBI policy on imaging data submissions](#).



BioImages
Biimage Archive Submission



Default
General submission



MicrobioRaman
Microbiological Raman Spectroscopy Data



BioImages
Biimage Archive Annotations Only Submission

[Cancel](#)[Add](#)

BioImage Archive

03-02-2026:08:34

A Crossref DOI will be assigned after submission automatically.

▼ Contacts

Name * ⓘ

E-mail * ⓘ

Organisation * ⓘ

Role ⓘ

ORCID ⓘ

|| Melina Velasquez

velasque@ebi.ac.uk

×

Add Row

Add Column

▼ Study

* Required

H Title * ⓘ

ReleaseDate * ⓘ

Select date

Description * ⓘ

Keywords * ⓘ

Acknowledgements ⓘ

License * ⓘ

CC0

Filelist

- List of all the files in your study component(s) and/or annotations component
- File lists are tabular data, either in tsv or Excel (.xlsx) format.
- The first column of the header has to be the word “Files”.
- If submitting annotations, the annotations file list should also include a “source_image” column.
- It has to have one file per line.
- Do not leave blank lines.
- Filenames are case-sensitive.
- File path separator must be forward slash “/”.
- You can assign metadata to an image level

Let's look at the file list of [S-BIAD29](#)

Exercise

BIA live submission

Mock dataset



<https://bit.ly/3VeOYJ3>

Exercise

BIA live submission

1. Create an account <https://wwwdev.ebi.ac.uk/biostudies/submissions/#/signup>
2. Prepare the file list
3. Upload the demo data <https://bit.ly/3VeOYJ3>
4. Create a draft and fill all the metadata fields that are mandatory or necessary for your submission
5. Select the file list
6. Submit

Making changes after submission

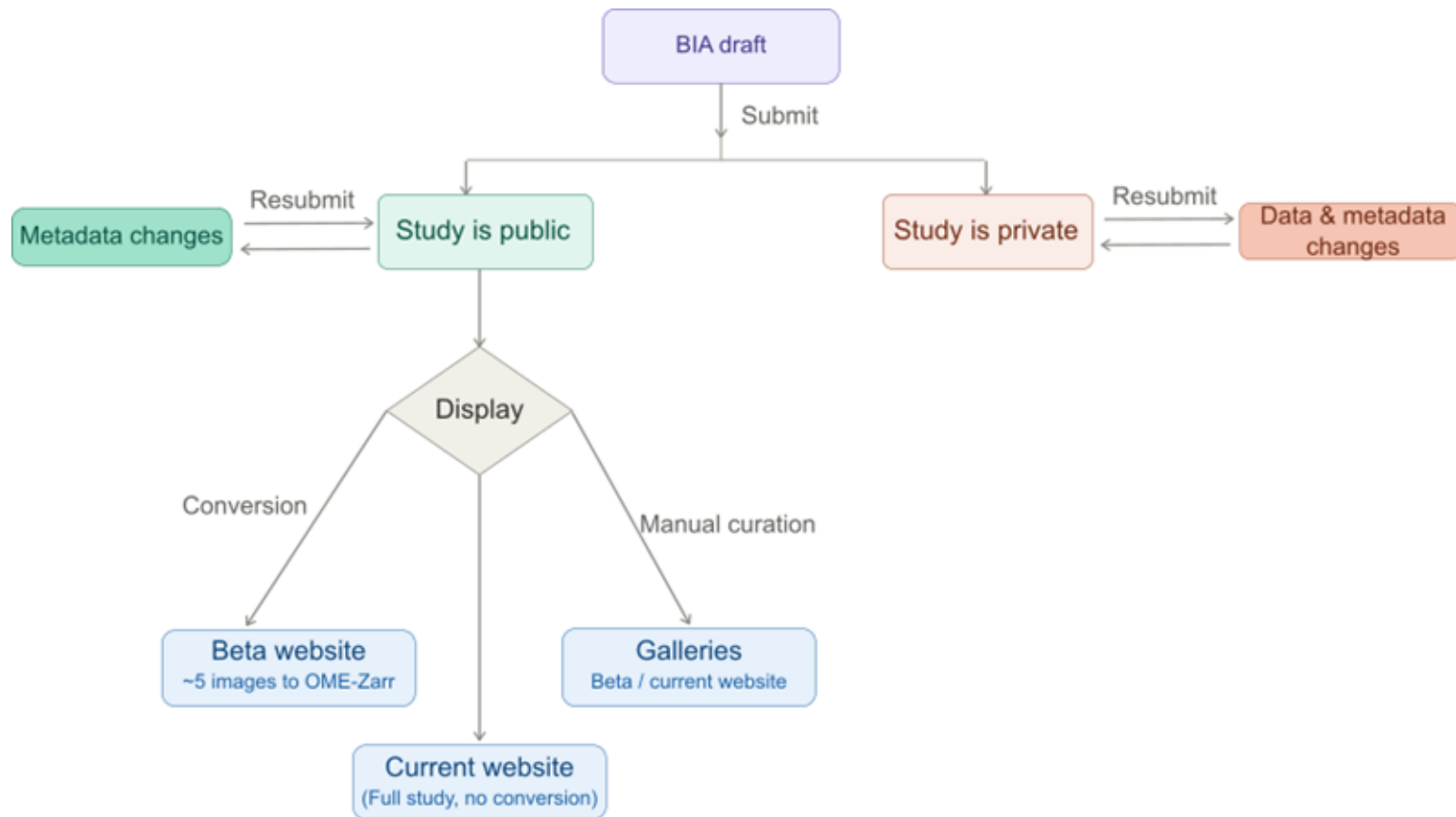
If the submission is still private, you can:

- Change metadata and data
- Add more data
- Change release date
- Delete the whole submission

If the submission is public:

- Change metadata
- Add more data
- You can't modify/delete data

Submission pipeline overview



Take home message

- **Plan and include data management human and infrastructure** needs in your plan and funding application
- Start **collecting metadata** early on, at the instrument if possible and systematically **throughout the data lifecycle**
- Metadata standardisation: Use **REMBI** and ontologies, controlled vocabularies
- Data standardisation: **Bio-Formats** and **OME-NGFF (OME-Zarr)**
- If you're publishing, submit data to **BIA, (IDR) or EMPIAR** (if not patient identifiable) or another FAIR public bioimage repository

Opportunities coming up



- Fellowship call mid-2026
- Pathfinder call late-2026
- [DisCouRSE flexible fund](#) 05/06/2026
- Fund call and Events
<https://casdar.ac.uk/funding/>

Moving forward

- Github repo for training materials:
<https://github.com/BiolImagingUK/Training>
 - We will add this slide deck
- Community Slack channel for knowledge and information sharing
- Feedback for this training: <https://forms.gle/ZBFYSTNDQt9G8Hvb7>

Join slack #



Give feedback

